# Tech pack: Pack 5200 physical test kit

BL-038: order of work, shopping list, guides

| Item | Value |
|---|---|
| generated by | designs/pack_5200/physical_test_kit.py |

# Pack 5200 physical test kit (BL-038)

Manufacturing-engineer, 2026-11-13. Generated by designs/pack_5200/physical_test_kit.py from designs/pack_5200/tests_physical/ (TEST_PLAN.md, test_register.csv, CA1_fatigue_rig.md). Re-run the script after the register changes; do not edit these files by hand.

## Start here

1. Read G1 today: order the long-lead parts and print the frame coupons (they need 7 days).

2. Buy the rest from shopping_list.md (total USD 2,670).

3. Run the sessions G2 to G7 in order. Each guide is one session, written for a phone.

4. Fill results_template.csv (columns measured, date, photo, notes) and send it after each session.

## What gets released, in this order

Frame release R1: H1 print files released for load-carrying prints (DECISIONS BL-027: MT-1 cinch, CA-1, T1/T2 gauges; CA1 doc: with IM3-EC and CA-5).

Tests: MT-1, CA-1, IM3-EC, CA-5, CLEVIS-SHARE, IM3-EP, W-PULL, MT-1cr, T9, T1, T1c-SW, T2, MT-1c

Bag release, step 1: cut and sew: every sewn load point and hardware item proven before the bag fabric is cut.

Tests: HE-FIT, PT-1, MT-2, MT-3, US-BS, PT-2, PT-3, PT-5, PT-6, LL-H, B-28 slip, PORT-1, TL-1, FS-O5, FRAY-1, WP-1

Bag release, step 2: batch: first-article bag checks before more bags are cut (release_tests.csv).

Tests: WP-3, SEW-F, WP-2, T9-R3a, FIT-US, T9-R3c, T9-R3b, T9-R3d, GT-1, FIT-R3, PT-7, WP-4, WP-5

BL-037 weight gates: the -119.1 g A_bl037 picks (weight_push.md).

Tests: SL-1, CS-1, BP-1, AB-1, BU-1, HS-1, ST-1, HB-1, HB-D2

Field release and the rest: first loaded carry, meat haul, belt and comfort tests.

Tests: HB-7, HB-6, CA-3, CA-4, HB-5, T6, T7, CA-2, HB-4, HB-T, PS-1, BP-2, HB-2, BP-3, RF-1, T3, CR-1, SP-1, T4a, T4b, T4c, T5, T5f, T2u, HB-3, HB-D1, HB-D3, FT-S2, MH-1, MH-2, MH-3, R28-1

Data only (no gate): measurements that credit a model; nothing waits on them (BL-041 TL-1b).

Tests: TL-1b

Frame release R1 needs, in order: CA-1 at 0.40 (3/3 run-outs), IM3-EC and CA-5 on the coupons; MT-1 on the bench; then on the first frames IM3-EP and W-PULL (P1), T9, T1, T2 and MT-1c (P2). R1 = CA-1 passes 0.40 with 3/3 run-outs AND MT-1c shows a cinch swing <= 117 % (peak-to-peak / mean) at a mean <= 150 N per anchor (structural countersign, adopted 2026-11-13, analysis/ca1_mt1c_confirm.md); if 0.40 fails: 3 XY bars at the table level for the measured swing, >= 0.313 (<= 0 %: 0.313 / <= 5 %: 0.316 / <= 10 %: 0.319 / <= 20 %: 0.325 / <= 30 %: 0.331 / <= 50 %: 0.345 / <= 70 %: 0.361 / <= 100 %: 0.386 / <= 115 %: 0.398). Swing > 115 %: STOP, no fallback level: the structural-fea-engineer re-runs the cinch case. The 0.276 / 0.404 levels are retired.

## Sessions

| session | what | when | hands-on h | machine h |
|---|---|---|---|---|
| G1 | Order the long-lead parts, print and condition the coupons | week 1, day 1 | 0.0 | 0 |

| session | what | when | hands-on h | machine h |
|---|---|---|---|---|
| G2 | Pull bench weekend: hooks, cams, bartacks, ladder locks | weekend 1 (week 1) | 25.0 | 7 |
| G3 | Wash and look: seams without binding | weekend 1, evenings | 3.0 | 9 |
| G4 | Lever rig: frame coupons (weeks 2-5) | build weekend 2, then runs day and night | 38.0 | 684 |
| G5 | Bench session 2: fabric, belt and back coupons | any weekend while the lever rig runs (weeks 2-4) | 9.9 | 8 |
| G6 | First-article frames on the stand (P1 then P2) | after G4's CA-1, IM3-EC and CA-5 pass (about week 5-6) | 58.7 | 37 |
| G7 | Complete pack: fit, field and first-article bag checks | after T1-T4 pass | 25.9 | 2 |
| | total | | 160.5 | 747 |

Hands-on and machine hours are the register's estimates (ASSUMPTION per row). The calendar is set by the lever rig (about 8 CA-1 runs, 3 IM3-EC, CA-5 and CA-3/CA-4 at 1 Hz, ~28 h each) and by the 4-week lab turnaround for AB-1.

## Order of work

Full list: order_of_work.csv. Inside each session the R1 tests run first, then the bag, then the BL-037 gates (tests that share a rig or a specimen batch stay together).

| # | session | test | releases |
|---|---|---|---|
| 1 | G1 | CA-1 (print only) | Frame release R1 |
| 2 | G1 | IM3-EC (print only) | Frame release R1 |
| 3 | G1 | CA-5 (print only) | Frame release R1 |
| 4 | G1 | CLEVIS-SHARE (print only) | Frame release R1 |
| 5 | G1 | CA-3 (print only) | Field release and the rest |
| 6 | G1 | CA-4 (print only) | Field release and the rest |
| 7 | G1 | HB-6 (print only) | Field release and the rest |
| 8 | G1 | T7 (print only) | Field release and the rest |
| 9 | G2 | MT-1 | Frame release R1 + Bag release, step 1: cut and sew |
| 10 | G2 | HE-FIT | Bag release, step 1: cut and sew |
| 11 | G2 | PT-1 | Bag release, step 1: cut and sew |
| 12 | G2 | MT-2 | Bag release, step 1: cut and sew |
| 13 | G2 | MT-3 | Bag release, step 1: cut and sew |
| 14 | G2 | US-BS | Bag release, step 1: cut and sew |
| 15 | G2 | PT-2 | Bag release, step 1: cut and sew |
| 16 | G2 | PT-3 | Bag release, step 1: cut and sew |
| 17 | G2 | PT-5 | Bag release, step 1: cut and sew |
| 18 | G2 | PT-6 | Bag release, step 1: cut and sew |
| 19 | G2 | LL-H | Bag release, step 1: cut and sew |
| 20 | G2 | B-28 slip | Bag release, step 1: cut and sew |
| 21 | G2 | PORT-1 | Bag release, step 1: cut and sew |
| 22 | G2 | TL-1 | Bag release, step 1: cut and sew |

| # | session | test | releases |
|---|---|---|---|
| 23 | G2 | FS-O5 | Bag release, step 1: cut and sew |
| 24 | G2 | WP-3 | Bag release, step 2: batch |
| 25 | G2 | SL-1 | BL-037 weight gates |
| 26 | G2 | HB-7 | Field release and the rest |
| 27 | G2 | TL-1b | Data only (no gate) |
| 28 | G3 | FRAY-1 | Bag release, step 1: cut and sew |
| 29 | G3 | WP-1 | Bag release, step 1: cut and sew |
| 30 | G4 | CA-1 | Frame release R1 |
| 31 | G4 | IM3-EC | Frame release R1 |
| 32 | G4 | CA-5 | Frame release R1 |
| 33 | G4 | CLEVIS-SHARE | Frame release R1 |
| 34 | G4 | SEW-F | Bag release, step 2: batch |
| 35 | G4 | CS-1 | BL-037 weight gates |
| 36 | G4 | BP-1 | BL-037 weight gates + Field release and the rest |
| 37 | G4 | HB-6 | Field release and the rest |
| 38 | G4 | CA-3 | Field release and the rest |
| 39 | G4 | CA-4 | Field release and the rest |
| 40 | G4 | HB-5 | Field release and the rest |
| 41 | G4 | T6 | Field release and the rest |
| 42 | G4 | T7 | Field release and the rest |
| 43 | G4 | CA-2 | Field release and the rest |
| 44 | G5 | WP-2 | Bag release, step 2: batch |
| 45 | G5 | AB-1 | BL-037 weight gates |
| 46 | G5 | HB-4 | Field release and the rest |
| 47 | G5 | HB-T | Field release and the rest |
| 48 | G5 | PS-1 | Field release and the rest |
| 49 | G5 | BP-2 | Field release and the rest |
| 50 | G5 | HB-2 | Field release and the rest |
| 51 | G6 | IM3-EP | Frame release R1 |
| 52 | G6 | W-PULL | Frame release R1 |
| 53 | G6 | MT-1cr | Frame release R1 |
| 54 | G6 | T9 | Frame release R1 + Bag release, step 2: batch |
| 55 | G6 | T9-R3a | Bag release, step 2: batch |
| 56 | G6 | FIT-US | Bag release, step 2: batch |
| 57 | G6 | T9-R3c | Bag release, step 2: batch |
| 58 | G6 | BP-3 | Field release and the rest |
| 59 | G6 | RF-1 | Field release and the rest |

| # | session | test | releases |
|---|---|---|---|
| 60 | G6 | T1 | Frame release R1 |
| 61 | G6 | T1c-SW | Frame release R1 |
| 62 | G6 | T3 | Field release and the rest |
| 63 | G6 | T2 | Frame release R1 |
| 64 | G6 | MT-1c | Frame release R1 |
| 65 | G6 | CR-1 | Field release and the rest |
| 66 | G6 | BU-1 | BL-037 weight gates |
| 67 | G6 | SP-1 | Field release and the rest |
| 68 | G6 | T4a | Field release and the rest |
| 69 | G6 | T4b | Field release and the rest |
| 70 | G6 | T4c | Field release and the rest |
| 71 | G6 | T5 | Field release and the rest |
| 72 | G6 | T5f | Field release and the rest |
| 73 | G6 | T2u | Field release and the rest |
| 74 | G7 | T9-R3b | Bag release, step 2: batch |
| 75 | G7 | T9-R3d | Bag release, step 2: batch |
| 76 | G7 | GT-1 | Bag release, step 2: batch |
| 77 | G7 | FIT-R3 | Bag release, step 2: batch |
| 78 | G7 | PT-7 | Bag release, step 2: batch |
| 79 | G7 | WP-4 | Bag release, step 2: batch |
| 80 | G7 | WP-5 | Bag release, step 2: batch |
| 81 | G7 | HS-1 | BL-037 weight gates |
| 82 | G7 | ST-1 | BL-037 weight gates |
| 83 | G7 | HB-1 | BL-037 weight gates + Field release and the rest |
| 84 | G7 | HB-D2 | BL-037 weight gates + Field release and the rest |
| 85 | G7 | HB-3 | Field release and the rest |
| 86 | G7 | HB-D1 | Field release and the rest |
| 87 | G7 | HB-D3 | Field release and the rest |
| 88 | G7 | FT-S2 | Field release and the rest |
| 89 | G7 | MH-1 | Field release and the rest |
| 90 | G7 | MH-2 | Field release and the rest |
| 91 | G7 | MH-3 | Field release and the rest |
| 92 | G7 | R28-1 | Field release and the rest |

## Value of the gram tests (re-scored, TEST_PLAN §5)

| rank | test | grams | g per $-eq | note |
|---|---|---|---|---|
| 1 | BP-1 | 49.9 | 2.67 | BL-005 strip with BP-2 (44.6) + BL-037 K-1 2 mm core |
| 2 | FT-S2 | 101.6 | 2.26 | owner's wing removal (BL-023) |
| 3 | CS-1 | 54.2 | 1.89 | BL-037 B-1 + H-1 (with HB-1 / HS-1) |
| 4 | HB-D3 | 12.6 | 1.68 | BL-006 f1 + f2 |
| 5 | HB-1 | 77.2 | 1.29 | front-strap pad (42.9) + BL-037 B-1, B-3 (with CS-1) |
| 6 | WP-3 | 5.4 | 1.20 | protects A-2 (already in B.2) |
| 7 | WP-1 | 36.4 | 1.04 | protects B-1 (already in B.2) |
| 8 | WP-5 | 12.8 | 1.02 | protects B-10 (already in B.2) |
| 9 | HB-D1 | 22.4 | 1.00 | BL-006 b |
| 10 | ST-1 | 6.0 | 0.98 | BL-037 H-2 |
| 11 | HB-D2 | 19.5 | 0.56 | BL-006 c2 + BL-037 B-2 0.5 mm |
| 12 | HS-1 | 22.3 | 0.46 | BL-037 H-1 |
| 13 | SL-1 | 15.0 | 0.46 | BL-037 G-1 + X-1 |
| 14 | WP-2 | 6.2 | 0.40 | C-6 lash tabs |
| 15 | HB-5 | 25.0 | 0.24 | package C plate; also a safety gate |
| 16 | BU-1 | 4.0 | 0.16 | BL-037 F-3 |
| 17 | RF-1 | 7.2 | 0.13 | C-8 rails |
| 18 | CA-1 | 9.9 | 0.09 | option F-1 only: CA-1 at 0.40 AND MT-1c static AND a buckling re-run (BL-037); not booked. CA-1 stays the top safety gate |
| 19 | FS-O5 | 7.0 | 0.08 | C-5 floor (alternate set); needs PT-3 on ULTRA too |
| 20 | AB-1 | 22.5 | 0.07 | BL-037 G-2 (Martindale lab fee dominates) |

## Results sheet

results_template.csv has 357 rows over 84 tests. Each row says how it is judged (check: max, min, band, yes, record) against limit. Type numbers in the unit shown; type Y or N for yes/no rows. Do not change the other columns. Read back with:

python -m huntkit.manufacturing.test_results outputs/release/test_kit/results_template.csv --md report.md

It reports each test as PASS / FAIL / PARTIAL / OPEN, each release, and each ASSUMPTION as VERIFIED / FAILED / OPEN (settles column).

## Files

- guides/G1.md ... guides/G7.md: one session each

- shopping_list.md, shopping_list.csv

- order_of_work.csv

- results_template.csv

- sketches/.png: rig sketches

- test_kit.pdf: everything above in one PDF

- US test piece (HE-FIT step 0, CLEVIS-SHARE): designs/pack_5200/print/frame_h1/test_pieces/us_route_stay_R.stl (+ .3mf, _frame_coords.stl) plus the bolt-on clevis designs/pack_5200/print/frame_h1/us2_clevis_R.stl (+ .3mf), both in the frame print folder (BL-049 / BL-050: R stay z 40-160 with the hinge fairlead and the US1 / US2 lane lugs; the stay piece is printed without the US2 outer cheek, bolt the clevis on with 2 Ti M5 x 35 DIN 7984, heads in the counterbores with no washer, 10 mm DIN 433 washers on the nut side only; BL-051 / 052). The old kit copy test_kit/print/ (BL-048 stations) is retired

- crown pawl for P1 and BU-1: the LOW-PROFILE pawl designs/pack_5200/print/frame_h1/crown_pawl_lp_fdm_pacf.stl (BL-047; the old crown_pawl_ does not fit H1)

- manifest.json

# Shopping list (BL-038)

Total USD 2,670.42. Prices from material records, cited shop pages (links), or marked ASSUMPTION (check at purchase). Cited or record-priced: USD 1,037.04; ASSUMPTION: USD 1,633.38. Prices checked 2026-11-10 (TEST_PLAN §4) and 2026-11-13.

Order these first (long lead):

- cam buckle 25 mm aluminium (Austere type) (MT-3, US-BS): LONG LEAD: RBTR all variants sold out, Extremtextil stock problems (2026-11-13). Order now; buy the pack's 5 cams in the same order

- YKK #10 coil slider, metal (cord pull fitted) (SL-1): LONG LEAD: Rockywoods listed sold out (2026-11-13); Shelby Outdoor EU stocks the YKK 10C slider

- ROHACELL 71 IG-F core for the 2 mm strip (BP-1): LONG LEAD: 2 mm is not a stock sheet (3 mm thinnest, back_system_integration.md); sand a 3 mm offcut or special order

- HyperD 300 printable, custom camo print (AB-1, FRAY-1): LONG LEAD: custom print, lead time not published

- Martindale abrasion test, accredited lab (2 samples + order fee) (AB-1): LONG LEAD: about 4 weeks after the samples arrive

- sacrificial P1 frame (IM3-EP, W-PULL, then T2u) (IM3-EP, W-PULL, T2u): print time: allow 2 weeks with P2

| kit | USD |
|---|---|
| Kit A pull bench | 246.38 |
| Kit B lever rig | 431.58 |
| Kit C frame stand | 227.39 |
| Kit D measuring | 140.46 |
| Specimens and lab | 1,624.61 |
| total | 2,670.42 |

Not in this list: the bag, harness and belt materials themselves (pack BOM, outputs/release/pack5200_corrie_f_rev3/manufacturing/bom.csv). Swatches are cut from that fabric order, so buy the bag fabric with the kit. You already own (kit E): washing machine, winter gloves, freezer, hose, tape, paint pen, phone.

## Kit A pull bench

| item | qty | USD | source | for |
|---|---|---|---|---|
| crane scale 300 kg / 660 lb, 0.1 kg, tare | 1 each | 59.00 | Fruugo US | all pulls |
| come-along, 2 ton cable puller (Big Red TR8021) | 1 each | 30.30 | Home Depot | all pulls |
| ratchet straps 1-1/2 in x 10 ft, 4-pack | 1 pack | 17.99 | Harbor Freight 61303 | rigging (never above 151 kg) |
| hardened dowel pins M5 x 22, 50-pack | 1 pack | 11.79 | Zoro | pin fixture |
| hardened dowel pin M5 x 50, DIN 6325 (m6), HRC 58-62 (NOT x 40: the hinge span is 40 mm since BL-049, review 5 m-4) | 3 each | 9.00 | Zoro / any DIN 6325 stockist; ASSUMPTION USD 3 each, as the x 40 (Heritage DOWMH-050-040 page refused fetch; the x 50 length is a standard DIN 6325 length, part number not verified); https://www.zoro.com/heritage-industrial-dowel-pin-m5-nominal-dia-alloy-steel-1-pk-dowmh-050-040/i/G100006107/ | PT-3, SEW-F (HE), HE-FIT: the fixture pin (BL-048 M-C) |
| US2 CLEVIS ONLY: Ti gr5 low-head socket cap screw M5 x 35, DIN 7984 (NOT M5 x 45) | 5 each | 12.50 | Ti fastener shop (e.g. eBay / AliExpress Ti gr5 DIN 7984); ASSUMPTION (shop price, check at purchase) | HE-FIT step 0 (2), CLEVIS-SHARE (2), spare (1) |
| US2 CLEVIS: M5 hex nut A2, ISO 4032 (8 AF x 4 mm; thin nuts do not fit the trap) | 5 each | 0.50 | hardware store; ASSUMPTION (shop price, check at purchase) | HE-FIT step 0, CLEVIS-SHARE |

| item | qty | USD | source | for |
|---|---|---|---|---|
| US2 CLEVIS ONLY: M5 washer 10 mm OD x 1 mm A2, DIN 433 (NOT the 12 mm z 234 washer; nut side only, BL-051) | 6 each | 0.30 | hardware store; ASSUMPTION (shop price, check at purchase) | HE-FIT step 0 (2), CLEVIS-SHARE (2), spare (2) |
| anchors: 4x4 lag-screwed into studs (or tow hitch / tree) | 1 set | 40.00 | lumber yard; ASSUMPTION (shop price, check at purchase) | bench |
| pin fixture: steel flat bar 40 x 6 x 300, M8 bolts | 1 set | 20.00 | hardware store; ASSUMPTION (shop price, check at purchase) | bench |
| bar clamps: 2 pairs Al 150 x 25 x 10, M6 bolts, grit cloth | 1 set | 15.00 | hardware store; ASSUMPTION (shop price, check at purchase) | PT-2, PT-3, FS-O5 |
| safety: glasses, gloves, moving blanket, back-up sling | 1 set | 20.00 | hardware store; ASSUMPTION (shop price, check at purchase) | all |
| calibration jug 20 L | 1 each | 10.00 | hardware store; ASSUMPTION (shop price, check at purchase) | scale check |

# Kit B lever rig

| item | qty | USD | source | for |
|---|---|---|---|---|
| z 234 JOINT ONLY: Ti gr5 low-head socket cap screw M5 x 45, DIN 7984 (NOT M5 x 35) | 8 each | 20.00 | Ti fastener shop; ASSUMPTION (shop price, check at purchase) | CA-5 (4 stub pairs x 2) |
| z 234 JOINT: M5 hex nut A2, ISO 4032 | 8 each | 0.80 | hardware store; ASSUMPTION (shop price, check at purchase) | CA-5 |
| z 234 JOINT ONLY: M5 washer 12 mm OD x 1 mm A2 (load-bearing; NOT the 10 mm clevis washer) | 16 each | 1.28 | hardware store; ASSUMPTION (shop price, check at purchase) | CA-5 |
| S-type load cell 100 kg | 1 each | 23.60 | Makerfabs | CA-*, IM3-EC, MT-1c |
| HX711 load-cell amplifier (SparkFun) | 1 each | 11.50 | SparkFun | CA-*, MT-1c |
| foil strain gauges 120 ohm, 1 mm grid, pre-wired, pack of 10 | 2 pack | 50.00 | electronics shop; ASSUMPTION (shop price, check at purchase) | T1, MT-1cr (US1 + US2 roots on P1 / P2), CLEVIS-SHARE (US2 root): 5 active + 5 dummy (BL-054 m-8) + spares |
| half-bridge completion: 120 ohm 0.1 % resistors (2 per channel) + 2nd and 3rd HX711 modules; the dummy gauges come from the same pack of 10 (one per active gauge, on an unloaded offcut of the same print) | 1 set | 30.00 | electronics shop; ASSUMPTION (shop price, check at purchase) | T1, MT-1cr, CLEVIS-SHARE (one channel per gauge; BL-054, review 5 m-8: temperature compensation) |
| cyanoacrylate strain-gauge adhesive + 400 grit paper + isopropanol | 1 set | 12.00 | electronics shop; ASSUMPTION (shop price, check at purchase) | gauge bonding on the sanded PAHT |
| Arduino Nano Every | 1 each | 15.81 | Arduino Store (EUR 14.50) | CA-*, MT-1c, FSR strips |
| gear motor 12 V 60 W 60 rpm 8 N m worm (uxcell) | 1 each | 54.99 | eBay | CA-* |
| 12 V 10 A power supply | 1 each | 20.00 | electronics shop; ASSUMPTION (shop price, check at purchase) | CA-* |
| MOSFET switch, hall sensor + magnets, AS5600 encoder | 1 set | 15.00 | electronics shop; ASSUMPTION (shop price, check at purchase) | CA-* |
| 6061-T6 flat bar 40 x 10 x 600 (lever) | 1 each | 18.00 | metal supplier; ASSUMPTION (shop price, check at purchase) | CA-* |
| 2 x 608 bearings, M8 shoulder bolt, 2 rod ends, M8 rod | 1 set | 25.00 | hardware store; ASSUMPTION (shop price, check at purchase) | CA-* |

| item | qty | USD | source | for |
|---|---|---|---|---|
| compression spring 10-12 N/mm, >= 350 N, washer cage | 1 each | 10.00 | hardware store; ASSUMPTION (shop price, check at purchase) | CA-* |
| turnbuckle M8, Al grip plates 4 x 50 x 40 x 6, M6 bolts | 1 set | 20.00 | hardware store; ASSUMPTION (shop price, check at purchase) | CA-* |
| plywood 18 mm + 2x4, screws (rig base) | 1 set | 20.00 | lumber yard; ASSUMPTION (shop price, check at purchase) | CA-* |
| S-type load cell 200 kg (CA-5 static pulls) | 1 each | 23.60 | Makerfabs (200 kg option) | CA-5 |
| 40 C box: cooler + 60 W heater + thermostat | 1 set | 35.00 | hardware store; ASSUMPTION (shop price, check at purchase) | CA-3, CA-4, CS-1 |
| 3-point bend fixture | 1 set | 10.00 | hardware store; ASSUMPTION (shop price, check at purchase) | CA-3, BP-1 |
| SD-card logger module + USB power bank (MT-1c, logs while the rig moves) | 1 set | 15.00 | electronics shop; ASSUMPTION (shop price, check at purchase) | MT-1c |

## Kit C frame stand

| item | qty | USD | source | for |
|---|---|---|---|---|
| hip log (4x4 in closed-cell foam to ~36 in) | 1 each | 10.00 | hardware store; ASSUMPTION (shop price, check at purchase) (load_standard.md 5.1) | T1-T5 |
| shoulder bar | 1 each | 10.00 | hardware store; ASSUMPTION (shop price, check at purchase) (load_standard.md 5.1) | T1-T5 |
| snap shackle, 316 swivel (drop release) | 1 each | 14.39 | Better Boat | T4a, MT-1c |
| sand tube bags 60 lb | 8 each | 48.00 | hardware store; ASSUMPTION (shop price, check at purchase) | T1-T5, T9, MT-1c, T2u |
| rubber gym mat 10-20 mm | 1 each | 30.00 | sports shop; ASSUMPTION (shop price, check at purchase) (load_standard.md 5.1) | T4 |
| 19 mm plywood sheet | 1 each | 40.00 | lumber yard; ASSUMPTION (shop price, check at purchase) | T4c |
| T5 crank: 12 V wiper motor, supply, crank, bungee, counter | 1 set | 70.00 | auto parts; ASSUMPTION (shop price, check at purchase) (load_standard.md 5.1) | T5, MT-1c |
| 22 mm hardwood dowel | 1 each | 5.00 | hardware store; ASSUMPTION (shop price, check at purchase) | SP-1 |

## Kit D measuring

| item | qty | USD | source | for |
|---|---|---|---|---|
| dial indicator 0-1 in with magnetic base (iGaging 34-101) | 1 set | 39.95 | Penn Tool | CA-5, T1, T4b |
| digital caliper 6 in | 1 each | 9.99 | Harbor Freight 63586 | CA-*, CS-1 |
| force-sensitive resistors (Adafruit 166) | 8 each | 31.60 | Adafruit | HB-1, HS-1, RF-1, BP-3 |
| 10x loupe | 1 each | 8.00 | hardware store; ASSUMPTION (shop price, check at purchase) | FRAY-1, WP-1, CA-* |
| conditioning box + magnesium nitrate | 1 set | 15.00 | chemical supplier; ASSUMPTION (shop price, check at purchase) | CA coupons |
| pocket scale 200 g x 0.01 g | 1 each | 13.30 | Ward's Science (United Scientific) | SL-1, T9-R3d |

| item | qty | USD | source | for |
|---|---|---|---|---|
| spring scale 15 kg (147 N), Kern R100150-S-W | 1 each | 7.62 | aio.lv (EUR 6.99) | T9-R3b, GT-1 |
| compression-set jig: 2 Al plates 100 x 100 x 6, 3.0 + 2.5 mm spacers, 4 M6 | 1 set | 10.00 | hardware store; ASSUMPTION (shop price, check at purchase) | CS-1 |
| stiff nylon scrub brush + 2 kg weight | 1 set | 5.00 | hardware store; ASSUMPTION (shop price, check at purchase) | AB-1 |
| phyphox app | 1 each | 0.00 | free | HB-2, T4a, GT-1 |

# Specimens and lab

| item | qty | USD | source | for | | | | |
|---|---|---|---|---|---|---|---|---|
| wiregate carabiner, CAMP Nano 22 class | 15 each | 150.00 | CAMP dealers (rack of 6); record carabiner_wiregate_22kn (cost is an ASSUMPTION there) | PT-1 (5), MT-2 (5), PT-6 (5) | | | | |
| G-hook 25 mm aluminium, AustriAlpin DY25K-G | 5 each | 20.00 | AustriAlpin / Extremtextil; record g_hook_25mm_aluminium (cost ASSUMPTION there) | MT-2 | | | | |
| cam buckle 25 mm aluminium (Austere type) LONG LEAD | 5 each | 127.50 | Ripstop by the Roll / Extremtextil; record cam_25mm_al | MT-3, US-BS | | | | |
| tri-glide 25 mm stainless | 3 each | 8.52 | Extremtextil art. 70460; record triglide_25mm (stainless line, EUR 3.10 incl. VAT) | US-BS | | | | |
| heavy ladder lock 25 mm (ITW Super Ladderloc) | 11 each | 23.10 | Liberty Mountain 6-pack / Buckleguy; record ladder_lock_25mm_heavy | LL-H (5), HB-7 (5), PT-5 (1) | | | | |
| tri-glide 25 mm nylon | 6 each | 2.22 | Extremtextil; record triglide_25mm | LL-H (5), PT-5 (1) | | | | |
| D-ring 25 mm aluminium | 5 each | 9.25 | Extremtextil; record d_ring_25mm_aluminium | HB-7 | | | | |
| YKK #10 coil slider, metal (cord pull fitted) LONG LEAD | 5 each | 9.95 | Rockywoods; alt. Shelby Outdoor EU; record slider_10 | SL-1 | | | | |
| #10 coil zipper chain | 0.4 m | 1.60 | Ripstop by the Roll; record coil_zipper_10 | SL-1 | | | | |
| Dyneema cord 3 mm (slider pulls) | 1 m | 0.90 | Ripstop by the Roll; record dyneema_cord_3mm | SL-1 | | | | |
| printed label (one, heat transfer) | 1 each | 5.00 | label printer; ASSUMPTION (shop price, check at purchase) | SL-1 | | | | |
| side-release buckle 25 mm | 2 each | 1.60 | Strapworks; record buckle_side_release_25mm | ST-1 | | | | |
| poly webbing 25 mm | 32 m | 22.40 | Strapworks; record poly_webbing_25mm | MT-1, MT-3 (dry/wet/cold), US-BS, LL-H (dry/wet/cold), SEW-F, HB-7, PT-5, PT-6, PORT-1, GT-1 | | | | |

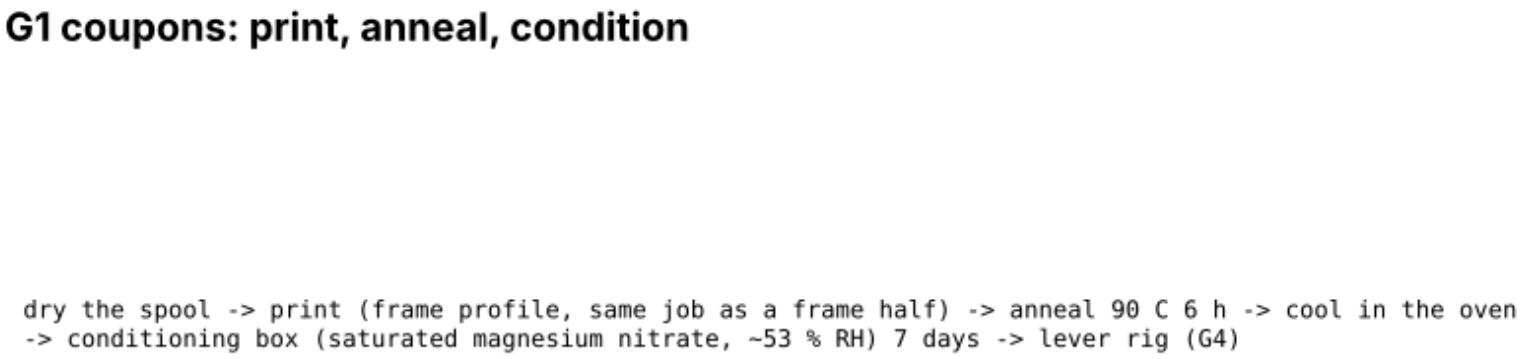
| item | qty | USD | source | for | | | | |
|---|---|---|---|---|---|---|---|---|
| poly webbing 38 mm, PLY 1.2-1.6 mm (1.5 nominal), break >= 6 kN: BUY A SAMPLE FIRST and caliper one ply at 5 places, reject > 1.6 mm (heavyweight / mil-spec rolls are 1.8-2.3 mm); fallback in the record notes | 3 m | 3.00 | Strapworks / extremtextil (see record); record poly_webbing_38mm (BL-048 M-B purchase spec) | PT-6 links, WP-3, HE-FIT, PT-3, SEW-F | | | | |
| XLPE L200 foam 5 + 6 mm | 0.35 m2 | 5.25 | Foam By Mail; record xlpe_foam_l200 | CS-1 (0.25), HS-1 (0.1) | | | | |
| EVA 45C foam 5 mm | 0.1 m2 | 2.50 | Seattle Fabrics; record eva_foam_45c | HS-1 | | | | |
| HDPE sheet 0.5 mm (BL-037 stiffener wing) | 0.05 kg | 0.30 | McMaster-Carr; record hdpe_sheet | HB-D2 | | | | |
| ROHACELL 71 IG-F core for the 2 mm strip LONG LEAD | 0.02 m2 | 2.40 | Rock West Composites; record rohacell_71_igf | BP-1 | | | | |
| 200 g 2x2 carbon twill + epoxy for the 2 mm BP-1 coupon skins | 0.028 m2 | 1.26 | left over from the spreader / strip layup (else any composites shop); record cfrp_t300_3k_woven_epoxy_ply (BL-039) | BP-1 | | | | |
| HyperD 300 printable, custom camo print LONG LEAD | 1 yd | 30.25 | Ripstop by the Roll (custom print, 1 yd min); record polyester_300d_printable | AB-1, FRAY-1 | | | | |
| shipping for the HyperD print order | 1 order | 12.00 | Ripstop by the Roll; ASSUMPTION SHIP (camo/cost_1_and_10_bags.csv) | AB-1, FRAY-1 | | | | |
| Martindale abrasion test, accredited lab (2 samples + order fee) LONG LEAD | 1 order | 249.61 | Measurlabs (EUR 66 / sample + EUR 97 fee) | AB-1 | | | | |
| specimens for PT-2: 5 swatches built like the pack, B-17 cut 70 mm tall | 1 lot | 15.00 | see register; test_register.csv PT-2: ASSUMPTION Hypalon + laminate offcuts | PT-2 | | | | |
| specimens for PT-3: 5 swatches sewn exactly as op 4.05 / SK-8, in the op 4.05 order (BL-041): B-01 ( | 1 lot | 15.00 | see register; test_register.csv PT-3: ASSUMPTION offcuts | PT-3 | | | | |
| specimens for WP-1: 4 seam pairings 330D | 330D, 330D | ULTRA 200, 500D | 330D, 100D | ULTRA; 301 + 2nd row | 1 lot | 5.00 | see register; test_register.csv WP-1: offcuts | WP-1 |
| specimens for CA-1: 12 XY + 4 Z + 3 spare bars, printed and annealed like the frame, 7 days at 23 C | 1 lot | 15.00 | see register; test_register.csv CA-1: CA1_fatigue_rig.md §5 (filament $80/kg) | CA-1 | | | | |

| item | qty | USD | source | for | | | | |
|---|---|---|---|---|---|---|---|---|
| specimens for IM3-EC: 3 eye coupons: 120 mm stay segment round z 520 with the IM3 eye and the 28 mm ch | 1 lot | 10.00 | see register; test_register.csv IM3-EC: ASSUMPTION: filament + 3 tongues | IM3-EC | | | | |
| specimens for WP-2: 5 laser-cut B-11 tabs | 1 lot | 5.00 | see register; test_register.csv WP-2: ASSUMPTION | WP-2 | | | | |
| specimens for FS-O5: 5 warp + 5 weft strips 50 mm wide (ISO 13934-1 / ASTM D5035 style) + 5 PT-3 swat | 1 lot | 35.00 | see register; test_register.csv FS-O5: ASSUMPTION fabric $25 + PT-3 set $10 | FS-O5 | | | | |
| specimens for HB-6: 5 inserts in printed, annealed PAHT-CF blocks | 1 lot | 5.00 | see register; test_register.csv HB-6: ASSUMPTION | HB-6 | | | | |
| specimens for CA-5: 4 stub pairs, one stay, 60 mm each side of z 234, real bosses, Ti bolts | 1 lot | 40.00 | see register; test_register.csv CA-5: frame_comp_verdict.md §6 (ASSUMPTION) | CA-5 | | | | |
| specimens for CA-3: 3 Z bars + 3 short-beam blocks 24 x 8 x 6 | 1 lot | 5.00 | see register; test_register.csv CA-3: ASSUMPTION | CA-3 | | | | |
| specimens for CA-4: 2 XY bars | 1 lot | 10.00 | see register; test_register.csv CA-4: frame_comp_verdict.md §6 (ASSUMPTION) | CA-4 | | | | |
| specimens for CA-2: 3 XY + 3 Z bars per anneal | 1 lot | 10.00 | see register; test_register.csv CA-2: ASSUMPTION | CA-2 | | | | |
| specimens for HB-4: 1 ear stitched into a wing-root coupon (3 recommended) | 1 lot | 15.00 | see register; test_register.csv HB-4: ASSUMPTION plate offcut + foam | HB-4 | | | | |
| specimens for HB-T: 5 tabs sewn like the belt: 50 mm webbing round a buckle bar, 25 mm anchor and lo | 1 lot | 5.00 | see register; test_register.csv HB-T: ASSUMPTION webbing | HB-T | | | | |
| specimens for TL-1: 5 x the real rev C crown top bar stock ([+-45/0x5/+-45], wall 1.065, same vendor | 1 lot | 100.00 | see register; test_register.csv TL-1: ASSUMPTION crown bar stock 2.0 m from the frame vendor (5 x 200 + 1 control x 200 + 2 TL-1 | TL-1 | | | | |
| specimens for HB-5: 1 plate on the Ti pin over a foam block | 1 lot | 45.00 | see register; test_register.csv HB-5: ASSUMPTION plate $25 + fixture $20 | HB-5 | | | | |
| specimens for PS-1: pad seat on a printed pan offcut | 1 lot | 5.00 | see register; test_register.csv PS-1: ASSUMPTION | PS-1 | | | | |

| item | qty | USD | source | for | | | | |
|---|---|---|---|---|---|---|---|---|
| specimens for BP-2: 100 mm printed H1 stay segment (z about 260) on a strip offcut, 10 mm foam under | 1 lot | 5.00 | see register; test_register.csv BP-2: ASSUMPTION carbon paper; Prescale roll about EUR 1275 not bought (§4.4) | BP-2 | | | | |
| specimens for T6: 3 corner coupons, 0.8 mm glass-bead bondline | 1 lot | 30.00 | see register; test_register.csv T6: ASSUMPTION tube offcuts + MMA | T6 | | | | |
| specimens for T7: 150 mm printed H1 sleeve segment + UHMWPE bushing + leg offcut | 1 lot | 10.00 | see register; test_register.csv T7: ASSUMPTION | T7 | | | | |
| specimens for SP-1: P2 spreader | 1 lot | 5.00 | see register; test_register.csv SP-1: ASSUMPTION dowel | SP-1 | | | | |
| specimens for HB-D2: belt B wings side by side: 0.5 mm and 0.75 mm HDPE stiffener (and the 1.0 mm ref | 1 lot | 5.00 | see register; test_register.csv HB-D2: ASSUMPTION 1.0 mm HDPE offcut | HB-D2 | | | | |
| specimens for RF-1: rails on P2, 3 testers | 1 lot | 10.00 | see register; test_register.csv RF-1: ASSUMPTION foam | RF-1 | | | | |
| specimens for WP-5: ports, hose frames, cam-cup base | 1 lot | 5.00 | see register; test_register.csv WP-5: ASSUMPTION | WP-5 | | | | |
| sacrificial P1 frame (IM3-EP, W-PULL, then T2u) LONG LEAD | 1 each | 437.00 | owner's print + parts; frame_h1.md 5 unit cost | IM3-EP, W-PULL, T2u | | | | |
| specimens for BU-1: 2 UHMWPE crown bushings cut 50 mm from the 70 mm stock, on P2 | 1 lot | 10.00 | see register; test_register.csv BU-1: ASSUMPTION spare UHMWPE tube length $10 (uhmwpe_sheet record has no tube price) | BU-1 | | | | |
| specimens for SEW-F: 3 x 25 mm poly webbing loops, box-X 21 x 25 mm (the IM / US box as drawn), 301, | 1 lot | 10.00 | see register; test_register.csv SEW-F: 25 mm webbing offcuts + thread (kit A stock); HE scrap + 3 fixture prints (book about 170 | SEW-F | | | | |
| specimens for FIT-R3: first sewn bag + lid on P2; 5.0 L of dry rice in a bag; the owner's bow (in IMP- | 1 lot | 2.00 | see register; test_register.csv FIT-R3: 5 L of rice | FIT-R3 | | | | |
| specimens for HE-FIT: 1 scrap HE sewn as op 4.05 (B-18 + B-01 offcut 150 x 120 + B-05 strip, floor sea | 1 lot | 15.00 | see register; test_register.csv HE-FIT: scrap + about 270 g PAHT-CF at USD 80/kg (CA1_fatigue_rig.md 5) with supports (BUILD_H1 se | HE-FIT | | | | |

| item | qty | USD | source | for | | | | |
|---|---|---|---|---|---|---|---|---|
| specimens for TL-1b: 2 lengths of the real rev C crown bar stock (>= 160 mm) with a 25 mm B-25-style | 1 lot | 20.00 | see register; test_register.csv TL-1b: ASSUMPTION crown bar offcut + dial bore gauge (shared with TL-1) | TL-1b | | | | |
| specimens for CLEVIS-SHARE: designs/pack_5200/print/frame_h1/test_pieces/us_route_stay_R.stl + designs/pa | 1 lot | 10.00 | see register; test_register.csv CLEVIS-SHARE: about 140 g PAHT-CF at USD 80/kg (us_route.json test_piece g_as_assembled) + the clevis bo | CLEVIS-SHARE | | | | |
| specimens for FIT-US: P1 frame with both US2 clevises bolted on (as built), the first sewn bag, US1 / | 1 lot | 2.00 | see register; test_register.csv FIT-US: lipstick / chalk paste; P1, bag and sand from kits C / D | FIT-US | | | | |
| specimens for MT-1cr: P1 frame (the sacrificial frame) on the kit C stand, bag OFF, both US2 clevises | 1 lot | 20.00 | see register; test_register.csv MT-1cr: strain gauges and bridge parts on the shopping list (kit B); P1, stand and sand from kits | MT-1cr | | | | |

# G1: Order the long-lead parts, print and condition the coupons




When: week 1, day 1

Rig: 3D printer (frame profile), oven, conditioning box (kit D)

Kits: B (filament), D

Start here, today. The frame coupons need 7 days in the conditioning box before the lever rig can test them, so they set the pace for frame release R1. Order the long-lead parts the same day.

## Safety (read before you start)

- Hot oven and hot nozzle: gloves. Magnesium nitrate is an oxidizer and an irritant: gloves, keep it away from heat sources and children, label the box.

## Rig sketch

dry the spool -> print (frame profile, same job as a frame half) -> anneal 90 C 6 h -> cool in the oven

-> conditioning box (saturated magnesium nitrate, ~53 % RH) 7 days -> lever rig (G4)

Picture: sketches/G1_rig.png

## Tests in this session, in order

| # | test | releases |
|---|---|---|
| 1 | CA-1 | Frame release R1 |
| 2 | IM3-EC | Frame release R1 |
| 3 | CA-5 | Frame release R1 |
| 4 | CLEVIS-SHARE | Frame release R1 |
| 5 | CA-3 | Field release and the rest |
| 6 | CA-4 | Field release and the rest |
| 7 | HB-6 | Field release and the rest |
| 8 | T7 | Field release and the rest |

## Print list (one job, same spool and profile as the frame)

| coupon | for | count |
|---|---|---|
| XY dog-bone bars 170 x 20 (gauge 10 x 4 x 60) | CA-1 | 12 + 2 spare |
| Z bars 100 x 20 (gauge 10 x 6 x 30), standing | CA-1 | 4 + 1 spare |
| Z bars + short-beam blocks 24 x 8 x 6 | CA-3 | 3 + 3 |
| XY bars | CA-4 | 2 |
| z 234 joint stub pairs (H1 bosses, traps) | CA-5 | 4 |
| IM3 eye coupons, 120 mm stay at z 520, 28 mm cheek (test_pieces/im3ec_cheek_R, SUPPORTS) | IM3-EC | 3 |
| US route piece test_pieces/us_route_stay_R (SUPPORTS) + US2 clevis us2_clevis_R (flat, no supports), a SECOND set (HE-FIT step 0 prints the first) | CLEVIS-SHARE | 1 + 1 |
| insert blocks (M5 heat-set) | HB-6 | 5 |
| 150 mm H1 sleeve segment | T7 | 1 |

Steps:

1. Dry the spool. Print with the frame profile (designs/pack_5200/print/frame_h1/BUILD_H1.md section 3): 0.6 mm hardened nozzle, 290 C, bed 100 C, 0.20 mm layers, 6 walls, 100 % infill. The flat bars need no supports. The frame-cut pieces (IM3-EC coupons, the CLEVIS-SHARE route piece) DO: SUPPORTS REQUIRED (it touches the plate over only 9 mm2, 3684 mm2 needs support, BL-053 print_settings_bl053.json): tree supports FROM THE BUILD PLATE under the whole piece, in PAHT-CF, top Z distance 0.20-0.25 mm, 2-3 interface layers, 45 deg threshold, support blockers painted inside every bore, pin hole, bolt hole, counterbore and nut trap. After printing break the supports off, sand the scars flush ALONG the stay, filament about 178 g, about 6.3 h. The US2 clevis prints flat with no supports. The CA-5 joint stubs and the T7 sleeve segment: turn on the same tree supports if the slicer's overhang view shows red under them (BL-054).

2. XY bars lie flat on their 10 mm face (4 mm is the print height), like the frame.

3. Anneal 90 C for 6 h on a flat plate; cool in the oven.

4. Into the conditioning box for 7 days (23 C, about 53 % RH). Write the date on the box.

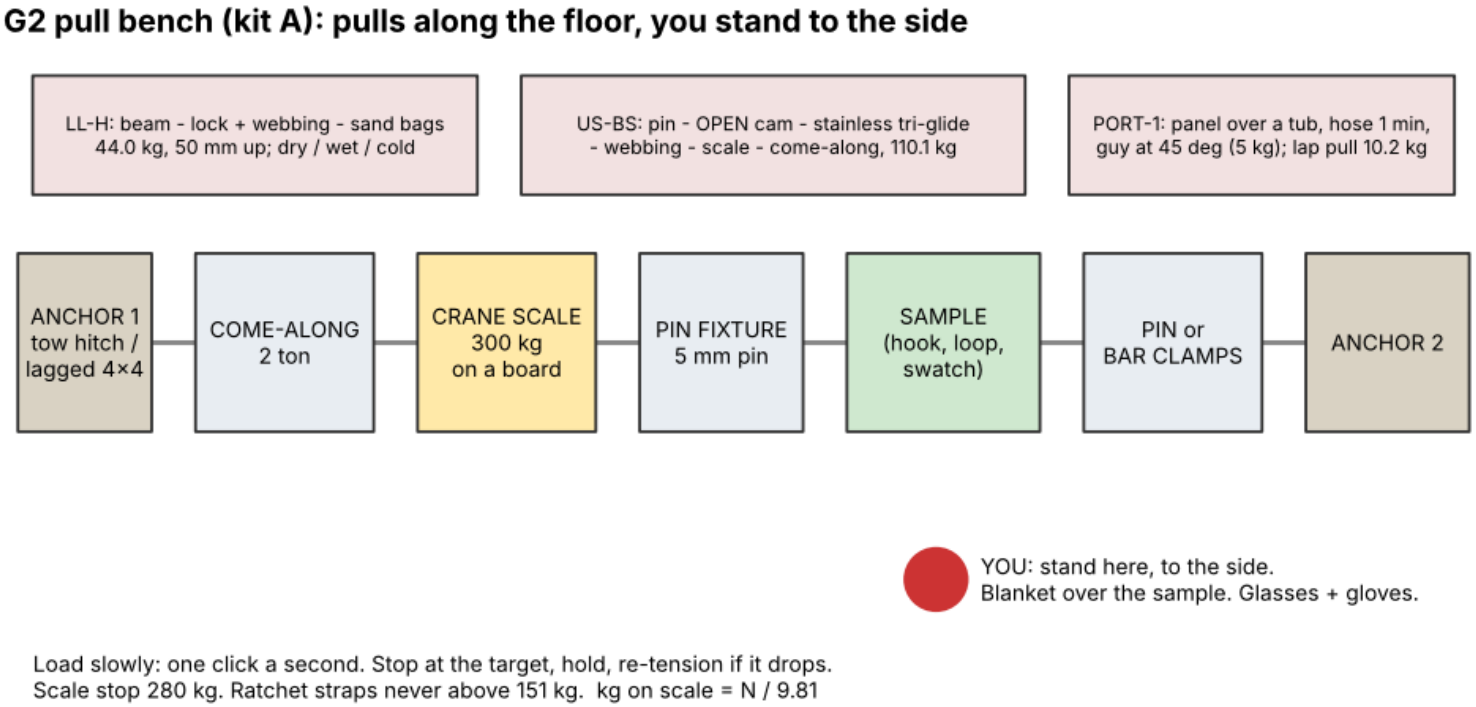
5. Measure each bar's gauge width and thickness at 3 points with the caliper; write the smallest on the bar.

## Order today (long lead)

- 25 mm aluminium cams (MT-3, US-BS) and the pack's own 5: both resellers out of stock on 2026-11-13.

- YKK #10 sliders (SL-1): Rockywoods sold out; Shelby Outdoor EU has them.
- HyperD 300 custom camo print (AB-1, FRAY-1) and the Martindale lab booking (about 4 weeks).
- 2 mm ROHACELL 71 (BP-1): not a stock sheet; sand a 3 mm offcut or special-order.
- Everything else in shopping_list.md.

# G2: Pull bench weekend: hooks, cams, bartacks, ladder locks




When: weekend 1 (week 1)

Rig: kit A pull bench (come-along, 300 kg crane scale, 5 mm pin fixture, bar clamps)

Kits: A, D (caliper, loupe, pocket scale), E

One bench, one weekend. MT-1 goes first because frame release R1 needs it; the rest are every sewn load point on the bag. After this weekend you may cut the bag (if G3's wash also passes).

About 25.0 h hands-on and 7 h of machine time.

## Safety (read before you start)

- Stand to the side of the load line, never in line with it. Moving blanket over every sample above 50 kg.

- Safety glasses and gloves on. Nobody near the line while it is loaded.

- Ratchet straps never in the load line above 151 kg (1480 N, ASSUMPTION: a third of the 1000 lb label).

- Crane scale stop: 280 kg (2750 N). PT-1 and PT-2 at 260.2 kg are the heaviest pulls: come-along only.

- LL-H hangs a dead load: nobody under it, 50 mm off the floor so a slip ends on the floor.

- If anything fails: stop, photograph, write it down, send the sheet. Do not repair and continue.

## Rig sketch

ANCHOR 1 -- sling -- COME-ALONG -- CRANE SCALE -- PIN FIXTURE -- SAMPLE -- PIN / BAR CLAMP -- sling -- ANCHOR 2

(tow hitch, tree, (2 ton) (300 kg, (5 mm hardened (5 mm pin, or the

lagged 4x4) on a board) dowel pin) bar clamps for swatches)

You stand HERE, to the side --> o (never in line with the pull; blanket over the sample)

LL-H dead hang: beam -- lock + webbing -- sand bags 44.0 kg (432 N), 50 mm above the floor; dry, wet, cold

US-BS: pin -- [open cam] -- [stainless tri-glide] -- webbing -- scale -- come-along

Picture: sketches/G2_rig.png

Sew before the weekend (about 5 h, same machine, thread and needle as the pack):

1. MT-1: 10 loops of 25 mm poly webbing; fold each end into a 20 mm loop and close it with a 25 x 25 mm box-X, stitch 301, 7 SPI, bonded nylon tex 135, Nm 140 round-point needle.

2. PT-2: 5 x T3 tongue on a 70 mm B-17 tab on a B-01 swatch about 200 x 200 mm (bag_attachment.md 5 T3).

3. HE-FIT first (print one z 60 hinge fairlead from print/frame_h1/test_pieces/hefit_hinge_fairlead_R.stl, sew 1 scrap hinge eye as op 4.05), then PT-3: 5 hinge-eye swatches sewn exactly as op 4.05 and sketch SK-8 of the release, IN ITS ORDER: B-18 rows first through B-18 + B-01 only, then the HE (folded in half, eye UP) and its box-X below the cheek line, B-18 and both HE raw ends in the B-01 | B-05 floor seam, no binding (register PT-3 specimen).

3a. TL-1 / TL-1b: buy 2.0 m of the real crown top bar stock from the frame vendor (1.52 m used); sew 5 x 38 mm HALF B-25 bridles (TL-1: sleeve, mitre, necked run, as SK-12) round 200 mm lengths and 2 x 25 mm sleeves (TL-1b) round 160 mm lengths, laps bartacked as at frame assembly; keep 1 200 mm control.

4. PT-6: 5 B-24 links to the cut list (carabiner + 3 loops).

5. US-BS: 3 x 600 mm of 25 mm webbing threaded through a cam with the stainless tri-glide behind it.

6. LL-H: 5 heavy ladder locks + nylon tri-glides and 15 x 600 mm of 25 mm webbing (5 dry, 5 wet, 5 cold); 2 of the locks on a hinge-band tab like the B-28 front lock. MT-3: 15 x 600 mm webbing for the 5 cams (dry / wet / cold). Put the cold set in the freezer 2 h before you need it.

7. PORT-1: one 300 x 300 mm B-01 panel with a B-27 lapped slit and B-27f facing (op sequence).

8. SL-1: a 400 mm #10 coil door sample on 330D tabs with 2 sliders; a 3 mm cord pull on each of 5 sliders.

9. Label every sample with a paint pen: test ID and sample number (PT-2-1 ... PT-2-5).

## Tests in this session, in order

| # | test | releases |
|---|---|---|
| 1 | MT-1 | Frame release R1 + Bag release, step 1: cut and sew |

| # | test | releases |
|---|---|---|
| 2 | HE-FIT | Bag release, step 1: cut and sew |
| 3 | PT-1 | Bag release, step 1: cut and sew |
| 4 | MT-2 | Bag release, step 1: cut and sew |
| 5 | MT-3 | Bag release, step 1: cut and sew |
| 6 | US-BS | Bag release, step 1: cut and sew |
| 7 | PT-2 | Bag release, step 1: cut and sew |
| 8 | PT-3 | Bag release, step 1: cut and sew |
| 9 | PT-5 | Bag release, step 1: cut and sew |
| 10 | PT-6 | Bag release, step 1: cut and sew |
| 11 | LL-H | Bag release, step 1: cut and sew |
| 12 | B-28 slip | Bag release, step 1: cut and sew |
| 13 | PORT-1 | Bag release, step 1: cut and sew |
| 14 | TL-1 | Bag release, step 1: cut and sew |
| 15 | FS-O5 | Bag release, step 1: cut and sew |
| 16 | WP-3 | Bag release, step 2: batch |
| 17 | SL-1 | BL-037 weight gates |
| 18 | HB-7 | Field release and the rest |
| 19 | TL-1b | Data only (no gate) |

# 1. MT-1: 25 mm box-X bartack proof at the BL-010 per-row loads

Why: every 25 mm box-X on IM1-IM3 and US1/US2; BL-003 (a) (A_imus)

Specimen: 14 specimens: 10 x 25 mm poly webbing, box-X 25 x 25, stitch 301, 7 SPI, bonded nylon tex 135; + 2 IMC cam anchor tabs cut to the pattern (BL-041; BL-043 m-4: 192 mm, 48 mm folds, 21 x 25 box-X both ends with 18 mm loops and each box 5 mm in from its raw end; cam bar in one loop, a 5 mm pin in the other); + 2 USA US-cam anchors cut to the pattern (BL-044: the same dog-bone, 202 mm, its pin loop 23 mm clear and NECKED to 18 mm, edges folded 3.5 mm in, BT-M per leg). The 5 US loops are sewn as the US eye: NECKED to 18 mm round the pin (BL-044, bag_attach.imus_eye_envelope)

| load | N | lb | kg on the scale |
|---|---|---|---|
| IM loops (5) + IMC tabs (2) | 908 | 204 | 92.6 |
| US loops (5) | 1080 | 243 | 110.1 |

Steps

1. Sew the loops exactly like the pack (necks folded and tacked first)

2. Loop over the 5 mm pin

3. 5 loops to 908 N (92.6 kg) and 5 to 1080 N (110.1 kg)

4. The 2 IMC tabs (cam bar on a 5 mm pin each end) to 908 N

5. The 2 USA anchors (necked loop on a 5 mm pin, cam bar in the other) to 1080 N

6. Hold 1 min each

Pass: 5 of 5 hold 908 N (IM) and 5 of 5 hold 1080 N (US) for 1 min, 2 of 2 IMC hold 908 N, 2 of 2 USA hold 1080 N, no broken stitch

Write in the results sheet (results_template.csv, column measured):

- MT-1-01 to MT-1-05 (IM#1 to IM#5): held 92.6 kg (908 N) 1 min, no broken stitch. Type Y or N.

- MT-1-06 to MT-1-10 (US#1 to US#5): held 110.1 kg (1080 N) 1 min, no broken stitch. Type Y or N.

- MT-1-11 to MT-1-12 (IMC#1 to IMC#2): IMC cam anchor tab held 92.6 kg (908 N) 1 min, both boxes whole (BL-041). Type Y or N.

- MT-1-13 to MT-1-14 (USA#1 to USA#2): USA US-cam anchor held 110.1 kg (1080 N) 1 min, both boxes whole, neck tacks whole (BL-044). Type Y or N.

Note: was 910 N on 38 mm loops. If BL-003 (a) fails, the 38 mm IM/US set comes back and the 38 mm box-X at 910 N applies. BL-052: this row is the bench box-X proof; the frame-side 'MT-1 cradle pull' named in h1_crown.md (US arm-root gauges, cradle belly) is row MT-1cr

Defined in: designs/pack_5200/analysis/loadpath_verification.md §8 (BOX25); designs/pack_5200/BUILD_PLAN.md §5a.0

## 2. HE-FIT: hinge eye 1:1 dry fit on the printed BL-049 z 60 hinge fairlead (eye on the pin, neck in the throat)

Why: HE pattern before the PT-3 specimens and before the bag is cut

Specimen: 1 scrap HE sewn as op 4.05 (B-18 + B-01 offcut 150 x 120 + B-05 strip, floor seam sewn); one hinge fairlead printed 1:1 from print/frame_h1/test_pieces/hefit_hinge_fairlead_R.stl (BL-049: z 60 R, 80 mm stay segment, 20 mm throat, 6.5 mm under the pin, 10 mm cheeks; ream the eye to 7.2) with its UHMWPE sleeve and the Ti-6Al-4V hinge pin (5 mm, 44 mm, BL-049; the hardened steel M5 x 50 dowel (DIN 6325 m6, HRC 58-62; NOT a 7075 frame pin; it stands about 5 mm proud of each hinge cheek so you can grip it) stands in for it in step 0)

| load | N | lb | kg on the scale |
|---|---|---|---|
| hang from the B-05 strip | 49 | 11 | 5.0 |

Steps

- Step 0. US ROUTE (BL-048 N-1, re-run on the BL-050 lanes; pass criterion below): print the US test piece designs/pack_5200/print/frame_h1/test_pieces/us_route_stay_R.stl (R stay z 40-160 with the hinge fairlead and the US1 / US2 lane lugs, cut from h1_assembled.stl by us_route.export_test_piece; BL-050: printed WITHOUT the US2 outer cheek) and the bolt-on clevis designs/pack_5200/print/frame_h1/us2_clevis_R.stl; print both as the H1 frame (BUILD_H1.md sections 3-5 and 10). The route piece: frame profile of BUILD_H1.md section 3 with SUPPORTS REQUIRED (it touches the plate over only 9 mm2, BL-053 print_settings_bl053.json): tree supports from the build plate in PAHT-CF, blockers in every bore, pin hole, bolt hole and trap, scars sanded flat (BUILD_H1 section 4), filament about 178 g, about 6.3 h. The clevis: frame profile of BUILD_H1.md section 3, NO supports (it prints flat), blockers in the holes, annealed 90 C 6 h, filament about 14 g, about 0.5 h. Then ream the clevis bolt holes 5.1 and bolt the clevis on with 2 Ti gr5 M5 x 35 DIN 7984 with the heads in the printed 9 mm counterbores and NO washer under the head, 2 A2 nuts in the side traps, 2 x 10 mm OD DIN 433 washers on the NUT side only (NOT the z 234 joint's M5 x 45 / 12 mm washers; shopping list), blue threadlocker, snug; check the heads sit below the clevis face (a steel rule across the face does not touch them; design 0.5 mm below) (BL-051 / 052); clamp the piece upright; sew two USA anchors (25 mm, as op 7.17: the piece is the R stay, where the cam anchors sit) and hang the HE specimen of this test on its hinge pin (hardened steel M5 x 50 dowel (DIN 6325 m6, HRC 58-62; NOT a 7075 frame pin; it stands about 5 mm proud of each hinge cheek so you can grip it) in each fairlead); USA on the US1 pin, its cam on a 25 mm strap laid down the B-01 offcut over the piece, round a 20 mm dowel at the floor-seam line (the floor corner, 25 mm under the hinge pin) to the crane scale, cinched to 444 N (45.3 kg); then the same from the US2 pin at 296 N (30.2 kg), the dowel at the strap's own lane (straight under its pin: the B-16 / B-16b channel ends are at the lane x, BL-050 B); repeat with the IME eye on the IM3-EC piece pulled at -82 to -88 deg (the sim directions); then slide the US2 eye by hand across the clevis seam in the throat floor under 5 kg (SEAM-FLOOR: it must not catch; chamfer 0.3 mm both edges if it does). PASS (gate US-ROUTE): at the cinched load, along its whole run from the eye to the dowel, no part of the strap, eye or box touches the hinge fairlead, the HE (eye, legs or box) or the other US fairlead: a 10 mm feeler block passes between them everywhere (the design clearance is >= 10 mm). FAIL = any contact or a gap under 10 mm. On the BL-050 print with the BL-050 B channel ends the model gives worst clearances US1 12.4 / US2 14.6 mm over the 15 / 19 / 23 in meat drapes, both sides (us_route.json 'now'; the BL-048 channel drawing at x +/-95 rode the hinge cheeks, bearing about 72 N); the bag is not cut until it passes

- Step 1. Re-caliper ONE ply of the 38 mm roll in hand at 5 places (the first caliper is at the PURCHASE, on a sample, shopping list; BL-048 M-B): every reading <= 1.6 mm, else reject the roll (at 1.6 mm a ply, 3 plies leave 1.09 mm under the pin on the BL-049 piece, he_stl_check.json, and a thicker web leaves less)

- Step 2. Clamp the printed stay segment upright

- Step 3. Slide the neck round the pin by hand: the pin goes through cheek, sleeve, eye, cheek with the eye seated, no forcing; then push the neck hard against each cheek in turn and slide it round the pin again (its folded edge passes the printed throat side: 0.69 mm clear on the model, 0.54 mm at the print tolerance, he_stl_check.json)

- Step 4. Hang 5 kg from the B-05 strip so the eye loads the pin

- Step 5. Caliper from the floor seam line to the pin centre

- Step 6. Feeler gauge between the box top and the cheeks' lower edge

- Step 7. Check the eye is not folded over the box and the neck is not pinched by the cheeks

- Step 8. BL-048 DECISION HE-NECK-GAUGE: 5 sewn necks (HE, IME, US) each pass the 18.6 mm go / no-go slot, and the printed throat admits a 19.6 mm gauge block (then the widest neck keeps >= 0.5 mm a side); photo

Pass: ply <= 1.6 mm; pin centre 22-29 mm above the floor seam line; >= 2 mm feeler between the box top and the cheeks; the eye slides round the pin by hand; neck alone in the 20 mm throat; eye points up (no fold-down)

Write in the results sheet (results_template.csv, column measured):

- HE-FIT-01 (#1): pin centre above the floor seam line (pass 22-29). Mm, pass if 22 to 29.

- HE-FIT-02 (#1): box top to the cheeks' lower edge (feeler). Mm, pass if at least 2.

- HE-FIT-03 (#1): webbing ply thickness, thickest of 5 readings. Mm, pass if at most 1.6.

- HE-FIT-04 (fit1): box top clear of the cheeks, neck alone in the throat, eye points up. Type Y or N.

Note: BL-041 (re-review X-2): a gate BEFORE PT-3 is sewn and before bag cut. BL-044 (third review Z-1 / m-1): on the printed hinge test piece (BL-049 size since BL-049). BL-054 (review 5 B-1 / m-3 / m-5): print text with supports, hinge numbers from he_stl_check.json, and the bench US-ROUTE check only screens the route (the dowel is set under its pin, so the 10 mm feeler mostly measures the lane offset): FIT-US step 8 re-checks it on the real bag. As before: the band is set by the cheek clearance (low end) and HE-PIN-HI (high end). A fail low: lengthen HE_EYE by the shortfall; a fail high: shorten it; a ply over 1.6 mm: change webbing

Defined in: designs/pack_5200/patterns_corrie_f_rev3.py DECISION HE-ENV (he_geometry, he_band, he_peel; bag_attach.HINGE_EYE_ENVELOPE), ASSUMPTIONS HE-CHEEK, HE-T, HE-PIN-HI, HE-NECK-TOL, HE-GAP-MARGIN; review_20261113_rev3_third.md Z-1 / m-1

## 3. PT-1: IM3 bag carabiner on the sewn T3 tongue, per-row proof (BL-010)

Why: IM3 bag hooks; BL-004 REC (A_att, IM2 hooks deleted so IM3 carries the top couple)

Specimen: 5 carabiners from the lot (CAMP Nano 22 class), each on a sewn T3 tongue

| load | N | lb | kg on the scale |
|---|---|---|---|
| proof, 60 s | 2553 | 574 | 260.2 |

Steps

1. Carabiner over the 5 mm steel pin

2. Sewn T3 tongue on the carabiner

3. Pull with the come-along to 2553 N (260.2 kg on the scale)

4. Hold 60 s

5. Release

6. Check the gate closes and look for set

Pass: 5 of 5 hold 2553 N for 60 s, gate closes after, no visible set

Write in the results sheet (results_template.csv, column measured):

- PT-1-01 to PT-1-05 (#1 to #5): held 260.2 kg (2553 N) 60 s, gate closes, no set. Type Y or N.

Note: was 2120 N on a 38 mm hook (bag_attachment.md); raised by BL-010 (1702 N x 1.5). 260 kg is under the 280 kg scale stop. Fallback if BL-004 REC fails: hood + IM2 hooks (PT-4 still applies)

Defined in: designs/pack_5200/analysis/loadpath_verification.md §5, §8 (CARA_PROOF); designs/pack_5200/analysis/bag_attachment.md §5

## 4. MT-2: meat-strap hook proof per strap (BL-010)

Why: IM1-IM3 free-end hooks; BL-003 (a) (A_imus)

Specimen: 5 x 25 mm rated G-hooks, 5 x IM3 carabiners (the PT-1 lot may be reused after PT-1 only if no set)

| load | N | lb | kg on the scale |
|---|---|---|---|
| G-hooks (5) | 645 | 145 | 65.7 |
| carabiners (5) | 910 | 205 | 92.8 |

Steps

1. As PT-1: G-hooks to 645 N (65.7 kg, covers IM1 410 N), carabiners to 910 N (92.8 kg)

2. Hold 1 min

Pass: 5 of 5 G-hooks hold 645 N and 5 of 5 carabiners hold 910 N for 1 min, gates close

Write in the results sheet (results_template.csv, column measured):

- MT-2-01 to MT-2-05 (G#1 to G#5): G-hook held 65.7 kg (645 N) 1 min, gate closes. Type Y or N.

- MT-2-06 to MT-2-10 (C#1 to C#5): carabiner held 92.8 kg (910 N) 1 min, gate closes. Type Y or N.

Note: BL-038: in the kit the MT-2 carabiners are a separate lot of 5 from PT-1

Defined in: designs/pack_5200/analysis/loadpath_verification.md §5, §8; designs/pack_5200/BUILD_PLAN.md §5a.0

## 5. MT-3: 25 mm cam buckle slip at the BL-010 per-row loads, dry, wet and cold

Why: IM and US cams; BL-003 (a) (A_imus)

Specimen: 5 x 25 mm Al cams, re-used for 3 conditions on FRESH 25 mm webbing each time (15 lengths of 600 mm)

| load | N | lb | kg on the scale |
|---|---|---|---|
| first hold | 908 | 204 | 92.6 |
| second hold | 1080 | 243 | 110.1 |

Steps

1. Thread the cam as in the pack and paint-pen mark at the cam mouth

2. DRY: pull to 908 N (92.6 kg), hold 1 min, measure

3. Then to 1080 N (110.1 kg), hold 1 min, measure. WET: fresh webbing, cam + webbing soaked 10 min, same two holds. COLD: fresh webbing, 2 h in the freezer, write the freezer reading, pull within 2 min of taking it out, same two holds. A home freezer at -18 C or colder is ACCEPTED as the record's -20 C (ASSUMPTION FREEZER)

Pass: mark moves < 3 mm at 908 N and at 1080 N, 5 of 5, in each of dry / wet / cold

Write in the results sheet (results_template.csv, column measured):

- MT-3-01 to MT-3-05 (dry-a#1 to dry-a#5): dry: mark moved at 92.6 kg (908 N). Mm, pass if at most 3.

- MT-3-06 to MT-3-10 (dry-b#1 to dry-b#5): dry: mark moved at 110.1 kg (1080 N). Mm, pass if at most 3.

- MT-3-11 to MT-3-15 (wet-a#1 to wet-a#5): wet: mark moved at 92.6 kg (908 N). Mm, pass if at most 3.

- MT-3-16 to MT-3-20 (wet-b#1 to wet-b#5): wet: mark moved at 110.1 kg (1080 N). Mm, pass if at most 3.

- MT-3-21 to MT-3-25 (cold-a#1 to cold-a#5): cold: mark moved at 92.6 kg (908 N). Mm, pass if at most 3.

- MT-3-26 to MT-3-30 (cold-b#1 to cold-b#5): cold: mark moved at 110.1 kg (1080 N). Mm, pass if at most 3.

Note: BL-038: price from the record (was ASSUMPTION $4). Cams LONG LEAD (both resellers out of stock 2026-11-13). BL-040 (review M-8): wet and -20 C conditions from the cam_25mm_al record's settled-by line

Defined in: designs/pack_5200/analysis/loadpath_verification.md §8 (CAM25); designs/pack_5200/BUILD_PLAN.md §5a.0

## 6. US-BS: US backstop: stainless tri-glide behind an open cam stops the cradle strap

Why: US1/US2 backstop (bag release)

Specimen: 3 x 25 mm poly webbing 600 mm, Al cam (the MT-3 cams, re-used) with the stainless 25 mm tri-glide threaded behind it as the op sequence

| load | N | lb | kg on the scale |
|---|---|---|---|
| US proof, cam held open | 1080 | 243 | 110.1 |

Steps

1. Thread cam and stainless tri-glide as in the pack

2. Paint-pen mark on the webbing at the tri-glide

3. Hold the cam lever open with a zip tie so it cannot grip

4. Pull slowly with the come-along to 110.1 kg (1080 N)

5. Hold 1 min

6. Measure how far the mark moved

Pass: tri-glide stops the strap within 10 mm and holds 1080 N for 1 min, 3 of 3

Write in the results sheet (results_template.csv, column measured):

- US-BS-01 to US-BS-03 (#1 to #3): strap stopped within. Mm, pass if at most 10.

- US-BS-04 to US-BS-06 (h#1 to h#3): held 110.1 kg (1080 N) 1 min. Type Y or N.

Note: BL-038: added from release_tests.csv. Run after MT-3 on the same cams

Defined in: outputs/release/pack5200_corrie_f_rev3/manufacturing/release_tests.csv; data/materials/hardware/triglide_25mm.yaml (stainless alternative); docs/DECISIONS.md 2026-11-13 orchestrator row

## 7. PT-2: T3 tongue on the 70 mm B-17 tab on a B-01 swatch (BL-010 load)

Why: IM3 bag attachment; BL-004 REC B-17 trim (A_att)

Specimen: 5 swatches built like the pack, B-17 cut 70 mm tall

| load | N | lb | kg on the scale |
|---|---|---|---|
| proof, 60 s | 2553 | 574 | 260.2 |

Steps

1. Clamp the B-01 swatch 120 mm wide between bar clamps

2. Tongue loop on the pin

3. Pull to 2553 N (260.2 kg)

4. Hold 60 s

Pass: 5 of 5 hold 2553 N, no stitch break, no laminate tear

Write in the results sheet (results_template.csv, column measured):

- PT-2-01 to PT-2-05 (#1 to #5): held 260.2 kg (2553 N) 60 s, no stitch break, no tear. Type Y or N.

Note: was 2120 N on the 110 mm tab

Defined in: designs/pack_5200/analysis/loadpath_verification.md §8; designs/pack_5200/analysis/bag_attachment.md §5; designs/pack_5200/analysis/attachment_diet.md (B-17 110 -> 70 mm)

## 8. PT-3: HE hinge eye on B-18 + B-01 + B-05 floor seam, sewn as op 4.05 (HE-ONE / HE-PIN), no binding

Why: hinge eyes; B-1 (no binding); C-5 if on ULTRA 200X

Specimen: 5 swatches sewn exactly as op 4.05 / SK-8, in the op 4.05 order (BL-041): B-01 (primary ULTRA 200) 150 x 120 mm; B-18 Hypalon 90 x 62 inside, its lower edge on the B-01 cut edge, its perimeter rows 3 / 9 mm on the top and both sides sewn FIRST through B-18 + B-01 only (rows first, before HE); HE 38 x 96 mm, neck edges folded 10 mm in + 1 BT-M per leg, folded in half, both raw ends on the B-01 cut edge, eye UP; box-X 34 x 7 mm, 4 mm above the seam line, through HE (2 plies) + B-01 + B-18 (301, 7 SPI, tex 135, Nm 140 R); stack basted at 6 mm in the allowance; a 150 x 80 mm B-05 (500D) strip joined by the SSa2 floor seam, sewn with the zipper / edge foot, allowances turned onto B-05; no binding

| load | N | lb | kg on the scale |
|---|---|---|---|
| proof, 60 s | 1530 | 344 | 156.0 |

Steps

1. FIXTURE: a fresh print of print/frame_h1/test_pieces/hefit_hinge_fairlead_R.stl (not the HE-FIT piece, BL-049 hinge size), eye reamed to 7.2, its UHMWPE sleeve, bolted upright to the bench plate through the stay segment

2. Flat spacer plate (PLA, or 6 mm ply shimmed) screwed to the printed stay segment, its face flush with the cheek tips (17.7 mm proud of the stay surface) from the cheeks down past the box, so B-01 is HELD at the cheek-tip plane and the box cannot swing in line: leg B leaves the box top at the worst peel angle (he_peel 'tip_plane'). The pin centre is then 7.6 mm toward the frame from the B-01 face and 14 mm above the box top (the bag geometry, DECISION HE-ENV). Put the hardened steel M5 x 50 dowel (DIN 6325 m6, HRC 58-62; NOT a

7075 frame pin; it stands about 5 mm proud of each hinge cheek so you can grip it) through cheek, sleeve, eye, cheek (only the 18 mm neck between the cheeks)

3. Clamp the B-05 strip ALONE in the bar clamp below

4. The B-01 top edge stays FREE (never clamped: a clamp there carries the load past the floor seam)

5. Pull DOWN along the spacer face (leg B leaves the box top at about 50 deg: the peel the frame gives) to 1530 N (156.0 kg)

6. Hold 60 s

7. With the loupe check the box TOP row first, then the neck tacks and the floor seam. PIN CHECK (BL-048 M-C): roll the dowel on the bench plate before and after the test: it rolls freely with no visible wobble and a 0.05 mm feeler does not pass under its middle

8. A bent pin voids the specimen. If the printed cheeks crack before the eye fails, the specimen is void: re-run it in a machined aluminium copy of the same section (ASSUMPTION PT3-PRINT). PRINT (BL-054, review 5 B-1): print the fixture as the H1 frame (designs/pack_5200/print/frame_h1/BUILD_H1.md sections 3-5 and 10): frame profile of BUILD_H1.md section 3 with SUPPORTS REQUIRED (it touches the plate over only 15 mm2, BL-053 print_settings_bl053.json): tree supports from the build plate in PAHT-CF, blockers in every bore, pin hole, bolt hole and trap, scars sanded flat (BUILD_H1 section 4), filament about 45 g, about 1.6 h, then ream as the frame.

Pass: 5 of 5 hold 1530 N 60 s: no stitch break (box top row included), no Hypalon or laminate tear, B-18 still in the floor seam, the eye still open (no row across it)

Write in the results sheet (results_template.csv, column measured):

- PT-3-01 to PT-3-05 (#1 to #5): held 156 kg (1530 N) 60 s, B-18 still in the floor seam, eye open (sewn as op 4.05, B-18 rows first). Type Y or N.

Note: BL-041 (re-review X-1 / X-2 / Y-1): specimen from the reordered op 4.05 (B-18 rows first). BL-044 (third review M-3): rig = the printed hinge fairlead (BL-049 size) with B-01 held at the cheek-tip plane (the box peels; it is not in line), B-05 alone in the clamp, B-01 top free. Run HE-FIT first

Defined in: designs/pack_5200/analysis/bag_attachment.md §5; designs/pack_5200/analysis/corrie_body.md §12.4 (rebuilt); designs/pack_5200/analysis/fabric_swap_evidence.md O-5

## 9. PT-5: guy ladder lock + tri-glide keeper slip

Why: TG guys (top anchor)

Specimen: 25 mm webbing, ladder lock + tri-glide keeper

| load | N | lb | kg on the scale |
|---|---|---|---|
| 10 cycles to | 170 | 38 | 17.3 |

Steps

1. Mark webbing

2. 10 cycles 0 -> 170 N -> 0 by hand on the come-along

3. Measure the mark

Pass: slip < 5 mm over 10 cycles

Write in the results sheet (results_template.csv, column measured):

- PT-5-01 (#1): total slip after 10 cycles to 17.3 kg. Mm, pass if at most 5.

Defined in: designs/pack_5200/analysis/bag_attachment.md §5

## 10. PT-6: B-24 float stand-off link

Why: float mode (D-33)

Specimen: 5 links built to the cut list (hook + 3 loops)

| load | N | lb | kg on the scale |
|---|---|---|---|
| proof, 60 s | 1410 | 317 | 143.7 |

Steps

1. Link hook on the pin

2. 38 mm loop through the 220 mm loop

3. Pull in line to 1410 N

4. Hold 60 s

Pass: 5 of 5 hold 1410 N, no box-X break, gate closes

Write in the results sheet (results_template.csv, column measured):

- PT-6-01 to PT-6-05 (#1 to #5): held 143.7 kg (1410 N) 60 s, no box-X break, gate closes. Type Y or N.

Defined in: designs/pack_5200/analysis/bag_attachment.md §5

## 11. LL-H: heavy 25 mm ladder lock slip at 432 N, dry, wet and cold (guys, W1-W3, B-28)

Why: guy ladder locks on B-22 (BL-004 REC, A_att); W1-W3 float clamp locks; B-28 crown strap locks (the B-28 slip gate, BL-031)

Specimen: 5 heavy ladder locks + tri-glides, re-used for 3 conditions on FRESH 25 mm webbing each time (15 lengths of 600 mm); 2 of the 5 threaded as a B-28 front lock on a hinge-band tab

| load | N | lb | kg on the scale |
|---|---|---|---|
| dead load: dry 1 h, wet 1 h, cold 10 min | 432 | 97 | 44.0 |

Steps

1. Thread as in the pack and paint-pen mark at the lock

2. DRY at room temperature: hang 44.0 kg (432 N, sand bags) 50 mm above the floor for 1 h, all 5 side by side, then measure the mark

3. WET: re-thread on fresh webbing, soak lock + webbing 10 min in water, hang the same load wet for 1 h, measure

4. COLD: re-thread on fresh webbing, 2 h in the freezer, write the freezer reading, load within 2 min of taking it out, hold 10 min, measure. The record asks for -20 C: a home freezer at -18 C or colder is ACCEPTED as the -20 C condition (ASSUMPTION FREEZER; dry ice in a cool box reaches -20 C if the owner wants the exact figure)

Pass: slip < 5 mm, 5 of 5, in each of dry / wet / cold at 432 N (covers the W 338 N proof); a fail in any condition: tell the pack-designer (record fallbacks: aluminium ladderlock)

Write in the results sheet (results_template.csv, column measured):

- LL-H-01 to LL-H-05 (dry-#1 to dry-#5): dry: slip after 1 h at 44.0 kg (432 N). Mm, pass if at most 5.

- LL-H-06 to LL-H-10 (wet-#1 to wet-#5): wet: slip after 1 h at 44.0 kg (432 N). Mm, pass if at most 5.

- LL-H-11 to LL-H-15 (cold-#1 to cold-#5): cold: slip after 10 min at 44.0 kg (432 N). Mm, pass if at most 5.

- LL-H-16 (cold): freezer reading. Type the value in C.

Note: BL-040 (review M-7, M-8): load 432 N (was 366 N = 1.5 x the old 244 N A3 load); wet and -20 C conditions added from the record

Defined in: designs/pack_5200/analysis/loadpath_verification.md §3 S9, A3, §5; designs/pack_5200/analysis/attachment_diet.md (guys 366 N)

## 12. B-28 slip: B-28 crown strap on its heavy ladder lock

Why: B-28 crown straps

Specimen: LL-H specimens (2 of the 5 threaded as a B-28 front lock on a hinge-band tab)

Nothing extra to run: covered by LL-H: same 25 mm heavy lock, 432 N proof (1.5 x the 288 N envelope; BL-031 dropped the 20 mm gate).

## 13. PORT-1: B-27 guy port mock-up: hose, 100 N lap pull, gloved lock use

Why: B-27 construction (bag release)

Specimen: 300 x 300 mm B-01 (ULTRA 200) panel with one B-27 lapped slit and its B-27f facing sewn as op sequence, 25 mm guy webbing through it

| load | N | lb | kg on the scale |
|---|---|---|---|
| guy at 45 deg during the hose | 49 | 11 | 5.0 |
| lap pull | 100 | 22 | 10.2 |

Steps

1. Clamp the panel over a plastic tub, outside face up, a dry paper towel in the tub

2. Hang 5 kg (49 N) on the guy so it pulls at 45 deg through the port

3. Hose from 1 m above for 1 min

4. Check the towel

5. Then pull the guy straight through the lap to 10.2 kg (100 N) on the crane scale and measure the lap gap

6. Then on the first sewn bag (or a hinge-band mock-up with B-22 + LLT sewn as op 3.05): with winter gloves, reach in through the open roll top, thread and adjust the guy ladder lock on its LLT tab at B-22 INSIDE the hinge band 5 times (DECISION GUY-LOCK-IN, BL-040 M-5: the lock is where the pattern and the ops put it, not at the port)

Pass: towel dry; lap gap <= 2 mm at 100 N (ASSUMPTION 'stays closed' = 2 mm); 5 of 5 gloved adjustments at B-22

Write in the results sheet (results_template.csv, column measured):

- PORT-1-01 (1): towel dry after 1 min hose. Type Y or N.

- PORT-1-02 (1): lap gap at 10.2 kg (100 N). Mm, pass if at most 2.

- PORT-1-03 (1): gloved lock adjustments OK out of 5. Count, pass if at least 5.

Note: BL-038: added from release_tests.csv / corrie_f_body 13.9

Defined in: designs/pack_5200/analysis/corrie_f_body.md 13.8, 13.9; outputs/release/pack5200_corrie_f_rev3/manufacturing/release_tests.csv

## 14. TL-1: guy + crown-strap toggles in the 38 mm B-25 sleeve on the real crown bar, 497 N

Why: B-25 bridle (38 mm, BL-041) and the guy / crown-strap toggles: bag cut

Specimen: 5 x the real rev C crown top bar stock ([+-45/0x5/+-45], wall 1.065, same vendor and clear coat; NOT a generic 18 mm offcut), each >= 200 mm, carrying a HALF B-25 as built (DECISION B25-TOPO, SK-12): 38 mm toggle tab (2 x 30) | sleeve round the bar at x 26.5-64.5 from the spine centre, lap bartacked as at frame assembly | the strip down the aft side, the 45 deg MITRE under the sleeve | the run necked to 25 mm, wrapped once round a 20 mm spine stub (crown spine tube offcut or 20 mm steel rod) bolted upright to the rig at x 0 and 25.0 mm FORE of the bar axis (the printed spine channel, BL-046), under a 6.0 mm collar spacer, its end clamped behind the stub, with the guy toggle AND the crown-strap toggle in the tab; 1 untested control length

| load | N | lb | kg on the scale |
|---|---|---|---|
| step 1 | 100 | 22 | 10.2 |
| step 2 | 200 | 45 | 20.4 |
| step 3 | 300 | 67 | 30.6 |
| step 4 | 400 | 90 | 40.8 |
| proof, hold 60 s | 497 | 112 | 50.7 |

Steps

1. Clamp the tube ends in V-blocks >= 40 mm outside the sleeve edges (the bar reacts as a beam), NO cradle or saddle under the sleeve

2. Anchor the run's spine end on the spine stub 25.0 mm fore of the bar axis under the 6.0 mm collar (BL-046 geometry; the run then pulls fore-down by itself)

3. Dial bore gauge 10-18 mm range, 0.01 mm, >= 110 mm reach (BL-043 m-7: basis A predicts dD at the LOOP CENTRE; the gauge goes in through the open outboard tube end, anvils on the pull axis at the sleeve centre, depth marked on the shaft; zero it on the unloaded tube)

4. TWO RIGS (BL-047 record, bl047_frame_items.json tl1.pull): specimens 1-3 on RIG B, 4-5 on RIG A, proof 497 N on the guy toggle (the larger of the BL-046 vector 462 N and BL-045's 497 N): RIG A (the governing BL-046 vector): guy toggle pulled 22 deg below horizontal, 0.5 % outboard (unit: down 0.38, aft 0.93, outboard 0.01; come-along anchor 1 m from the toggle at 377 mm below, 926 mm aft, 5 mm outboard)

5. RIG B (run loaded, BL-045's worst guy): 58 deg below horizontal, 13.6 % outboard (down 0.85, aft 0.51, outboard 0.14; anchor at 849 below, 510 aft, 136 outboard)

6. In BOTH rigs the crown-strap toggle carries 43 N (4.4 kg on a cord over a pulley, up and aft as the crown strap) through the proof

7. The rig's design guy load is 454 N, proofed to 497 N

8. Check each angle with a phone inclinometer on the line

9. Pull in steps 100 / 200 / 300 / 400 / 497 N (50.7 kg), reading the gauge at each step

10. Hold 497 N 60 s, listening (phone microphone on the tube)

11. Unload, remove the sleeve, micrometer the loop centre at 0 and 90 deg

12. Check the mitre and the run did not slip (pen marks across the fold and at the stub)

13. Loupe + tap test against the control

14. Repeat 1 sample after 4 h in the freezer (POM)

Pass: 5 of 5: (1) holds 497 N 60 s, toggle does not pull through or crack, no broken stitch; (2) no crackle; (3) gauge at 497 N within 1.15 x the straight line through the 100-300 N readings, no knee; (4) ovalisation at the LOOP CENTRE (bore gauge, BL-043 m-7) <= 0.40 mm at 497 N (basis A 0.39, B 0.2, C 0.005 mm, all at the loop centre); (5) out-of-round after unloading <= 0.02 mm more than before; (6) no crush mark, whitening or resin crack at 10x, tap test clean; the cold sample holds

Write in the results sheet (results_template.csv, column measured):

- TL-1-01 to TL-1-05 (oval#1 to oval#5): ovalisation at 497 N. Mm, pass if at most 0.40.

- TL-1-06 to TL-1-10 (set#1 to set#5): permanent out-of-round after unloading (increase). Mm, pass if at most 0.02.

- TL-1-11 to TL-1-15 (hold#1 to hold#5): held 497 N 60 s, linear (no knee), silent, no crush mark / whitening at 10x, toggles whole. Type Y or N.

- TL-1-16 (cold1): cold sample (4 h freezer) held the proof. Type Y or N.

Note: BL-041 (crown_tube_guy_loop.md 6, orchestrator 2026-11-13): 497 N on real bar stock with the 38 mm B-25 (was 432 N on a 25 mm loop round an offcut, which basis A puts above the bare bar's 454 N). TL-1b is the bare-bar data test. BL-043 (third review M-4, m-7, m-8): specimen with the real turn and run (B25-TOPO), ovalisation read at the loop centre with a bore gauge, stock 2.0 m

Defined in: designs/pack_5200/analysis/crown_tube_guy_loop.md sections 5-6 (adopted 2026-11-13); designs/pack_5200/analysis/loadpath_verification.md §3 A3

## 15. FS-O5: ULTRA 200X 50 mm strip tensile (C-5 / O-5)

Why: C-5 bag bottom 500D -> ULTRA 200X

Specimen: 5 warp + 5 weft strips 50 mm wide (ISO 13934-1 / ASTM D5035 style) + 5 PT-3 swatches on ULTRA 200X

Steps

1. Clamp each strip in the 50 mm bar grips, 200 mm gauge

2. Pull slowly to break

3. Record the peak

Pass: mean >= 1840 N/50 mm, minimum >= 1195 N/50 mm

Write in the results sheet (results_template.csv, column measured):

- FS-O5-01 (1): passes the register criterion: mean >= 1840 N/50 mm, minimum >= 1195 N/50 mm. Type Y or N.

Note: between 1195 and 1840: swap allowed, report to structural

Defined in: designs/pack_5200/analysis/fabric_swap_evidence.md O-5

## 16. WP-3: gloved tail on W1-W3 ladder locks

Why: A-2 wrap tails 250 -> 150 mm

Specimen: ladder lock on 38 mm webbing with a 150 mm tail

Steps

1. Winter gloves on

2. Release and re-tension, loaded (10 kg hanging) and slack, 5 times each

Pass: release and re-tension in one grab every time

Write in the results sheet (results_template.csv, column measured):

- WP-3-01 (1): passes the register criterion: release and re-tension in one grab every time. Type Y or N.

Note: A-2 is already applied in B.2: a fail adds 5.4 g back

Defined in: designs/pack_5200/analysis/corrie_body.md §12.4

## 17. SL-1: weigh the cord-pull door sliders and the printed label; gloved door cycles

Why: BL-037 G-1 (-8.0 g) and X-1 (-7.0 g)

Specimen: 5 YKK #10 coil slider bodies with a 3 mm cord pull each; the printed label; a 400 mm #10 coil door sample on 330D tabs with 2 sliders

Steps

1. Fit the cord pull to each slider

2. Weigh each on the 0.01 g scale

3. Weigh the label

4. With winter gloves open and close the door sample 20 times

Pass: each slider with pull <= 15.0 g; label <= 1.0 g; 20 of 20 without a snag

Write in the results sheet (results_template.csv, column measured):

- SL-1-01 to SL-1-05 (#1 to #5): slider + cord pull mass. G, pass if at most 15.0.

- SL-1-06 (label): printed label mass. G, pass if at most 1.0.

- SL-1-07 (door): gloved open / close without a snag, out of 20. Count, pass if at least 20.

Note: BL-038: added (BL-037). Sliders are LONG LEAD (Rockywoods sold out 2026-11-13)

Defined in: designs/pack_5200/analysis/weight_push.md 2 (G-1, X-1), 7 SLIDER-CORD, LABEL

## 18. HB-7: hipbelt heavy ladder-lock slip, corrected 2:1 routing

Why: belt B 2:1 pull

Specimen: 5 heavy 25 mm ladder locks on 25 mm webbing, routed anchor -> D-ring -> lock

| load | N | lb | kg on the scale |
|---|---|---|---|
| hold 1 min, no slip | 426 | 96 | 43.4 |
| slip must be above | 490 | 110 | 49.9 |

Steps

1. Mark webbing

2. Pull the lock leg to 426 N, hold 1 min

3. Then pull on until it slips and record the load

Pass: no slip at 426 N (1.5 x 284 N); slip load >= 490 N keeps the record

Write in the results sheet (results_template.csv, column measured):

- HB-7-01 to HB-7-05 (#1 to #5): no slip at 43.4 kg (426 N), 1 min. Type Y or N.

- HB-7-06 to HB-7-10 (s#1 to s#5): slip load. Kg, pass if at least 50.0.

Note: pass number derived here: 1.5 x 284 N = 426 N (doc gives only 'as the record notes')

Defined in: designs/pack_5200/analysis/hipbelt_structural.md §5, §8

# 19. TL-1b: bare crown bar under today's 25 mm sleeve: step load, ovalisation vs load (which ring model is real)

Why: none (data, not a gate); TL-1 is the gate

Specimen: 2 lengths of the real rev C crown bar stock (>= 160 mm) with a 25 mm B-25-style sleeve (lap bartacked), both toggles in the tab

| load | N | lb | kg on the scale |
|---|---|---|---|
| step 1 | 100 | 22 | 10.2 |
| step 2 | 200 | 45 | 20.4 |
| step 3 | 288 | 65 | 29.4 |
| step 4 | 300 | 67 | 30.6 |
| step 5 | 400 | 90 | 40.8 |
| last step (stop at a knee) | 497 | 112 | 50.7 |

Steps

1. Same rig as TL-1 (V-blocks >= 40 mm outside the sleeve, no cradle under it)

2. Dial bore gauge 10-18 mm range, 0.01 mm, >= 110 mm reach (BL-043 m-7: basis A predicts dD at the LOOP CENTRE; the gauge goes in through the open outboard tube end, anvils on the pull axis at the sleeve centre, depth marked on the shaft; zero it on the unloaded tube)

3. Step 100 / 200 / 288 / 300 / 400 / 497 N, read the gauge at each step

4. Stop at the first knee or sound

Pass: record only: dD at 288 N against the predictions A 0.35 / B 0.18 / C 0.004 mm; holds 497 N linear with dD at or below the B line = basis B credited (tell the structural-fea-engineer)

Write in the results sheet (results_template.csv, column measured):

- TL-1b-01 (1): ovalisation at the loop centre at 100 N (bore gauge). Type the value in mm.

- TL-1b-02 (1): ovalisation at the loop centre at 200 N (bore gauge). Type the value in mm.

- TL-1b-03 (1): ovalisation at the loop centre at 288 N (bore gauge). Type the value in mm.

- TL-1b-04 (1): ovalisation at the loop centre at 300 N (bore gauge). Type the value in mm.

- TL-1b-05 (1): ovalisation at the loop centre at 400 N (bore gauge). Type the value in mm.

- TL-1b-06 (2): ovalisation at the loop centre at 100 N (bore gauge). Type the value in mm.

- TL-1b-07 (2): ovalisation at the loop centre at 200 N (bore gauge). Type the value in mm.

- TL-1b-08 (2): ovalisation at the loop centre at 288 N (bore gauge). Type the value in mm.

- TL-1b-09 (2): ovalisation at the loop centre at 300 N (bore gauge). Type the value in mm.

- TL-1b-10 (2): ovalisation at the loop centre at 400 N (bore gauge). Type the value in mm.

- TL-1b-11 to TL-1b-12 (knee1 to knee2): load at the first knee or sound (none: write the last step). Type the value in N.

Note: BL-041 (crown_tube_guy_loop.md, orchestrator 2026-11-13): new

Defined in: designs/pack_5200/analysis/crown_tube_guy_loop.md section 6 (TL-1b), ASSUMPTIONS RING_SLICE, BASIS_A

## After the session

1. Fill measured and date for every row you ran; leave the rest empty.
2. Photograph every failed or doubtful sample; put the photo number in photo.
3. Send the sheet. The engineers read it back with huntkit.manufacturing.test_results.

# G3: Wash and look: seams without binding

## G3 wash swatches

```
mesh laundry bag + towel + all swatches  ->  normal cold cycle  ->  line dry  ->  x3  ->  loupe on every edge
```

When: weekend 1, evenings

Rig: washing machine, mesh laundry bag, 10x loupe

Kits: E, D (loupe)

Both swatch sets go in the same wash. Three cold cycles, line dry, then look at every edge.

About 3.0 h hands-on and 9 h of machine time.

## Safety (read before you start)

- Nothing loaded. Keep the hot knife off unless a swatch fails.

## Rig sketch

mesh laundry bag + towel + all swatches  ->  normal cold cycle  ->  line dry  ->  x3  ->  loupe on every edge

Picture: sketches/G3_rig.png

Sew first: the 6 FRAY-1 SSa2 swatches (rotary-cut, edges NOT seared) and the 4 WP-1 swatches (seared), 200 mm each, as listed under each test below.

## Tests in this session, in order

| # | test | releases |
|---|---|---|
| 1 | FRAY-1 | Bag release, step 1: cut and sew |
| 2 | WP-1 | Bag release, step 1: cut and sew |

# 1. FRAY-1: wash-swatch fray of the un-bound SSa2 body seam (rev 3 fabrics, as cut)

Why: un-bound body seams (bag release)

Specimen: 6 SSa2 swatches 200 mm long, rotary-cut edges NOT seared (as the prototype cutting plan cuts them): ULTRA 200|330D, 330D|330D, 500D|330D, 630D printable|330D, HyperD 300|330D, 100D|330D; 301, 8 SPI, tex 90, Nm 110 round point

Steps

1. Sew each swatch as op 3.07 (SSa2)

2. Wash with the WP-1 swatches: 3 normal cold cycles in a mesh bag with a towel, line dry

3. Look at every edge with the loupe

Pass: no allowance frayed past the 2nd row on any swatch; a fail on one pairing: hot-knife that cut edge or bind that seam (20 mm grosgrain)

Write in the results sheet (results_template.csv, column measured):

- FRAY-1-01 to FRAY-1-06 (ULTRA|330D to 100D|330D): no fray past the 2nd row after 3 washes. Type Y or N.

Note: BL-038: added; same wash load as WP-1 (WP-1 is the B.2 seared seam)

Defined in: outputs/release/pack5200_corrie_f_rev3/manufacturing/release_tests.csv (RD-17); designs/pack_5200/mfg_corrie_f_rev3.py ASSUMPTION NO-BIND; docs/DECISIONS.md 2026-11-13 orchestrator call (3)

# 2. WP-1: wash-swatch fray, seams without binding

Why: B-1 (no binding on body seams)

Specimen: 4 seam pairings 330D|330D, 330D|ULTRA 200, 500D|330D, 100D|ULTRA; 301 + 2nd row 6 mm in, trimmed, hot-knife seared

Steps

1. Sew one 200 mm swatch per pairing

2. 3 machine-wash cycles (R-12), dry

3. Look at each edge

Pass: no fray past the second row on any swatch

Write in the results sheet (results_template.csv, column measured):

- WP-1-01 to WP-1-04 (330D|330D to 100D|ULTRA): no fray past the 2nd row after 3 washes. Type Y or N.

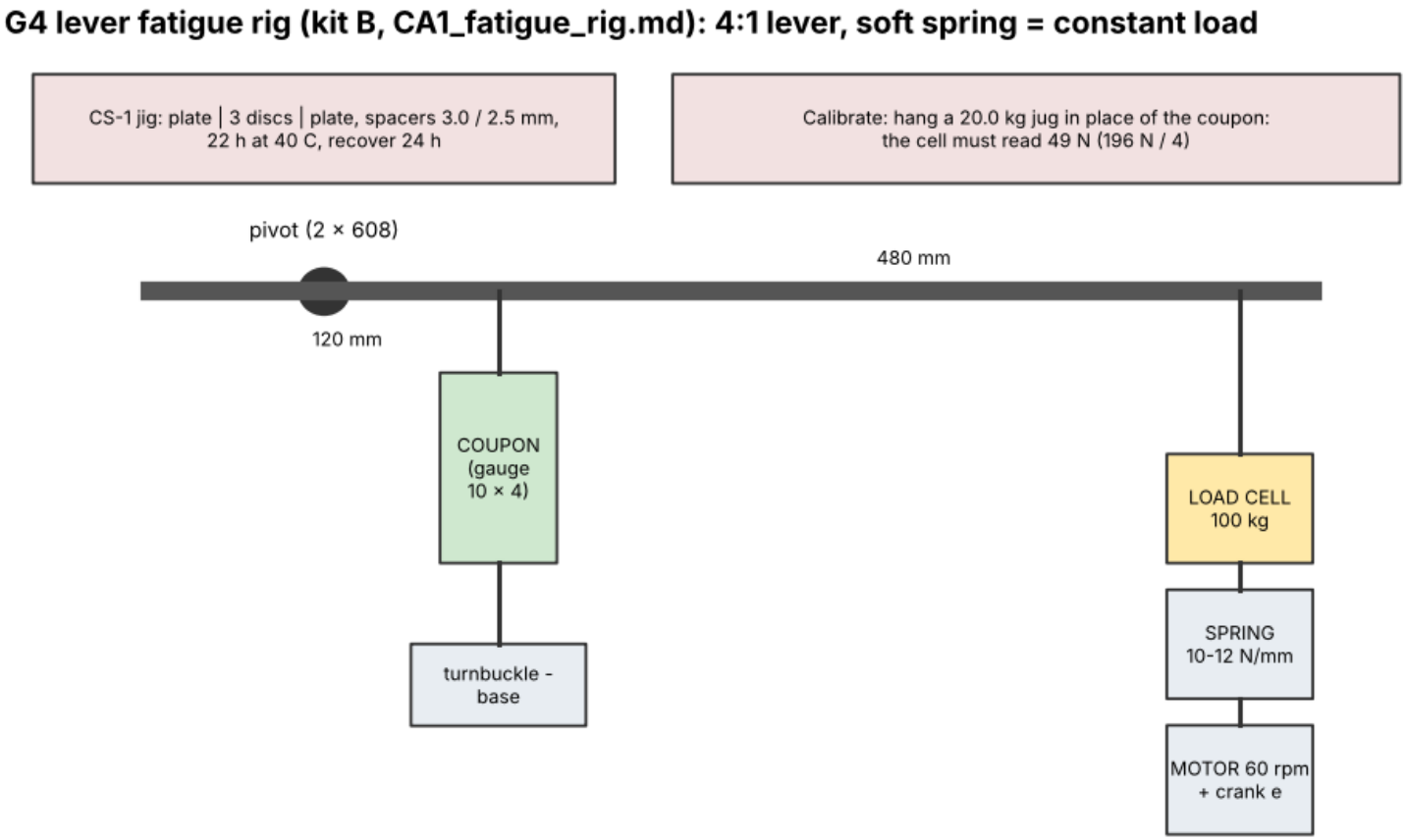
Note: B-1 is already in B.2: a fail adds 36.4 g back. R-12 / S-03 gate is the same test (merged)

Defined in: designs/pack_5200/analysis/corrie_body.md §12.4; designs/pack_5200/analysis/weight_pass_corrie.md B-1; designs/pack_5200/analysis/build_simplicity.md S-03

## After the session

1. Fill measured and date for every row you ran; leave the rest empty.
2. Photograph every failed or doubtful sample; put the photo number in photo.
3. Send the sheet. The engineers read it back with huntkit.manufacturing.test_results.

# G4: Lever rig: frame coupons (weeks 2-5)




When: build weekend 2, then runs day and night

Rig: kit B lever fatigue rig (CA1_fatigue_rig.md), 40 C box, 3-point bend fixture

Kits: B, D

Build the rig once; every frame coupon runs on it. CA-1 at the release level comes first: frame release R1 cannot happen without it. The 40 C box also runs CS-1 (BL-037) beside CA-4.

About 38.0 h hands-on and 684 h of machine time.

## Safety (read before you start)

- The crank and lever move for days: guard the crank with a box, keep hands out while it runs, unplug before touching a coupon.

- 40 C box: thermostat set and checked with a thermometer before you leave it.

- CA-5 static pulls reach 4.4 kN at the stub: blanket over the rig, stand to the side.

## Rig sketch

```
            load cell (S-type 100 kg) in the drive link
  motor + crank --> rod --> [spring cage] --> [LC] --> lever end (480 mm from pivot)
                                          |
  pivot (M8 shoulder bolt, 2 x 608) O----------------------+-------- lever 6061 40 x 10 x 600
                                | 120 mm
                          upper grip
                          [coupon]       ratio 480 / 120 = 4
                          lower grip -- turnbuckle -- base
```

CS-1 jig:  [plate]  disc disc disc  [plate]  spacers 3.0 mm (6 mm discs) / 2.5 mm (5 mm discs)

4 x M6 bolts, tighten until the plates sit on the spacers, then 22 h in the 40 C box

Picture: sketches/G4_rig.png

Before: the coupons have had their 7 days (G1). Build the rig (one weekend, CA1_fatigue_rig.md 4) and calibrate it with the 20 L jug. Sew the 3 SEW-F box-X loops with the MT-1 loops (G2 prep).

## Tests in this session, in order

| # | test | releases |
|---|---|---|
| 1 | CA-1 | Frame release R1 |
| 2 | IM3-EC | Frame release R1 |
| 3 | CA-5 | Frame release R1 |
| 4 | CLEVIS-SHARE | Frame release R1 |
| 5 | SEW-F | Bag release, step 2: batch |
| 6 | CS-1 | BL-037 weight gates |
| 7 | BP-1 | BL-037 weight gates + Field release and the rest |
| 8 | HB-6 | Field release and the rest |
| 9 | CA-3 | Field release and the rest |
| 10 | CA-4 | Field release and the rest |
| 11 | HB-5 | Field release and the rest |
| 12 | T6 | Field release and the rest |
| 13 | T7 | Field release and the rest |
| 14 | CA-2 | Field release and the rest |

# 1. CA-1: PAHT-CF fatigue coupons, 0.40 (fallback >= 0.313)

Why: H1 print release R1: R1 = CA-1 passes 0.40 with 3/3 run-outs AND MT-1c shows a cinch swing <= 117 % (peak-to-peak / mean) at a mean <= 150 N per anchor (structural countersign, adopted 2026-11-13, analysis/ca1_mt1c_confirm.md); 0.40 also feeds option F-1 (BL-037)

Specimen: 12 XY + 4 Z + 3 spare bars, printed and annealed like the frame, 7 days at 23 C / ~50 % RH

| load | N | lb | kg on the scale |
|---|---|---|---|
| XY bar peak at the R1 level 0.40 | 1126 | 253 | 114.8 |
| XY bar valley (R 0.1) | 113 | 25 | 11.5 |
| load cell reads (4:1 lever) | 282 | 63 | 28.7 |
| XY bar peak at the lowest fallback level 0.313 (only if 0.40 fails) | 881 | 198 | 89.8 |

R1 rule (adopted 2026-11-13, structural countersign): R1 = CA-1 passes 0.40 with 3/3 run-outs AND MT-1c shows a cinch swing <= 117 % (peak-to-peak / mean) at a mean <= 150 N per anchor (structural countersign, adopted 2026-11-13, analysis/ca1_mt1c_confirm.md). Run 3 XY bars at 0.40. Then, if 0.40 fails: 3 XY bars at the table level for the measured swing, >= 0.313 (<= 0 %: 0.313 / <= 5 %: 0.316 / <= 10 %: 0.319 / <= 20 %: 0.325 / <= 30 %: 0.331 / <= 50 %: 0.345 / <= 70 %: 0.361 / <= 100 %: 0.386 / <= 115 %: 0.398). Swing > 115 %: STOP, no fallback level: the structural-fea-engineer re-runs the cinch case. The BL-038 / BL-039 levels 0.276 and 0.404 are retired: the 0.276 rule left out the cinch mean stress and the R 0 (frame) vs R 0.1 (coupon) gap (body Goodman flip 0.2926, margin 0.943 at 0.276).

Steps

1. Calibrate: hang the 20.0 kg jug in place of the coupon; the cell must read 49 N. Write the reading.

2. Static (5 bars): put the turnbuckle in place of the spring; clamp an XY bar 35 mm deep in each grip; turn the turnbuckle about 1 turn every 10 s until the bar breaks; peak cell force x 4 / smallest gauge area = strength in MPa. 3 XY bars, then 2 Z bars (CA1_fatigue_rig.md 6 step 1).

3. UTS_meas = mean of the 3 XY strengths (cap 88 MPa). Work out your own peak: F_max = 0.4 x 0.8 x UTS_meas x area (at 88 MPa and 40 mm2: 1126 N; the cell then reads 282 N). F_min = 0.1 x F_max.

4. Set the crank throw and the turnbuckle so the cell swings between F_min / 4 and F_max / 4; check over 20 slow turns by hand before the motor runs.

5. R1 level 0.40, 3 XY bars one at a time: run to 100,000 cycles (27.8 h at 1 Hz). The rig stops itself if the bar breaks, its stiffness drops more than 10 % or the force drifts more than 5 %.

6. After each run-out look at the gauge and fillets with the loupe; then pull that bar to failure (step 2): it must keep at least 0.9 x UTS_meas.

7. Then 2 Z bars at 0.40 (0.40 x 0.8 x the Z strength).

8. Only if a 0.40 bar failed: 3 XY bars at the countersign table level for MT-1c's measured swing, at least 0.313 (881 N peak at 88 MPa at 0.313); see the R1 rule above.

9. 2 of 3 at a level is inconclusive: run 3 more; a level passes only with 5 of 6.

10. Send the rig's CSV log with the results sheet.

Pass: 3/3 run-outs to 1e5 at 0.40, stiffness loss <= 10 %, no crack; static XY >= 79.2 MPa, Z >= 42.3 MPa. R1 also needs MT-1c swing <= 117 % at mean <= 150 N. The 0.276 / 0.404 levels are retired (structural countersign 2026-11-13)

Write in the results sheet (results_template.csv, column measured):

- CA-1-01 to CA-1-03 (XY#1 to XY#3): static XY UTS. MPa, pass if at least 79.2.

- CA-1-04 to CA-1-05 (Z#1 to Z#2): static Z UTS. MPa, pass if at least 42.3.

- CA-1-06 to CA-1-08 (RXY#1 to RXY#3): cycles at the R1 level 0.40 (run-out 100000). Cycles, pass if at least 100000.

- CA-1-09 to CA-1-11 (SXY#1 to SXY#3): stiffness loss at run-out. %, pass if at most 10.

- CA-1-12 to CA-1-13 (RZ#1 to RZ#2): Z cycles at 0.40. Cycles, pass if at least 100000.

- CA-1@fallback-01 to CA-1@fallback-03 (FXY#1 to FXY#3): only if 0.40 failed: cycles at the table level for the MT-1c swing (run-out 100000). Cycles, pass if at least 100000.

- CA-1@fallback-04 (level): fallback level used (table, >= 0.313). Type the value in -.

Note: BL-040: R1 rule from the adopted structural countersign (ca1_mt1c_confirm.countersign). BL-038: release level from h1_buckling.json (BL-027 body flip) and im3_body_simloads.json (28 mm cheek), replacing the 0.310 placeholder (CA1_fatigue_rig.md 1 asked for this). Grams: BL-037 F-1 (-9.9 g, an option that also needs a buckling re-run), not -50..-70 g. Machine: 8 runs x 27.8 h. CT-B1 merged here

Defined in: designs/pack_5200/tests_physical/CA1_fatigue_rig.md; designs/pack_5200/analysis/frame_comp_verdict.md §6

## 2. IM3-EC: IM3 eye cycle coupons, 3 x 1e5 at >= 1.15 x 1024 N

Why: H1 print release (IM3 eye); P_cheek

Specimen: 3 eye coupons: 120 mm stay segment round z 520 with the IM3 eye and the 28 mm cheek (BL-021), printed from print/frame_h1/test_pieces/im3ec_cheek_R.stl (BL-042: z 520 R, eye printed 7.0 / reamed 7.2), printed with the CA-1 bars. frame profile of BUILD_H1.md section 3 with SUPPORTS REQUIRED (it touches the plate over only 729 mm2, BL-053 print_settings_bl053.json): tree supports from the build plate in PAHT-CF, blockers in every bore, pin hole, bolt hole and trap, scars sanded flat (BUILD_H1 section 4), filament about 71 g, about 2.5 h

| load | N | lb | kg on the scale |
|---|---|---|---|
| eye cycle peak (1.15 x 1024 N) | 1178 | 265 | 120.1 |
| after run-out, proof pull | 2553 | 574 | 260.2 |

Steps

1. Sewn T3 tongue + IM3 carabiner through the eye, led 32 deg aft of the stay axis (the DIR_SCATTER angle: computed 22.3 deg + 10), coupon clamped by the stay ends

2. Lever rig at R about 0.02 (F_min / peak): 24 -> 1178 N (F_min only keeps the rig taut), 1 Hz, to 1e5

3. Loupe check every 25,000

4. Then pull to 2553 N on the pull bench. (Equivalent alternative: 1272 N peak at R 0.1)

Pass: 3/3 run-outs to 1e5 at 1178 N, no crack at 10x; residual pull 2553 N with no crack

Write in the results sheet (results_template.csv, column measured):

- IM3-EC-01 to IM3-EC-03 (#1 to #3): cycles at 24 -> 1178 N (run-out 100000). Cycles, pass if at least 100000.

- IM3-EC-04 to IM3-EC-06 (p#1 to p#3): no crack at 10x and held 2553 N after. Type Y or N.

Note: BL-040: angle 32 deg and R about 0.02 from the adopted structural countersign. BL-038: cheek 27 -> 28 mm (BL-021 adopted). BL-039: load 1178 N from the BL-021 sim demand (was 1017 N = 1.15 x 1.6 x 552.4 N, BL-010's 15 in IM3 force, below today's demand)

Defined in: designs/pack_5200/tests_physical/CA1_fatigue_rig.md §6a; designs/pack_5200/analysis/im3_body_simloads.md (28 mm cheek, governing hook + strap); designs/pack_5200/analysis/loadpath_verification.md §4.1, §8

## 3. CA-5: dry z 234 joint stub pulls + 1e5 cycle

Why: H1 print release

Specimen: 4 stub pairs, one stay, 60 mm each side of z 234, real bosses, Ti bolts

| load | N | lb | kg on the scale |
|---|---|---|---|
| static mean must reach | 4400 | 989 | 448.5 |
| lowest static | 3666 | 824 | 373.7 |
| cycle force | 1543 | 347 | 157.3 |

Steps

1. 3 static eccentric pulls to failure (200 kg cell or 6:1 hole)

2. 1 stub 1e5 cycles at the 1.6 g force with a dial across the joint

Pass: static mean >= 4400 N, lowest >= 3666 N (1.5 / 1.25 x 2933 N); cycle: no crack at the boss root, joint step <= 0.1 mm

Write in the results sheet (results_template.csv, column measured):

- CA-5-01 to CA-5-03 (#1 to #3): static stub peak. N, pass if at least 3666.

- CA-5-04 (mean): mean of the 3 static peaks. N, pass if at least 4400.

- CA-5-05 (cyc): joint step after 1e5 cycles at 1543 N. Mm, pass if at most 0.1.

- CA-5-06 (cyc): no crack at the boss root after 1e5 cycles. Type Y or N.

Note: CONFLICT: verdict uses 2894 N and 0 -> 1543 N (team A); CA1 doc uses 2933 N (H1). 2933 N used. H1 cycle force not restated: 1543 N used until structural gives the H1 value. Fail -> B 8-bolt flange, +17 to +33 g. CT-B3 merged

Defined in: designs/pack_5200/tests_physical/CA1_fatigue_rig.md §7; designs/pack_5200/analysis/frame_h1.md §8; designs/pack_5200/analysis/frame_comp_verdict.md §6

## 4. CLEVIS-SHARE: US2 bolt-on clevis joint: 720 N cradle pull with the eye on the outer cheek, then 1e5 cycles at 480 N; no joint opening

Why: frame release R1 (the US2 lug as built from two printed parts)

Specimen: designs/pack_5200/print/frame_h1/test_pieces/us_route_stay_R.stl + designs/pack_5200/print/frame_h1/us2_clevis_R.stl (a SECOND print of each: HE-FIT step 0 uses the first) printed with the CA-1 bars and conditioned 7 days. The route piece: frame profile of BUILD_H1.md section 3 with SUPPORTS REQUIRED (it touches the plate over only 9 mm2, BL-053 print_settings_bl053.json): tree supports from the build plate in PAHT-CF, blockers in every bore, pin hole, bolt hole and trap, scars sanded flat (BUILD_H1 section 4), filament about 178 g, about 6.3 h. The clevis: frame profile of BUILD_H1.md section 3, NO supports (it prints flat), blockers in the holes, annealed 90 C 6 h, filament about 14 g, about 0.5 h. Clevis holes reamed 5.1, bolted with 2 Ti gr5 M5 x 35 DIN 7984 with the heads in the printed 9 mm counterbores and NO washer under the head, 2 A2 nuts in the side traps, 2 x 10 mm OD DIN 433 washers on the NUT side only (NOT the z 234 joint's M5 x 45 / 12 mm washers; shopping list), blue threadlocker, snug; check the heads sit below the clevis face (a steel rule across the face does not touch them; design 0.5 mm below) (no preload credited, CLEVIS-DRY); strain gauges (1 mm foil, 120 ohm, cyanoacrylate on the sanded PAHT) on the US2 arm root (R side, bag face, 3 mm off the stay surface, gauge axis along the arm); root stress = strain x 3.86 GPa (youngs_modulus_xy_mpa, record pa12_cf_fdm_paht); TEMPERATURE (BL-054, review 5 m-8): wire each gauge as a HALF bridge with a dummy gauge of the same lot bonded on an unloaded offcut of the same print, lying beside the specimen (it cancels the PAHT-CF's thermal strain); write the room temperature at every reading; with no dummy, re-zero the bridge unloaded at every stop and log the temperature (REQUIRED since BL-052); one sewn US eye (25 mm, necked as op 'US', 18.6 mm slot) on hardened steel M5 x 50 dowel (DIN 6325 m6, HRC 58-62; NOT a 7075 frame pin; it stands about 5 mm proud of each hinge cheek so you can grip it) in the US2 throat with its UHMWPE sleeve

Steps

- Set-up. Clamp the piece by its stay ends on the lever rig base (as IM3-EC)

- Set-up. 1 mm PTFE shim between the eye and the INNER cheek so the eye bears on the OUTER cheek (the clevis: CLEVIS-SHARE worst case)

- Set-up. Lead the US eye's strap aft and down at 31 deg below the bag direction (the lug check's cradle line; the lane span gives 23.0 deg for US2, which loads the joint less in shear-down) through the rig link

- Set-up. Pencil lines across the joint face top and bottom, a 0.05 mm feeler at hand

- Step 1. Static: turnbuckle in place of the spring, pull in steps to 720 N (73.4 kg), hold 60 s; at 720 N try the 0.05 mm feeler at the joint face from both sides and both ends; unload, check the pencil lines

- Step 2. Cycles: spring back in, R about 0.1 (48 -> 480 N), 1e5 cycles, stop at 1e3, 1e4 and 1e5 to check the feeler at 480 N and the pencil lines

- Step 3. At the end re-check the bolts' torque mark and look at the arm root, the throat floor and each head counterbore (no pull-in, no crack round the head: HEAD-BEAR) with the loupe; STRAIN GAUGE (required, BL-052): read the US2 arm-root gauge at 240 / 480 / 720 N in step (1) and at the peak at each stop in step (2); record strain and root stress (results rows)

Pass: no joint opening: the 0.05 mm feeler does not enter the joint face at 720 N or at 480 N at any stop; pencil lines step <= 0.05 mm (none visible under the loupe); no crack at the cheek root, the arm root, round the bolt holes or at the head counterbores, no head pull-in; the bolt torque marks have not moved; 1e5 cycles run out; US2 root stress (gauge) <= the h1_crown lug model's root stress at the same load (structural-fea-engineer reads it off; ASSUMPTION STAY-STUB / CROWN-SECTION)

Write in the results sheet (results_template.csv, column measured):

- CLEVIS-SHARE-01 (static): no joint opening at 720 N (0.05 mm feeler does not enter, both sides, both ends). Type Y or N.

- CLEVIS-SHARE-02 (static): pencil-line step across the joint after 720 N. Mm, pass if at most 0.05.

- CLEVIS-SHARE-03 to CLEVIS-SHARE-05 (1000 cycles to 100000 cycles): no joint opening at 480 N (feeler), no crack (loupe). Type Y or N.

- CLEVIS-SHARE-06 (end): bolt torque marks unmoved, no crack at the cheek root / arm root / bolt holes. Type Y or N.

- CLEVIS-SHARE-07 (end): no head pull-in, no crack at the head counterbores (BL-051 heads, no washer). Type Y or N.

- CLEVIS-SHARE-08 to CLEVIS-SHARE-10 (US2 root 240 N to US2 root 720 N): US2 arm-root strain (gauge, BL-052). Type the value in microstrain.

Note: BL-050 B (orchestrator): new. BL-052: M5 x 35, no head washer, 2 nut-side washers (BL-051 counterbore); the US2 root gauge is required. The US test piece printed for HE-FIT step 0 can be re-used AFTER step 0 only if this test is run on a second print (this test may crack it)

Defined in: designs/pack_5200/analysis/h1_bedfit.md 2 and 7 (ASSUMPTIONS CLEVIS-SHARE, CLEVIS-DRY, CLEVIS-PRINT); designs/pack_5200/analysis/h1_crown.md 3a, 4, 7 (BL-051: counterbored heads, HEAD-BEAR, the US2 root gauge); huntkit/gendesign/h1_bedfit.py JOINT; us_route.json cradle_span (BL-050 B)

## 5. SEW-F: box-X fatigue on the lever rig: 25 mm box-X loops, 1e5 cycles 44 -> 442 N (R 0.1); HE hinge eyes 62 -> 625 N on the printed fairlead (BL-044)

Why: IM1-IM3 end loops (S1), US1/US2 sewn eyes (S6), belt anchor box-X (H1), HE hinge-eye 34 x 7 box (BL-044, review M-3): loaded carry and the bag batch

Specimen: 3 x 25 mm poly webbing loops, box-X 21 x 25 mm (the IM / US box as drawn), 301, 7 SPI, tex 135, Nm 140 R, sewn with the MT-1 loops; + 3 HE hinge-eye specimens sewn exactly as the PT-3 specimens (op 4.05), on a third print of the HE-FIT fairlead with the PT-3 spacer plate

| load | N | lb | kg on the scale |
|---|---|---|---|
| loop peak (1.15 scatter x 1.6 g x the S6 1 g load) | 442 | 99 | 45.1 |
| valley (R 0.1) | 44 | 10 | 4.5 |
| HE peak (1.15 x 1.6 g x the rev C hinge 1 g load, BL-044) | 625 | 141 | 63.7 |
| HE valley (R 0.1) | 62 | 14 | 6.3 |
| HE after run-out, PT-3 proof | 1530 | 344 | 156.0 |

| load | N | lb | kg on the scale |
|---|---|---|---|
| load cell reads at the peak (4:1 lever) | 110 | 25 | 11.3 |
| after run-out, MT-1 US proof | 1080 | 243 | 110.1 |

Steps

1. Loop over a 5 mm pin on the lever (in place of a coupon), the free end in the bar clamp

2. Set the crank so the cell swings 11 -> 110 N (4:1 lever = 44 -> 442 N on the loop)

3. 1 Hz to 100,000 cycles (27.8 h)

4. Loupe check at 25,000 / 50,000 / 75,000

5. After run-out pull the loop on the G2 bench to 1080 N (the MT-1 US proof), 1 min. HE (BL-044; mounting BL-048 m-6, G4 rig sketch): the printed fairlead + spacer (the PT-3 fixture, hardened steel M5 x 50 dowel (DIN 6325 m6, HRC 58-62; NOT a 7075 frame pin; it stands about 5 mm proud of each hinge cheek so you can grip it)) bolted to the lever rig's FIXED base plate at the coupon station, in place of the coupon block, stay segment upright under the lever's clamp

6. The B-05 strip in the lever's moving bar clamp (the fixture grip never holds the fabric), the pull straight down the spacer face as PT-3

7. Cell 16 -> 156 N (62 -> 625 N on the eye), 1 Hz to 100,000 cycles, loupe on the box TOP row

8. After run-out, the PT-3 proof 1530 N 60 s. PIN CHECK (BL-048 M-C): roll the dowel on the bench plate before and after the test: it rolls freely with no visible wobble and a 0.05 mm feeler does not pass under its middle

9. A bent pin voids the specimen. PRINT (BL-054, review 5 B-1): print the fixture as the H1 frame (designs/pack_5200/print/frame_h1/BUILD_H1.md sections 3-5 and 10): frame profile of BUILD_H1.md section 3 with SUPPORTS REQUIRED (it touches the plate over only 15 mm2, BL-053 print_settings_bl053.json): tree supports from the build plate in PAHT-CF, blockers in every bore, pin hole, bolt hole and trap, scars sanded flat (BUILD_H1 section 4), filament about 45 g, about 1.6 h, then ream as the frame.

Pass: loops: 3 of 3 run-outs to 1e5 at 442 N, no broken stitch, then hold 1080 N 1 min; HE: 3 of 3 run-outs to 1e5 at 625 N with no broken stitch in the box top row, then hold 1530 N 60 s (a 2 of 3 result in either set: run 3 more of that set, pass with 5 of 6)

Write in the results sheet (results_template.csv, column measured):

- SEW-F-01 to SEW-F-03 (#1 to #3): cycles at 44 -> 442 N (run-out 100000). Cycles, pass if at least 100000.

- SEW-F-04 to SEW-F-06 (HE#1 to HE#3): HE cycles at 62 -> 625 N on the printed fairlead (run-out 100000). Cycles, pass if at least 100000.

- SEW-F-07 to SEW-F-09 (p#1 to p#3): no broken stitch at run-out and held 1080 N 1 min after. Type Y or N.

- SEW-F-10 to SEW-F-12 (HEtop#1 to HEtop#3): HE: no broken stitch in the box TOP row at run-out (loupe). Type Y or N.

- SEW-F-13 to SEW-F-15 (HEproof#1 to HEproof#3): HE: then held 156.0 kg (1530 N) 60 s, eye open, fixture dowel straight after (BL-048 m-6 / M-C). Type Y or N.

Note: BL-040 (review M-9): added. Runs on the G4 lever rig between coupons; a loop is not a frame coupon so it can run beside CA-4 / CS-1. BL-044 (third review M-3): + 3 HE specimens at 1.15 x 1.6 g x the rev C hinge load, on the printed fairlead at the real offset (the box top row peels; ASSUMPTION SEWF-HE-RIG: the lever rig carries 625 N on the eye, 156 N at the cell)

Defined in: designs/pack_5200/analysis/loadpath_verification.md S1, S6, H1 ('PASS after ... SEW-F (lever rig)'), ASSUMPTION SEW_FAT; designs/pack_5200/analysis/review_20261113_rev3_release.md M-9

## 6. CS-1: XLPE L200 compression set, 50 % for 22 h at 40 C

Why: BL-037 B-1 (-31.9 g) and H-1 (-22.3 g)

Specimen: 3 discs 30 mm diameter of XLPE L200 6 mm (belt) and 3 of 5 mm (straps)

Steps

1. Measure each disc's thickness t0 with the caliper (light touch, 3 points, mean)

2. Put the discs between the two jig plates with 3.0 mm spacers (6 mm discs) or 2.5 mm spacers (5 mm discs)

3. Tighten until the plates sit on the spacers

4. 22 h in the 40 C box

5. Take out, unclamp, let them recover 24 h on a flat table

6. Measure tf the same way

7. Set % = (t0 - tf) / t0 x 100

Pass: every disc set <= 15 % (the EVA 45C record value)

Write in the results sheet (results_template.csv, column measured):

- CS-1-01 to CS-1-03 (6mm#1 to 6mm#3): compression set, 6 mm disc. %, pass if at most 15.

- CS-1-04 to CS-1-06 (5mm#1 to 5mm#3): compression set, 5 mm disc. %, pass if at most 15.

Note: BL-038: added (BL-037). Shares the CA-4 40 C box (ASTM D3575 style)

Defined in: designs/pack_5200/analysis/weight_push.md 2 (B-1, H-1), 7 XLPE-SET; data/materials/foams/xlpe_foam_l200.yaml (12 %), eva_foam_45c.yaml (15 %)

## 7. BP-1: sandwich backer strip 3-point bend coupon

Why: BL-005 sandwich strip (A_back); BL-037 K-1 (2 mm ROHACELL 71 core)

Specimen: 70 x 200 mm offcuts of the strip layup: 3 mm and 2 mm ROHACELL 71 core (2 mm: sand a 3 mm sheet or special order)

Steps

1. 100 mm span on two 10 mm rods, load at mid-span with a 10 mm rod

2. Dead weight 3.06 kg (30 N)

3. Dial indicator at mid-span

4. 3 loadings

Pass: mean mid-span deflection 0.205-0.277 mm (model 0.241 mm +/- 15 %)

Write in the results sheet (results_template.csv, column measured):

- BP-1-01 (3mm): 3 mm core strip passes BP-1. Type Y or N.

- BP-1-02 (2mm): 2 mm core strip passes BP-1 (SF >= 1 at the factored strip load). Type Y or N.

Note: model 0.241 mm = 0.134 bending (D11 66,647 N mm) + 0.107 core shear (S 100.2 N/mm), per mm width x 70 mm, back_system2 RECOMMENDED (computed by BL-023). Fail: BP-08 twill backer back (+44.6 g)

Defined in: designs/pack_5200/analysis/back_system_integration.md §6

## 8. HB-6: M5 heat-set insert pull-out in printed PAHT-CF

Why: hip pivot mount on H1

Specimen: 5 inserts in printed, annealed PAHT-CF blocks

Steps

1. Bolt an M5 eye into each insert

2. Pull along the insert axis to 1.5 kN

Pass: >= 1.5 kN, 5 of 5

Write in the results sheet (results_template.csv, column measured):

- HB-6-01 (1): passes the register criterion: >= 1.5 kN, 5 of 5. Type Y or N.

Note: D-I1 (frame D) and CT-B5 (frame B) merged here

Defined in: designs/pack_5200/analysis/hipbelt_structural.md §8; designs/pack_5200/analysis/fabric_swap_evidence.md (release gate)

## 9. CA-3: interlayer after 40 C water soak

Why: FI interlayer term; plug shear-out

Specimen: 3 Z bars + 3 short-beam blocks 24 x 8 x 6

Steps

1. 48 h in 40 C water

2. Static Z pull

3. 3-point short-beam shear, span 5 x thickness

Pass: Z >= 37.6 MPa; interlayer shear >= 22.6 MPa

Write in the results sheet (results_template.csv, column measured):

- CA-3-01 to CA-3-03 (Z#1 to Z#3): Z strength after 48 h at 40 C water. MPa, pass if at least 37.6.

- CA-3-04 to CA-3-06 (SB#1 to SB#3): short-beam interlayer shear. MPa, pass if at least 22.6.

Note: CT-B2 merged

Defined in: designs/pack_5200/tests_physical/CA1_fatigue_rig.md §7; designs/pack_5200/analysis/frame_comp_A.md

## 10. CA-4: creep-rupture 40 C 8 h

Why: creep margin

Specimen: 2 XY bars

| load | N | lb | kg on the scale |
|---|---|---|---|
| dead load at the lever end (4:1) | 282 | 63 | 28.7 |

Steps

1. Dead weight 29 kg on the lever end (282 N link)

2. Bars in a 40 C box for 8 h

3. Log stretch

Pass: no rupture

Write in the results sheet (results_template.csv, column measured):

- CA-4-01 to CA-4-02 (#1 to #2): no rupture after 8 h at 40 C (282 N link). Type Y or N.

Defined in: designs/pack_5200/tests_physical/CA1_fatigue_rig.md §7

## 11. HB-5: 7075-T6 1.2 bow-tie plate proof + 1e5 cycles

Why: C-1 + C-2 + C-3

Specimen: 1 plate on the Ti pin over a foam block

Steps

1. Template-check end slope 14.6 +/- 2 deg

2. 1.0 kN per ear + 1.6 kN vertical with a scribed grid

3. Then 1e5 walking cycles

4. Re-check slope

Pass: no yield (grid unchanged), slope within tolerance after cycling, no crack

Write in the results sheet (results_template.csv, column measured):

- HB-5-01 (1): passes the register criterion: no yield (grid unchanged), slope within tolerance after cycling, no crack. Type Y or N.

Note: CONFLICT: two definitions (6061 1.5 vs 7075 1.2); 7075 adopted 2026-11-09

Defined in: designs/pack_5200/analysis/weight_c_structural.md §7 (supersedes designs/pack_5200/analysis/hipbelt_structural.md §8 6061 version)

## 12. T6: crown-corner bond coupons (0.8 mm bondline)

Why: carbon crown corners + extension-post socket (P-1)

Specimen: 3 corner coupons, 0.8 mm glass-bead bondline

Steps

1. Build like the frame

2. Cure

3. Pull to failure

Pass: mean >= 1.5 x, lowest >= 1.25 x the largest FEA limit end force (frame_sizing_cases_19in.md); cohesive failure

Write in the results sheet (results_template.csv, column measured):

- T6-01 (1): passes the register criterion: mean >= 1.5 x, lowest >= 1.25 x the largest FEA limit end force (frame_sizing_cases_19in.md); cohesive failure. Type Y or N.

Note: type (b) collar-on-stay retired: H1 stays are printed. Numeric pass force not restated in any test doc (FLAG)

Defined in: designs/pack_5200/analysis/load_standard.md §5.3; designs/pack_5200/print/frame/PRODUCTION.md (T6 note)

## 13. T7: bushing crush coupon

Why: crown bushing ring crush credit

Specimen: 150 mm printed H1 sleeve segment + UHMWPE bushing + leg offcut

Steps

1. Press the leg sideways at the bushing centre to 1.5 x the max FEA bushing force, unload, measure

2. Then to failure

Pass: no crack; ovalisation <= 0.2 mm after unload at 1.5 x

Write in the results sheet (results_template.csv, column measured):

- T7-01 (1): passes the register criterion: no crack; ovalisation <= 0.2 mm after unload at 1.5 x. Type Y or N.

Note: CONFLICT: defined on a CFRP stay with a wrap collar; H1 carries the bushing in a printed sleeve, so the specimen is revised. Force value not restated (FLAG)

Defined in: designs/pack_5200/analysis/load_standard.md §5.3

## 14. CA-2: anneal 90 C vs 120 C

Why: nothing until a gain is measured

Specimen: 3 XY + 3 Z bars per anneal

Steps

1. Static pulls of both sets

Pass: credit only a measured gain (0-11 % expected)

Write in the results sheet (results_template.csv, column measured):

- CA-2-01 (1): passes the register criterion: credit only a measured gain (0-11 % expected). Type Y or N.

Note: grams not quantified anywhere

Defined in: designs/pack_5200/analysis/frame_comp_A.md; docs/research/frame_comp_A.md

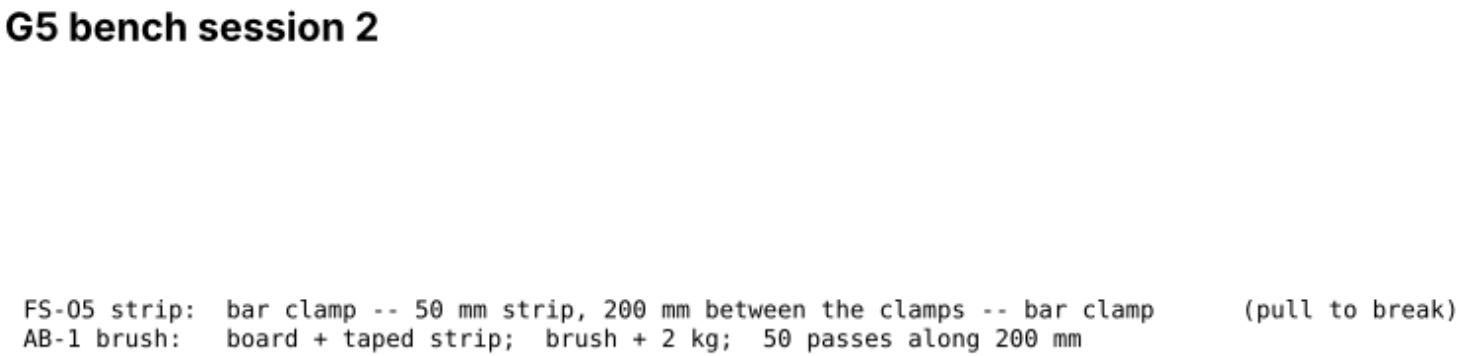
## After the session

1. Fill measured and date for every row you ran; leave the rest empty.

2. Photograph every failed or doubtful sample; put the photo number in photo.

3. Send the sheet. The engineers read it back with huntkit.manufacturing.test_results.

# G5: Bench session 2: fabric, belt and back coupons




When: any weekend while the lever rig runs (weeks 2-4)

Rig: kit A pull bench + kit D

Kits: A, D

Sewn and laminated coupons that do not need the frame. Send the AB-1 lab samples on the first day: the lab takes about 4 weeks.

About 9.9 h hands-on and 8 h of machine time.

## Safety (read before you start)

- Same bench rules as G2. FS-O5 strips break suddenly: blanket over the clamps.

## Rig sketch

FS-O5 strip: bar clamp -- 50 mm strip, 200 mm between the clamps -- bar clamp (pull to break)

AB-1 brush: board + taped strip; brush + 2 kg; 50 passes along 200 mm

Picture: sketches/G5_rig.png

Make first: the specimens listed under each test (register specimen). Ship the AB-1 lab samples on day 1.

## Tests in this session, in order

| # | test | releases |
|---|---|---|
| 1 | WP-2 | Bag release, step 2: batch |
| 2 | AB-1 | BL-037 weight gates |
| 3 | HB-4 | Field release and the rest |
| 4 | HB-T | Field release and the rest |
| 5 | PS-1 | Field release and the rest |
| 6 | BP-2 | Field release and the rest |
| 7 | HB-2 | Field release and the rest |

# 1. WP-2: TPU 840D lash-tab slot pull (C-6 / O-15)

Why: C-6 lash tabs Hypalon -> TPU 840D

Specimen: 5 laser-cut B-11 tabs

Steps

1. Strap through the slot pulled in its working direction to 450 N

2. Hold 60 s

Pass: 5 of 5 hold 450 N, no slot tear

Write in the results sheet (results_template.csv, column measured):

- WP-2-01 (1): passes the register criterion: 5 of 5 hold 450 N, no slot tear. Type Y or N.

Note: slot direction still to be confirmed on the tab drawing (materials-engineer flag)

Defined in: designs/pack_5200/analysis/corrie_body.md §12.4; designs/pack_5200/analysis/weight_pass_corrie.md C-6

# 2. AB-1: HyperD 300 vs 630D printable: Martindale 8000 rev, brush drag, print swatch

Why: BL-037 G-2 (-22.5 g)

Specimen: 1 yd HyperD 300 printable with the owner's camo (custom print); a 630D printable offcut; 2 lab samples 300 x 300 mm

Steps

1. Order the HyperD print (an 8 x 8 in swatch first if offered)

2. Send one 300 x 300 mm sample of each fabric to a textile lab for Martindale abrasion (ASTM D4966 or ISO 12947-2) to 8000 rev, end point a hole or 2 broken yarns

3. At home tape a 50 x 300 mm strip of each to a board and drag a stiff nylon brush loaded with 2 kg along 200 mm, 50 passes

4. Look with the loupe

5. Judge the printed camo in daylight at 1 m and 10 m

Pass: HyperD: no hole at 8000 rev; brush strip no worse than the 630D strip; owner OK on the print in writing

Write in the results sheet (results_template.csv, column measured):

- AB-1-01 (lab): HyperD 300: no hole at 8000 rev. Type Y or N.

- AB-1-02 (lab): 630D: rev at end point (reference). Type the value in rev.

- AB-1-03 (brush): HyperD strip no worse than 630D after 50 passes. Type Y or N.

- AB-1-04 (print): owner OK on the camo print (in writing). Type Y or N.

Note: BL-038: added (BL-037). LONG LEAD: lab turnaround about 4 weeks after receipt; custom print lead time unknown

Defined in: designs/pack_5200/analysis/weight_push.md 2 (G-2), 7 HYPERD-ABRASION; data/materials/textiles/polyester_300d_printable.yaml

## 3. HB-4: hipbelt stitched-ear joint

Why: belt B wing joint

Specimen: 1 ear stitched into a wing-root coupon (3 recommended)

Steps

1. Clamp the wing coupon

2. Pull the ear at 20 deg to 1.5 kN

3. Measure the holes

Pass: no thread break; hole elongation < 0.5 mm

Write in the results sheet (results_template.csv, column measured):

- HB-4-01 (1): passes the register criterion: no thread break; hole elongation < 0.5 mm. Type Y or N.

Note: defined on the 6061 1.5 mm plate; repeat on the adopted 7075 1.2 mm plate

Defined in: designs/pack_5200/analysis/hipbelt_structural.md §8

## 4. HB-T: 50 mm buckle tab + 25 mm anchor and lock-tab box-X pull

Why: belt B 50 mm buckle tabs (C8 booking)

Specimen: 5 tabs sewn like the belt: 50 mm webbing round a buckle bar, 25 mm anchor and lock-tab box-X

Steps

1. Tab round a steel bar (no buckle) on the pin fixture

2. Pull with the come-along to 1500 N (152.9 kg)

3. Hold 60 s

Pass: 5 of 5 hold 1500 N for 60 s, no broken stitch

Write in the results sheet (results_template.csv, column measured):

- HB-T-01 (1): passes the register criterion: 5 of 5 hold 1500 N for 60 s, no broken stitch. Type Y or N.

Note: 1500 N is above the 1480 N ratchet-strap limit: come-along only

Defined in: designs/pack_5200/analysis/loadpath_verification.md §3 H1, §8

## 5. PS-1: pad-seat backer sag

Why: C-4 correction (pad seat)

Specimen: pad seat on a printed pan offcut

Steps

1. 413 N (42.1 kg) on a 150 x 190 foam block for 8 h

2. Then 661 N (67.4 kg)

3. Dial in the channel

Pass: sag < 3 mm; no contact with a dummy plate spine at 4.5 mm

Write in the results sheet (results_template.csv, column measured):

- PS-1-01 (1): passes the register criterion: sag < 3 mm; no contact with a dummy plate spine at 4.5 mm. Type Y or N.

Note: CONFLICT: corrie_body has 413 N only and 'no contact'; weight_c adds 661 N and < 3 mm. Stricter weight_c version used

Defined in: designs/pack_5200/analysis/weight_c_structural.md §7; designs/pack_5200/analysis/corrie_body.md §12.4

## 6. BP-2: stay contact width on the backer strip (pressure film)

Why: BL-005 sandwich strip (A_back)

Specimen: 100 mm printed H1 stay segment (z about 260) on a strip offcut, 10 mm foam under the strip

Steps

1. Carbon paper + white paper (or a Prescale LW sheet) between stay and strip

2. Dead load 25.5 kg (2.5 N/mm x 100 mm) for 2 min

3. Measure the imprint width at 5 stations

Pass: imprint width >= 8 mm at all 5 stations

Write in the results sheet (results_template.csv, column measured):

- BP-2-01 (1): passes the register criterion: imprint width >= 8 mm at all 5 stations. Type Y or N.

Note: ASSUMPTION: a carbon-paper imprint stands in for pressure film; if the width is within 2 mm of 8 mm, buy one Prescale LW sheet and repeat. Fail: structural re-checks core crush on the measured width

Defined in: designs/pack_5200/analysis/back_system_integration.md §6

## 7. HB-2: belt face friction

Why: silicone grip print (A-1; OD-HB4)

Specimen: belt face fabric with and without the grip print; polyester and merino base layers, dry and sweaty

Steps

1. Base layer taped to a board

2. 500 g weighted sled with the face fabric on it

3. Raise one end slowly until it slides

4. Read the angle in phyphox

5. Mu = tan(angle)

6. 5 runs each

Pass: mu >= 0.45 credits the grip print

Write in the results sheet (results_template.csv, column measured):

- HB-2-01 (1): passes the register criterion: mu >= 0.45 credits the grip print. Type Y or N.

Note: ASSUMPTION: tilt-board static mu replaces the ASTM D1894-style sled pull (no low-range force gauge in the kits); grip print adds 4.9 g if adopted

Defined in: designs/pack_5200/analysis/hipbelt_fit.md §10

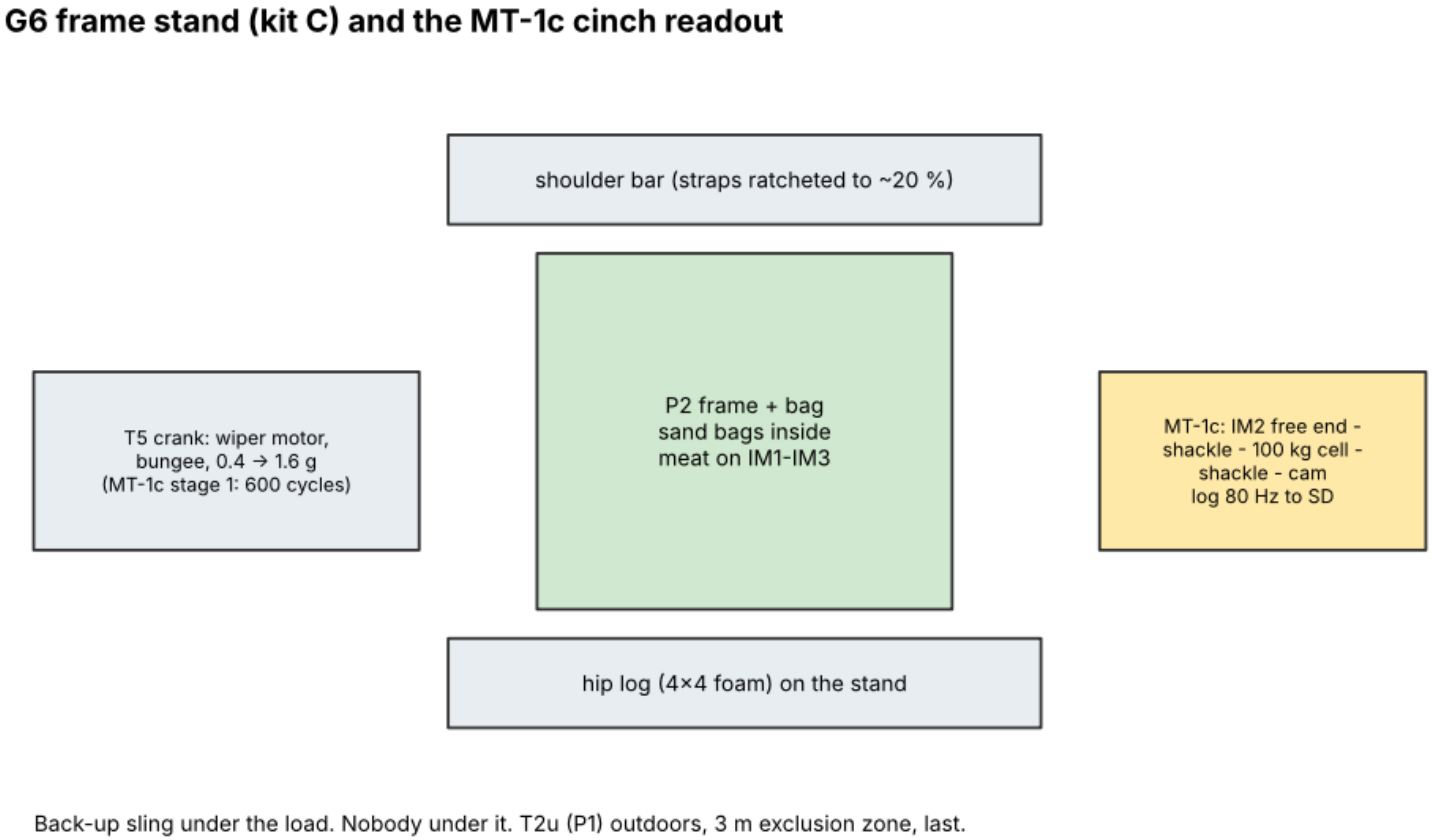
## After the session

1. Fill measured and date for every row you ran; leave the rest empty.

2. Photograph every failed or doubtful sample; put the photo number in photo.

3. Send the sheet. The engineers read it back with huntkit.manufacturing.test_results.

# G6: First-article frames on the stand (P1 then P2)




When: after G4's CA-1, IM3-EC and CA-5 pass (about week 5-6)

Rig: kit C frame test stand (hip log, shoulder bar, sand bags, T5 crank) + kit A scale and come-along

Kits: C, A, B (load cell), D

Frame release R1 completes here: IM3-EP on P1, then T9, T1, T2 and MT-1c on P2. Nobody carries the frame until T1 to T4 have passed. T2u destroys P1 and is always last.

About 58.7 h hands-on and 37 h of machine time.

## Safety (read before you start)

- Back-up sling under every load. Nobody under the load. T2u (P1, 204 kg) outdoors with a 3 m exclusion zone.
- Nobody carries the frame until T1, T3, T2 and T4 have passed. MT-1c stage 1 runs on the stand for that reason.

- T4c: -20 C parts are brittle; gloves and glasses.

## Rig sketch

crown top bar --- shoulder bar (straps ratcheted to carry ~20 %)

|

[ P2 frame + bag, sand bags inside ]

|

hip log (foam 4x4) on the stand  come-along + crane scale pull down for T1 / T2 / T3

|

T5 crank: wiper motor -- crank -- bungee -- bag  (0.4 -> 1.6 g)

MT-1c:  IM2 strap free end -- [snap shackle] -- [100 kg S cell] -- [snap shackle] -- cam

HX711 + Nano + SD logger taped to the frame, 80 Hz

Picture: sketches/G6_rig.png

Before: CA-1 at the release level, IM3-EC and CA-5 have passed (G4), and MT-1 (G2). Print P1 and P2 from the released files, same spool and profile as the coupons. Sew the bag (G2 + G3 passed).

## Tests in this session, in order

| # | test | releases |
|---|---|---|
| 1 | IM3-EP | Frame release R1 |
| 2 | W-PULL | Frame release R1 |
| 3 | MT-1cr | Frame release R1 |
| 4 | T9 | Frame release R1 + Bag release, step 2: batch |
| 5 | T9-R3a | Bag release, step 2: batch |
| 6 | FIT-US | Bag release, step 2: batch |
| 7 | T9-R3c | Bag release, step 2: batch |
| 8 | BP-3 | Field release and the rest |
| 9 | RF-1 | Field release and the rest |
| 10 | T1 | Frame release R1 |
| 11 | T1c-SW | Frame release R1 |
| 12 | T3 | Field release and the rest |
| 13 | T2 | Frame release R1 |
| 14 | MT-1c | Frame release R1 |
| 15 | CR-1 | Field release and the rest |
| 16 | BU-1 | BL-037 weight gates |
| 17 | SP-1 | Field release and the rest |

| # | test | releases |
|---|---|---|
| 18 | T4a | Field release and the rest |
| 19 | T4b | Field release and the rest |
| 20 | T4c | Field release and the rest |
| 21 | T5 | Field release and the rest |
| 22 | T5f | Field release and the rest |
| 23 | T2u | Field release and the rest |

# 1. IM3-EP: IM3 eye pull on the first printed frame

Why: H1 release; P_cheek

Specimen: P1 frame (the sacrificial frame), both IM3 eyes, each with its 38 mm IME eye sewn as op 'IME fairlead eyes' (BL-044: NECKED to 18 mm round the pin, the 20 mm throat of the BL-045 fairlead)

| load | N | lb | kg on the scale |
|---|---|---|---|
| eye pull | 2553 | 574 | 260.2 |

Steps

1. Frame clamped to the kit C stand

2. Sewn T3 tongue + carabiner through the eye

3. Pull along the rev C IM3 vector (frame_h1_results.json) to 2553 N (260.2 kg)

4. Hold 60 s

5. Loupe

Pass: both eyes hold 2553 N for 60 s, no crack, no whitening at the cheek root

Write in the results sheet (results_template.csv, column measured):

- IM3-EP-01 (1): passes the register criterion: both eyes hold 2553 N for 60 s, no crack, no whitening at the cheek root. Type Y or N.

Note: on P1 before T2u; P2 is not pulled. The throat fit check (two carabiner rods) is done here too

Defined in: designs/pack_5200/analysis/loadpath_verification.md §8 (FAIRLEAD_DIR), §4.1

# 2. W-PULL: W1-W3 lash points on the first frame

Why: W1-W3 frame anchors (BL-003)

Specimen: P1 frame, 6 lash points (z 80 / 250 / 420, both sides)

Steps

1. 25 mm webbing loop through each lash point

2. Pull along the strap line to 338 N (34.5 kg)

3. Hold 60 s

Pass: 6 of 6 hold 338 N, no crack, no set

Write in the results sheet (results_template.csv, column measured):

- W-PULL-01 (1): passes the register criterion: 6 of 6 hold 338 N, no crack, no set. Type Y or N.

Note: same session as IM3-EP

Defined in: designs/pack_5200/analysis/loadpath_verification.md §3 S8, S9, §8

## 3. MT-1cr: MT-1 cradle pull on P1: cradle belly set by the user card (60 mm minimum, strap gauge) and US1 / US2 arm-root strain at 240 / 480 / 720 N per pin

Why: frame release R1 (US1 arm root: stub model 0.95 at a 50 mm belly) and the user card's belly rule

Specimen: P1 frame (the sacrificial frame) on the kit C stand, bag OFF, both US2 clevises bolted on as built, the US1 / US2 straps with their sewn gauge tacks and USA anchors; strain gauges (1 mm foil, 120 ohm, cyanoacrylate on the sanded PAHT) on the US1 and US2 arm roots (R side, bag face, 3 mm off the stay surface, gauge axis along the arm); root stress = strain x 3.86 GPa (youngs_modulus_xy_mpa, record pa12_cf_fdm_paht); TEMPERATURE (BL-054, review 5 m-8): wire each gauge as a HALF bridge with a dummy gauge of the same lot bonded on an unloaded offcut of the same print, lying beside the specimen (it cancels the PAHT-CF's thermal strain); write the room temperature at every reading; with no dummy, re-zero the bridge unloaded at every stop and log the temperature

| load | N | lb | kg on the scale |
|---|---|---|---|
| (A) sand quarter in the cradle | 196 | 44 | 20.0 |
| (B) pin pull, step 1 | 240 | 54 | 24.5 |
| (B) pin pull, step 2 | 480 | 108 | 48.9 |
| (B) pin pull, step 3 | 720 | 162 | 73.4 |

Steps

- Step A. BELLY: rig the bag-off cradle round a 20 kg sand bag as a quarter; a helper who has not read this sheet cams the US1 and US2 straps following the packaging card only (3 tries, re-rigged each time); lay a straight stick across each strap's two pins and measure the strap's drop at mid-span with a rule; record 3 x 2 bellies and the gauge readings

- Step B. PIN PULLS: per strap, eye on its pin, the strap led along its lane leg (45 deg below the pin line for US1, 23 deg for US2, the 70 mm design belly; h1_crown.json belly_sweep) to a turnbuckle and the crane scale anchored to the stand; zero the gauges; pull to 240 / 480 / 720 N (24.5 / 48.9 / 73.4 kg), hold 10 s at each, read both gauges; unload, read the zero again; loupe on both arm roots

Pass: (A) every measured belly >= 60 mm (6 of 6); (B) measured root stress at each step <= the h1_crown lug model's root stress at that load (the structural-fea-engineer reads it off), zero returns within 5 % of the 720 N reading, no crack or whitening at either root

Write in the results sheet (results_template.csv, column measured):

- MT-1cr-01 to MT-1cr-06 (try 1 US1 to try 3 US2): cradle belly below the pin line, set by the packaging card. Mm, pass if at least 60.

- MT-1cr-07 to MT-1cr-09 (US1 240 N to US1 720 N): US1 arm-root strain at the pin pull (gauge). Type the value in microstrain.

- MT-1cr-10 to MT-1cr-12 (US2 240 N to US2 720 N): US2 arm-root strain at the pin pull (gauge). Type the value in microstrain.

- MT-1cr-13 (roots): no crack or whitening at the US1 / US2 arm roots; zero returns within 5 %. Type Y or N.

Note: BL-052: new. The 'MT-1 cradle pull' of h1_crown.md (the register's MT-1 is the bench box-X proof). Run on P1 before T1 on P2, so a root that reads above the model stops the carry tests

Defined in: designs/pack_5200/analysis/h1_crown.md 4 and 7 (ASSUMPTIONS BELLY-MIN, US1-STUB-GOVERNS, CROWN-SECTION); designs/pack_5200/analysis/corrie_f_body.md 26 (BL-052: BELLY-GAUGE)

## 4. T9: loaded CG (+ 15 in, 30 kg meat check); CORRIE F rev 3 items in rows T9-R3a..d (BL-038 split)

Why: HA-LS-1 rev C basis; before T1; CORRIE F rev 3 (LID-SIM, CROWN_PRE_N, ROLL-PATH, LID-SEAT, LID_SLIDER5_G, LID_THREAD)

Specimen: P2 frame + bag (rev C CG at 15 / 19 / 23 in); the rev 3 items are rows T9-R3a..d

| load | N | lb | kg on the scale |
|---|---|---|---|
| fill | 667 | 150 | 68.0 |
| meat check | 294 | 66 | 30.0 |

Steps

1. 68 kg fill and meat on IM1-IM3 at 15/19/23 in

2. Hang from haul loop, then from a hip pin, plumb both lines

3. Add 30 kg on IM1-IM3 at 15 in

4. Then rows T9-R3a (port guys, S kit CG), T9-R3c (crown strap path) in the same session

5. T9-R3b (cinch after a walk) and T9-R3d (lid) after T1-T4

Pass: bag CG <= 205 mm at z <= 590; meat <= 190; tall <= 185; rev 3 items: rows T9-R3a..d

Write in the results sheet (results_template.csv, column measured):

- T9-01 to T9-03 (15in to 23in): meat CG aft of the stay plane. Mm, pass if at most 190.

- T9-04 (fill): bag CG aft (uniform fill). Mm, pass if at most 205.

- T9-05 (tall): tall-load CG aft. Mm, pass if at most 185.

Note: corrie 11.4 plumb check and MOD-5 merged here; BL-038 split the rev 3 items (BL-030) into T9-R3a..d and moved the crown straps to BL-031's 25 mm per-kit cuts

Defined in: designs/pack_5200/analysis/load_standard.md §5.3; designs/pack_5200/analysis/corrie_body.md §11.4; designs/pack_5200/analysis/corrie_f_body.md §14.10, §15.6 (BL-030)

## 5. T9-R3a: T9 rev 3 (a): port guys hold the S-kit top, meat CG at 15 in

Why: CORRIE F rev 3 top as drawn (bag batch release)

Specimen: P2 frame, S crown kit (15 in), sewn rev 3 bag, port guys fitted, lid on with 1 kg inside

| load | N | lb | kg on the scale |
|---|---|---|---|
| fill incl. 30 kg meat | 667 | 150 | 68.0 |

Steps

1. Top packed full above the guys

2. 30 kg sand meat on IM1-IM3, fill to 68 kg

3. Hang from the haul loop and plumb a line

4. Hang from a pin through the hip pivot and plumb a second line

5. The CG is where the lines cross

6. Measure its distance aft of the stay plane

Pass: meat CG <= 190 mm aft (record against 179.3 mm)

Write in the results sheet (results_template.csv, column measured):

- T9-R3a-01 (S15): meat CG aft, S kit, top full, lid + 1 kg (sim 179.3). Mm, pass if at most 190.

Note: BL-038: split out of T9 (was T9 rev 3 item a); sim value updated to BL-033's 179.3 mm

Defined in: designs/pack_5200/analysis/corrie_f_body.md 14.5, 14.10; docs/DECISIONS.md 2026-11-13 BL-033 row (179.3 mm)

## 6. FIT-US: US fit on P1 with the bag: clevis bolt heads and clevis face vs the bag at every crown kit end (+ CLEVIS-FACE wear inspection); US straps on their lanes; US2 eye over the clevis seam

Why: bag first article (BAG-BATCH): a FAIL means a frame-side head change (counterbore or cap, owner decision) before more bags

Specimen: P1 frame with both US2 clevises bolted on (as built), the first sewn bag, US1 / US2 straps rigged (bag on), crown kits S, M, L

Steps

- Step 1. Colour each clevis bolt head AND each clevis outer face (x +/-183, z 127-157) with lipstick or chalk paste, a different colour on the face

- Step 2. Mount the bag; 30 kg sand meat on IM1-IM3, cinch the US straps; set the crown to 15 in (kit S shortest), run the T5 crank 200 cycles (or walk 5 min if T1-T4 have passed)

- Step 3. Take the bag off and look for colour on B-01, B-03 and the B-18 / floor corner; photo

- Step 4. Repeat at 19 in and 23 in (kits M, L)

- Step 5. Bag-off cradle: rig the US straps round a 20 kg sand bag as a quarter, check the US2 strap and its tail clear the heads

- Step 6. Slide the US2 eye across the clevis seam in the throat floor under load (SEAM-FLOOR)

- Step 7. CLEVIS-FACE INSPECTION (BL-052, ASSUMPTION CLEVIS-FACE), after step 4 and again after the T9 / T9-R3a walk on the same bag: bag off, lay it flat, find the face-colour patch on B-03 / the B-01 | B-03 corner (frame z 127-157 = about 92-122 mm above the floor seam, at the side seam), look at it under the 10x loupe against an unworn spot of the same panel: fuzz, pilled or broken face yarns, shine, coating worn through (hold to a light: pin-holes); photo both sides; record; IN THE SAME LOOK (review 5 m-9 / M-6): the US2 strap where it passes the clevis block's lower aft edge and the US1 strap where it leaves the US1 fairlead floor's lower end: fuzz, whitening or glazing on the strap, and whitening or a worn edge on the print; photo

- Step 8. US STRAP CLEARANCE ON THE REAL BAG (BL-054, review 5 m-5: the HE-FIT bench check sets the dowel under its pin, so it cannot show the bag pulling the strap sideways): bag ON, 30 kg sand meat on IM1-IM3, both US cams cinched tight as the user card says; at 15, 19 and 23 in (kits S, M, L) slide a 10 mm (0.39 in) feeler block between each US strap and the hinge fairlead cheeks and between each US strap and the HE (eye, legs, box), both sides, along the whole strap run you can reach; record the smallest gap per strap per setting

Pass: no colour from the HEADS on the bag at any crown setting; (8) the 10 mm feeler passes between every US strap and the hinge cheeks / HE at S, M and L, both sides (a gap under 10 mm or contact = FAIL, photo to the pack-designer); the strap and its tail clear the heads in the cradle; the US2 eye slides across the seam without catching; CLEVIS-FACE: face colour on the panel is expected, but no fuzz, pilling, broken yarn, shine or pin-hole there after the crank cycles and after the T9 walk; FAIL: photo to the pack-designer (panel let out about 7 mm over that band, or a wear patch) and the structural-fea-engineer

Write in the results sheet (results_template.csv, column measured):

- FIT-US-01 to FIT-US-03 (S (15 in) to L (23 in)): no colour from the clevis bolt heads on the bag. Type Y or N.

- FIT-US-04 (cradle): US2 strap and tail clear the bolt heads in the bag-off cradle. Type Y or N.

- FIT-US-05 (seam): US2 eye slides across the clevis seam under load without catching. Type Y or N.

- FIT-US-06 to FIT-US-07 (face after FIT-US to face after the T9 walk): bag panel over the clevis face (z 127-157): no fuzz, pilling, broken yarn, shine or pin-hole under the 10x loupe (face colour itself is expected). Type Y or N.

Note: BL-050 B (orchestrator): new. Run with T9-R3a on P1 / the first sewn bag. BL-052: heads counterbored (BL-051), the clevis face colour and the CLEVIS-FACE wear inspection (step 7) added. BL-054 (review 5 m-5 / m-9): step 8 (US strap clearance on the real bag, every kit) and the strap / lip look in step 7

Defined in: designs/pack_5200/analysis/h1_bedfit.md 7 (ASSUMPTIONS HEAD-PROUD, SEAM-FLOOR); designs/pack_5200/analysis/corrie_f_body.md 25-26 (BL-050 B: us_route.head_clearance; BL-052); designs/pack_5200/analysis/h1_crown.md 3a, 7 (ASSUMPTION CLEVIS-FACE)

## 7. T9-R3c: T9 rev 3 (c): crown strap free length (ROLL-PATH, TORSO-LIN)

Why: B-28 cut per crown kit (bag batch release)

Specimen: sewn rev 3 bag on P2 with the S kit at its short end (15 in) packed full, collar standing; then the L kit at 21.5 in, 6-day load

Steps

1. Cinch both crown straps

2. Measure the strap path from the bar tab to the ladder lock along the strap with a tape (both straps)

3. Repeat at the L kit 6-day setting

Pass: S kit full: path <= 968.1 mm (cut 1068.1 - 2 x 50 mm grip tail; sim 928.1 mm); L kit 6-day: record against the sim's 438.8 mm

Write in the results sheet (results_template.csv, column measured):

- T9-R3c-01 to T9-R3c-02 (S-full-L to S-full-R): crown strap path, S kit full. Mm, pass if at most 968.1.

- T9-R3c-03 (L-6day): crown strap path, L kit 6-day (sim 438.8). Type the value in mm.

Note: BL-038: split out of T9 (item c). BL-030's 1138 mm single 20 mm cut is replaced by BL-031's per-kit 25 mm cuts. BL-039 added 40 mm spare to every kit (BL-031 had none: the S sim path 928.1 mm equalled its limit); a path above 968.1 mm still means a longer S cut

Defined in: designs/pack_5200/analysis/corrie_f_body.md 15.6, 16.2, 16.5

## 8. BP-3: back pressure on the rails, below and above z 190

Why: back system release; BL-005 (A_back) skin model

Specimen: P2 frame + bag + Sandwich v2 back, crown kits S / M / L (15 / 19 / 23 in), 3 testers

Steps

1. Calibrate 8 FSRs on a 12.7 mm puck with 0.5 / 1 / 2 kg

2. Tape 4 per side on the stay line at z 100, 160 (below 190) and z 260, 400

3. T9 loaded state, 68 kg meat on IM1-IM3

4. Stand 2 min and walk 5 min at each torso

5. Read peaks

Pass: every sensor <= 14 kPa at 15, 19 and 23 in; record whether the z 100 / 160 sensors read > 1 kPa (rails bear below z 190)

Write in the results sheet (results_template.csv, column measured):

- BP-3-01 (1): passes the register criterion: every sensor <= 14 kPa at 15, 19 and 23 in; record whether the z 100 / 160 sensors read > 1 kPa (rails bear below z 190). Type Y or N.

Note: run in the T9 session with RF-1. Fail: lumbar-pad shim (structural to size; ASSUMPTION fallback, mass not yet booked) and re-test. FSR scatter about 10 % (ASSUMPTION): a reading within 1.4 kPa of the limit is repeated

Defined in: designs/pack_5200/analysis/back_system_integration.md §6, §8; designs/pack_5200/analysis/review_20261111.md S4

## 9. RF-1: back rail fit 130/70 on 3 backs (C-8)

Why: C-8

Specimen: rails on P2, 3 testers

Steps

1. FSRs under the rails at the stay line

2. 68 kg

3. 3 testers

Pass: peak <= 14 kPa and comfort OK

Write in the results sheet (results_template.csv, column measured):

- RF-1-01 (1): passes the register criterion: peak <= 14 kPa and comfort OK. Type Y or N.

Note: run with BP-3 (same FSR set-up, same session)

Defined in: designs/pack_5200/analysis/corrie_body.md §12.4; designs/pack_5200/analysis/weight_pass_corrie.md C-8

## 10. T1: static proof 150 / 300 / 225 lb (+ joint marks, yoke dial)

Why: first loaded carry

Specimen: P2 frame + bag

| load | N | lb | kg on the scale |
|---|---|---|---|
| (a) | 667 | 150 | 68.0 |
| (b) | 1334 | 300 | 136.0 |
| (c) hipbelt off | 1001 | 225 | 102.0 |

Steps

1. (a) 150 lb, (b) 300 lb, (c) belt off 225 lb 2 min

2. Dial on crown, lumbar and the yoke tip

3. Pencil marks across z 234

4. STRAIN GAUGES (BL-052): before loading, strain gauges (1 mm foil, 120 ohm, cyanoacrylate on the sanded PAHT) on the US1 and US2 arm roots (R side, bag face, 3 mm off the stay surface, gauge axis along the arm)

5. Root stress = strain x 3.86 GPa (youngs_modulus_xy_mpa, record pa12_cf_fdm_paht)

6. TEMPERATURE (BL-054, review 5 m-8): wire each gauge as a HALF bridge with a dummy gauge of the same lot bonded on an unloaded offcut of the same print, lying beside the specimen (it cancels the PAHT-CF's thermal strain)

7. Write the room temperature at every reading

8. With no dummy, re-zero the bridge unloaded at every stop and log the temperature

9. Zero both with the frame unloaded on the stand

10. Read both at 0 / 150 / 300 lb in (a) / (b) (and at the end of (c)) and record strain and root stress (results rows)

Pass: no crack, tap clean, crown set <= 1 mm, crown slip 0, re-load within 10 %, FEA +/-30 %; z 234 step <= 0.5 mm after unload; US roots: measured root stress (gauge) <= the h1_crown lug model's root stress at that load (the structural-fea-engineer reads it off; ASSUMPTION US1-STUB-GOVERNS / STAY-STUB / CROWN-SECTION)

Write in the results sheet (results_template.csv, column measured):

- T1-01 ((a) US1 root): US1 arm-root strain at 150 lb (gauge, BL-052). Type the value in microstrain.

- T1-02 ((a) US2 root): US2 arm-root strain at 150 lb (gauge, BL-052). Type the value in microstrain.

- T1-03 ((b) US1 root): US1 arm-root strain at 300 lb (gauge, BL-052). Type the value in microstrain.

- T1-04 ((b) US2 root): US2 arm-root strain at 300 lb (gauge, BL-052). Type the value in microstrain.

- T1-05 ((c) US1 root): US1 arm-root strain at 225 lb belt off, end (gauge, BL-052). Type the value in microstrain.

- T1-06 ((c) US2 root): US2 arm-root strain at 225 lb belt off, end (gauge, BL-052). Type the value in microstrain.

- T1-07 (all): T1 pass rules (no crack, crown set <= 1 mm, z 234 step <= 0.5 mm, ...). Type Y or N.

Note: D-T1 torsion check is referenced in frame_h1.md §8 but never defined (FLAG). BL-052: US1 / US2 arm-root strain gauges added (h1_crown.md 4)

Defined in: designs/pack_5200/analysis/load_standard.md §5.3; designs/pack_5200/analysis/frame_h1.md §8; designs/pack_5200/analysis/weight_c_structural.md §7; designs/pack_5200/analysis/h1_crown.md 4 (BL-052: US arm-root gauges)

## 11. T1c-SW: T1(c) hipbelt-off frame test with sway gauges (crown buckling check of the GNL model)

Why: frame release R1 (crown buckling, BL-047 path (a))

Specimen: P2 frame + bag, crown kit L at 23 in (kit S 15 in too if available)

| load | N | lb | kg on the scale |
|---|---|---|---|
| 25 % | 249 | 56 | 25.4 |
| 50 % | 498 | 112 | 50.8 |
| 75 % | 752 | 169 | 76.6 |
| 100 % hipbelt off, hold 2 min | 1001 | 225 | 102.0 |

Steps

1. As T1c_hipbelt_off_frame.md: hipbelt unbuckled, lumbar pad on the hip log

2. Dial gauges D1 / D2 on the stay feet (fore-aft), D3 crown top bar sideways, D4 crown top fore-aft

3. Crane scales S1 lifter, S2 lower strap

4. 56 / 112 / 169 / 225 lb (25 / 51 / 77 / 102 kg) steps, 1 min each, hold 225 lb 2 min

5. Unload

6. Run 2 lifters cinched

7. Optional free-hanging variant to 150 lb only

Pass: T1 rules; sway |D1-D2|/2 <= 3 mm at 225 lb, returns within 0.5 mm; sway(225) <= 2.5 x sway(112); Southwell P_cr >= 340 lb if readable; D3 <= 3 mm; record S1, S2, D4

Write in the results sheet (results_template.csv, column measured):

- T1c-SW-01 (run1): sway |D1-D2|/2 at 225 lb, lifters snug. Mm, pass if at most 3.

- T1c-SW-02 (run1): sway after unloading. Mm, pass if at most 0.5.

- T1c-SW-03 (run1): sway(225) / sway(112). Ratio, pass if at most 2.5.

- T1c-SW-04 (run1): Southwell P_cr (if sway > 0.3 mm). Lb, pass if at least 340.

- T1c-SW-05 (run1): D3 crown sideways at 225 lb. Mm, pass if at most 3.

- T1c-SW-06 (run2): sway |D1-D2|/2 at 225 lb, lifters cinched. Mm, pass if at most 3.

- T1c-SW-07 (run1): S1 lifter tension at 225 lb. Type the value in N.

- T1c-SW-08 (run1): S2 lower strap tension at 225 lb. Type the value in N.

Note: BL-048: register row added at the orchestrator's request (BL-047's test doc); runs after T1 (a)/(b) on P2

Defined in: designs/pack_5200/tests_physical/T1c_hipbelt_off_frame.md (BL-047); designs/pack_5200/analysis/crown_gnl.md; load_standard.md 5.3 C8

## 12. T3: side-hill 240 lb at 25 deg

Why: first loaded carry

Specimen: P2 frame

| load | N | lb | kg on the scale |
|---|---|---|---|
| side-hill 25 deg | 1068 | 240 | 108.8 |

Steps

1. Tilt rig 25 deg, downhill strap at 25 % tension, 240 lb pushed 30 % downhill

Pass: as T1; crown legs slide freely after

Write in the results sheet (results_template.csv, column measured):

- T3-01 (1): passes the register criterion: as T1; crown legs slide freely after. Type Y or N.

Defined in: designs/pack_5200/analysis/load_standard.md §5.3

## 13. T2: meat / tall-load proof 300 lb

Why: first meat haul

Specimen: P2 frame + bag

| load | N | lb | kg on the scale |
|---|---|---|---|
| meat / tall load | 1334 | 300 | 136.0 |

Steps

1. 300 lb sandbags in game bags inside the bag on IM1-IM3

2. Tall variant lashed to crown bar

Pass: as T1

Write in the results sheet (results_template.csv, column measured):

- T2-01 (1): passes the register criterion: as T1. Type Y or N.

Defined in: designs/pack_5200/analysis/load_standard.md §5.3

## 14. MT-1c: meat-strap cinch readout: does the IM cinch cycle with the walking g (CINCH_STATIC)

Why: frame release R1: R1 = CA-1 passes 0.40 with 3/3 run-outs AND MT-1c shows a cinch swing <= 117 % (peak-to-peak / mean) at a mean <= 150 N per anchor (structural countersign, adopted 2026-11-13, analysis/ca1_mt1c_confirm.md)

Specimen: P2 frame + sewn bag, 30 kg sand meat on IM1-IM3 cinched by hand as in the field; kit B 100 kg S-type cell + HX711 inline in the IM2 strap's free end (two snap shackles), logging to an SD card

| load | N | lb | kg on the scale |
|---|---|---|---|
| meat on IM1-IM3 | 294 | 66 | 30.0 |
| pack fill | 667 | 150 | 68.0 |

Steps

1. Cinch IM1-IM3 by hand as in the field and lock the cams

2. Log the IM2 strap tension at 80 Hz

3. Stage 1 on the stand: run the T5 crank 600 cycles (10 min, 0.4 -> 1.6 g) with the meat in

4. Stage 2 after T1-T4 pass: a 10 min loaded walk on rough ground

5. For each stage compute the MEAN tension and the mean peak-to-peak swing per cycle / step

6. Send the log

Pass: record BOTH the mean tension (N) and the larger peak-to-peak swing / mean of the two stages (%). R1 needs swing <= 117 % at a mean <= 150 N per anchor (with CA-1 passed at 0.40); if 0.40 fails: 3 XY bars at the table level for the measured swing, >= 0.313 (<= 0 %: 0.313 / <= 5 %: 0.316 / <= 10 %: 0.319 / <= 20 %: 0.325 / <= 30 %: 0.331 / <= 50 %: 0.345 / <= 70 %: 0.361 / <= 100 %: 0.386 / <= 115 %: 0.398). Swing > 115 %: STOP, no fallback level: the structural-fea-engineer re-runs the cinch case. A mean above 150 N: tell the structural-fea-engineer (the countersign table is at 150 N)

Write in the results sheet (results_template.csv, column measured):

- MT-1c-01 (stand): mean cinch tension per anchor, T5 crank 600 cycles. N, pass if at most 150.

- MT-1c-02 (stand): peak-to-peak swing / mean tension. %, pass if at most 117.

- MT-1c-03 (walk): mean cinch tension per anchor, 10 min walk. N, pass if at most 150.

- MT-1c-04 (walk): peak-to-peak swing / mean tension. %, pass if at most 117.

Note: BL-038: added. The register's MT-1 is the box-X proof; the R1 hold named 'MT-1' in DECISIONS is this cinch readout (h1_buckling.md). Stage 1 replaces the walk for R1 (no person under an unreleased frame); stage 2 confirms

Defined in: designs/pack_5200/analysis/h1_buckling.md (CINCH_STATIC); designs/pack_5200/analysis/h1_sleeve.md; docs/DECISIONS.md 2026-11-12 BL-024 / BL-027 rows (R1 waits on MT-1)

## 15. CR-1: crown retention hang 136 kg

Why: torso adjust

Specimen: P2 frame

| load | N | lb | kg on the scale |
|---|---|---|---|
| crown hang | 1334 | 300 | 136.0 |

Steps

1. Run the crown through full travel (pawl clicks)

2. Hang 136 kg from the crown top bar, frame upside down then upright

Pass: no slip either way

Write in the results sheet (results_template.csv, column measured):

- CR-1-01 (1): passes the register criterion: no slip either way. Type Y or N.

Defined in: designs/pack_5200/BUILD_PLAN.md §5

## 16. BU-1: 50 mm crown bushings: kit change and pawl travel, gloved

Why: BL-037 F-3 (-4.0 g)

Specimen: 2 UHMWPE crown bushings cut 50 mm from the 70 mm stock, on P2

Steps

1. Fit the 50 mm bushings

2. Winter gloves on

3. Change the crown kit S -> M -> L -> S and run the torso pawl (the LOW-PROFILE pawl crown_pawl_lp, BL-047) through every stop, 20 cycles

4. Note any stick

Pass: 20 of 20 without binding (slides by hand, no tool); the pawl engages every stop

Write in the results sheet (results_template.csv, column measured):

- BU-1-01 (1): gloved kit / pawl cycles without binding, out of 20. Count, pass if at least 20.

Note: BL-038: added (BL-037); CR-1 session

Defined in: designs/pack_5200/analysis/weight_push.md 2 (F-3), 7 BUSHING-LD; designs/pack_5200/analysis/crown_diet (e)

## 17. SP-1: spreader bone-end proof 400 N

Why: spreader release

Specimen: P2 spreader

| load | N | lb | kg on the scale |
|---|---|---|---|
| bone-end proof | 400 | 90 | 40.8 |

Steps

1. 400 N through a 22 mm dowel at window centre, US2 corner relief, IM1 notch root

2. Coin tap before/after

Pass: no crack, no skin-core disbond, deflection inside the 14 mm standoff

Write in the results sheet (results_template.csv, column measured):

- SP-1-01 (1): passes the register criterion: no crack, no skin-core disbond, deflection inside the 14 mm standoff. Type Y or N.

Note: T8 (same idea for the rejected d6L) retired

Defined in: designs/pack_5200/print/frame/PRODUCTION.md (QC proof load)

## 18. T4a: drop-step 2.0 g, 20 drops

Why: HA-LS-1 stumble level

Specimen: P2

| load | N | lb | kg on the scale |
|---|---|---|---|
| pack load, 20 drops at 2.0 g | 667 | 150 | 68.0 |

Steps

1. 150 lb

2. Lift by haul loop h mm, release on the snap shackle

3. Raise h in 5 mm steps to 2.0 g on phyphox

4. 20 drops

Pass: no damage; if 2.0 g needs < 10 mm, revisit the standard

Write in the results sheet (results_template.csv, column measured):

- T4a-01 (1): passes the register criterion: no damage; if 2.0 g needs < 10 mm, revisit the standard. Type Y or N.

Defined in: designs/pack_5200/analysis/load_standard.md §5.3

## 19. T4b: single-foot set-down (+ dial on the branch)

Why: branch buckling

Specimen: P2

Steps

1. 150 lb upright, raised 50 mm, dropped onto one foot on the gym mat, 5 x each foot

2. Dial on the upper branch

Pass: no damage at foot/lumbar, tap clean; record peak g and branch deflection

Write in the results sheet (results_template.csv, column measured):

- T4b-01 (1): passes the register criterion: no damage at foot/lumbar, tap clean; record peak g and branch deflection. Type Y or N.

Defined in: designs/pack_5200/analysis/load_standard.md §5.3; designs/pack_5200/analysis/frame_h1.md §8

## 20. T4c: empty abuse drop 122 cm at -20 C

Why: cold-weather use

Specimen: P2, empty, bag on

Steps

1. 4 h freezer soak

2. Drop 122 cm onto 19 mm plywood on concrete, 6 orientations

Pass: crown still telescopes and locks, no crack, tap clean

Write in the results sheet (results_template.csv, column measured):

- T4c-01 (1): passes the register criterion: crown still telescopes and locks, no crack, tap clean. Type Y or N.

Note: CONFLICT: load_standard has no temperature; three later docs say -20 C. -20 C used. ASSUMPTION household freezer ~-18 C

Defined in: designs/pack_5200/analysis/load_standard.md §5.3; designs/pack_5200/analysis/frame_comp_verdict.md §6; designs/pack_5200/analysis/printed_frame_study.md

## 21. T5: cycle 0.4 -> 1.6 g, 1e5 cycles (H1 body)

Why: frame release

Specimen: P2

| load | N | lb | kg on the scale |
|---|---|---|---|
| dead load | 267 | 60 | 27.2 |
| peak | 1068 | 240 | 108.8 |

Steps

1. 60 lb dead + crank/bungee to 240 lb peak, ~50/min

2. Stop every 25,000: T1(a) re-check + tap

Pass: no crack/debond, stiffness change <= 10 %, crown slip < 1 tooth

Write in the results sheet (results_template.csv, column measured):

- T5-01 (1): passes the register criterion: no crack/debond, stiffness change <= 10 %, crown slip < 1 tooth. Type Y or N.

Note: was 'on the C-9 body'; H1 replaced C-9

Defined in: designs/pack_5200/analysis/load_standard.md §5.3; designs/pack_5200/analysis/weight_c_structural.md §7

## 22. T5f: float US1 share 56 kg

Why: float meat limit 53 -> 56 kg

Specimen: frame + cradle, 56 kg wet sandbags

| load | N | lb | kg on the scale |
|---|---|---|---|
| float meat | 549 | 124 | 56.0 |

Steps

1. Scale in series with US1

2. Crank 0 -> 1.6 g

3. Record US1 share

Pass: share <= 0.95 x the no-friction value

Write in the results sheet (results_template.csv, column measured):

- T5f-01 (1): passes the register criterion: share <= 0.95 x the no-friction value. Type Y or N.

Note: optional; only if the owner wants 56 kg

Defined in: designs/pack_5200/analysis/float_fea.md §3; designs/pack_5200/analysis/load_standard.md §4.3

## 23. T2u: ultimate demo 450 lb on P1 (destroys P1)

Why: none after it (last test)

Specimen: P1 frame (sacrificial)

| load | N | lb | kg on the scale |
|---|---|---|---|
| ultimate, 30 s | 2002 | 450 | 204.0 |

Steps

1. Outdoors, backup sling, exclusion zone

2. Step to 450 lb, hold 30 s

Pass: still carries the load at 30 s; record first noise

Write in the results sheet (results_template.csv, column measured):

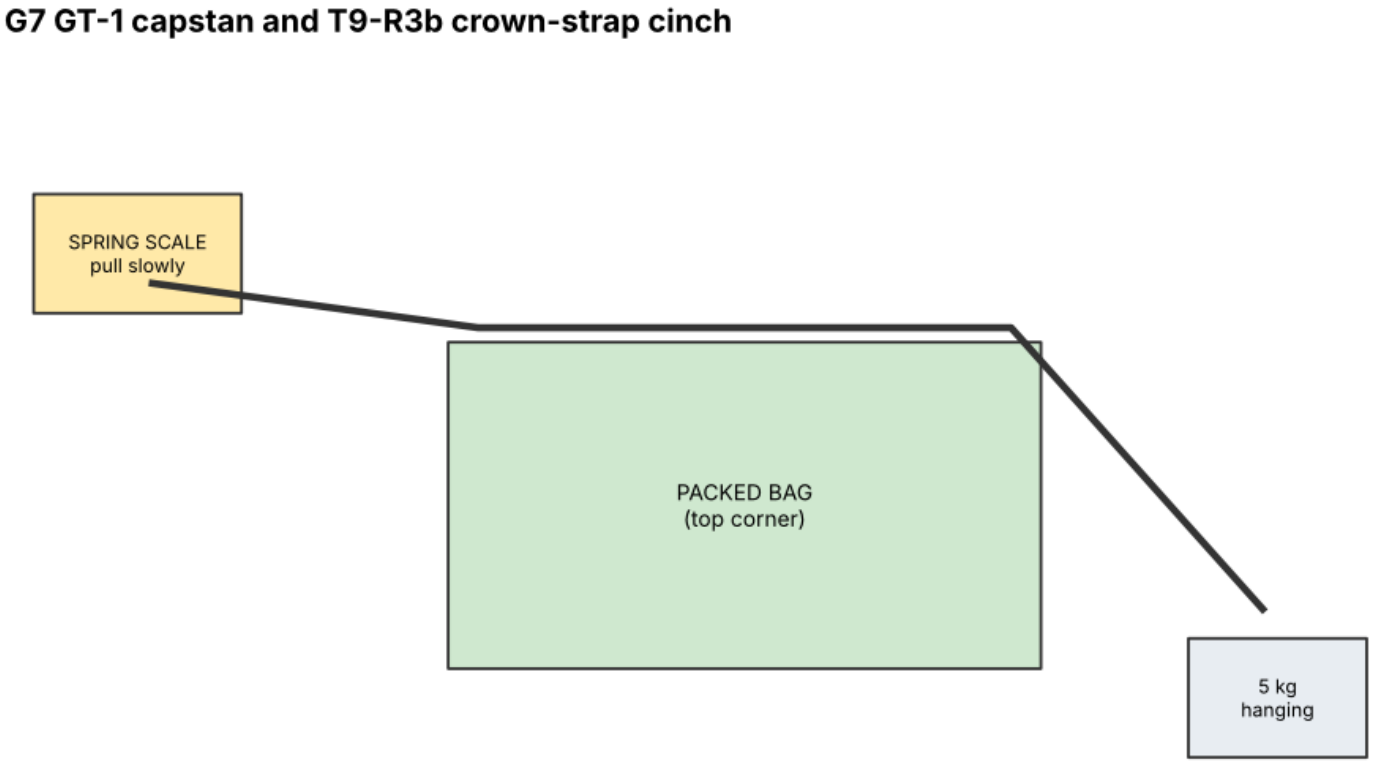
- T2u-01 (1): passes the register criterion: still carries the load at 30 s; record first noise. Type Y or N.

Defined in: designs/pack_5200/analysis/load_standard.md §5.3

## After the session

1. Fill measured and date for every row you ran; leave the rest empty.
2. Photograph every failed or doubtful sample; put the photo number in photo.
3. Send the sheet. The engineers read it back with huntkit.manufacturing.test_results.

# G7: Complete pack: fit, field and first-article bag checks




When: after T1-T4 pass

Rig: the complete pack, sand bags, spring scale, FSR strips, phone (phyphox)

Kits: D, E

The bag's batch release (T9-R3b/d, GT-1, PT-7) comes first in this session, then the BL-037 comfort gates, then the field tests.

About 25.9 h hands-on and 2 h of machine time.

## Safety (read before you start)

- Loaded walks: a second person nearby, flat ground first, stop at any crack noise. Float meat limit 53 kg (56 kg only after T5f).

## Rig sketch

GT-1 capstan:  spring scale <-- guy webbing over the packed bag's top corner --> 5 kg hanging

mu = ln(T / 49 N) / wrap angle (rad)

T9-R3b:  spring scale on each crown strap tail: pull to 8.2 kg (80 N), lock

Picture: sketches/G7_rig.png

Before: T1, T3, T2 and T4 have passed on P2 (G6). Then the pack may be carried.

## Tests in this session, in order

| # | test | releases |
|---|---|---|
| 1 | T9-R3b | Bag release, step 2: batch |
| 2 | T9-R3d | Bag release, step 2: batch |
| 3 | GT-1 | Bag release, step 2: batch |
| 4 | FIT-R3 | Bag release, step 2: batch |
| 5 | PT-7 | Bag release, step 2: batch |
| 6 | WP-4 | Bag release, step 2: batch |
| 7 | WP-5 | Bag release, step 2: batch |
| 8 | HS-1 | BL-037 weight gates |
| 9 | ST-1 | BL-037 weight gates |
| 10 | HB-1 | BL-037 weight gates + Field release and the rest |
| 11 | HB-D2 | BL-037 weight gates + Field release and the rest |
| 12 | HB-3 | Field release and the rest |
| 13 | HB-D1 | Field release and the rest |
| 14 | HB-D3 | Field release and the rest |
| 15 | FT-S2 | Field release and the rest |
| 16 | MH-1 | Field release and the rest |
| 17 | MH-2 | Field release and the rest |
| 18 | MH-3 | Field release and the rest |
| 19 | R28-1 | Field release and the rest |

## 1. T9-R3b: T9 rev 3 (b): crown strap cinch force and retention (CROWN_PRE_N, MU_LID)

Why: CORRIE F rev 3 crown straps (bag batch release)

Specimen: sewn rev 3 bag on P2, lid on, full top; 0-15 kg spring scale

| load | N | lb | kg on the scale |
|---|---|---|---|
| cinch | 80 | 18 | 8.2 |
| after the walk, at least | 40 | 9 | 4.1 |

Steps

1. Hook the spring scale on each B-28 tail

2. Pull to 8.2 kg (80 N) and lock

3. Walk 30 min loaded (only after T1-T4 pass)

4. Read the tension again (pull the tail until it just moves)

5. Set the pack down hard from 0.5 m on its base

6. Read again

7. Check the lid is still seated

Pass: after the walk >= 40 N, or the lid still seated on the full top (report both values)

Write in the results sheet (results_template.csv, column measured):

- T9-R3b-01 (L): left strap tension after the walk. Type the value in N.

- T9-R3b-02 (R): right strap tension after the walk. Type the value in N.

- T9-R3b-03 (both): >= 40 N after the walk OR lid still seated. Type Y or N.

Note: BL-038: split out of T9 (item b)

Defined in: designs/pack_5200/analysis/corrie_f_body.md 14.9, 15.6

## 2. T9-R3d: T9 rev 3 (d): sewn lid mass and seat (LID_SLIDER5_G, LID_THREAD, LID-SEAT)

Why: B-29 lid mass booking (bag batch release)

Specimen: first sewn B-29 lid; 0.01 g pocket scale (lid fits on a 200 g scale)

Steps

1. Weigh the lid alone 3 times

2. Photograph it seated on the packed 6-day bag and the full bag, from the rear 3/4 and the side

Pass: lid 47.8 g +/- 5 g (ledger); lid edge on the hinge-band seam, camo caps over the blades

Write in the results sheet (results_template.csv, column measured):

- T9-R3d-01 (lid): sewn lid mass (ledger booking). G, pass if 42.8 to 52.8.

- T9-R3d-02 (lid): lid seated as drawn (photo). Type Y or N.

Note: BL-038: split out of T9 (item d); mass from the ledger at run time

Defined in: designs/pack_5200/analysis/corrie_f_body.md 14.9, 14.10; huntkit/mass/ledger.py adopted_bag_rows (crown lid)

## 3. GT-1: capstan pull: guy webbing over the packed bag's corner (MU_WEB_FAB)

Why: guy friction share in the CG runs (bag batch release)

Specimen: first sewn bag, packed full; a 25 mm guy webbing

| load | N | lb | kg on the scale |
|---|---|---|---|
| hanging weight | 49 | 11 | 5.0 |

Steps

1. Lay the guy webbing over the packed bag's top corner

2. Hang 5 kg (49 N) on one end

3. Measure the wrap angle with phyphox (tilt of the free leg minus tilt of the hanging leg, in degrees)

4. Pull the other end slowly with the spring scale until the weight starts to rise

5. Read the peak T

6. 3 pulls

7. Mu = ln(T / 49) / (angle x pi / 180)

Pass: mu inside 0.20-0.40 on each pull (record the value); outside: tell the pack-designer (CG runs re-checked)

Write in the results sheet (results_template.csv, column measured):

- GT-1-01 to GT-1-03 (#1 to #3): mu = ln(T / 49 N) / wrap angle (rad). -, pass if 0.20 to 0.40.

Note: BL-038: added. Needs the first sewn bag

Defined in: designs/pack_5200/analysis/corrie_f_body.md 13.2, 13.8, 13.9; outputs/release/pack5200_corrie_f_rev3/manufacturing/release_tests.csv

## 4. FIT-R3: rev 3 first-article fit checks (yes / no): lid dart, B-11 loadouts, IM2 notch, carabiner gate on T3, roll clip with rifle + W3, muzzle gap

Why: CORRIE F rev 3 first article (bag batch release)

Specimen: first sewn bag + lid on P2; 5.0 L of dry rice in a bag; the owner's bow (in IMP-8) and rifle; the IM3 carabiners

Steps

- Step 1. LID-DART: fill the sewn lid with 5.0 L of dry rice, set it on a flat table: height 85 mm +/- 5, no pucker at the 4 corner darts (photo)

- Step 2. B11-PLACE: carry the bow in IMP-8 with the 2 BRL loops round the riser, and the rifle with the muzzle in the B-11 muzzle keeper, both sides: they sit and do not touch the frame (photo)

- Step 3. IM2 notch fit: IM2 strap girth-hitched on its eye seats in the fairlead notch (photo)

- Step 4. Carabiner gate: the IM3 carabiner clips the T3 loop and the gate closes fully, 5 of 5

- Step 5. ROLL CLIP + RIFLE + W3 (BL-041 RT-FWD, BL-043 W3-OVER / RT-ORIGIN): 6-day load, rifle muzzle in the keeper, roll clipped, W3 cinched OVER everything, both sides: with W3's G-hook off the roll clip opens and closes gloved without touching the rifle, the strap leaves the roll at its END (not out of a gusset tuck) and the buckle body clears the muzzle tab (photo)

- Step 6. MK-GAP: the barrel sits between the two muzzle tabs (30 mm clear) with the keeper cord closed round it

- Step 7. W3 SWEEP (BL-043 M-1), at BOTH ends of the owner's crown kit (shortest and longest torso setting): W3 cinched lies on webbing, cord or the roll only, never on the buckle body or the hose (measure the gap from W3's lower edge to the buckle top: >= 10 mm at the shortest setting), and the hose leaves its slot below W3 (photo each end)

Pass: 7 of 7 yes (any no: photo to the pack-designer)

Write in the results sheet (results_template.csv, column measured):

- FIT-R3-01 (lid): LID-DART: 5.0 L rice, height 85 +/- 5 mm, no pucker. Type Y or N.

- FIT-R3-02 (bow): B11-PLACE: bow in IMP-8 + BRL loops sits clear of the frame. Type Y or N.

- FIT-R3-03 (rifle): B11-PLACE: rifle muzzle keeper holds, clear of the frame. Type Y or N.

- FIT-R3-04 (IM2): IM2 strap seats in the fairlead notch. Type Y or N.

- FIT-R3-05 (gate): IM3 carabiner gate closes on the T3 loop, out of 5. Count, pass if at least 5.

- FIT-R3-06 (clip): roll clip opens / closes gloved with W3 off; strap leaves the roll END. Type Y or N.

- FIT-R3-07 (W3 short): shortest kit setting: gap W3 lower edge -> buckle top. Mm, pass if at least 10.

- FIT-R3-08 (W3 long): longest kit setting: W3 on webbing / cord / roll only, hose clear. Type Y or N.

Note: BL-040 (review m-18): added; the four fits were settled-by items with no row in the kit. BL-041 (re-review Y-2): rows 5-6 added. BL-043 (third review M-1 / M-2): row 5 for W3-OVER / RT-ORIGIN, row 7 (W3 at both ends of the owner's crown kit) added

Defined in: designs/pack_5200/patterns_corrie_f_rev3.py ASSUMPTIONS LID-DART, B11-PLACE; designs/pack_5200/analysis/loadpath_verification.md S1 (IM2 notch fit); data/materials/hardware/carabiner_wiregate_22kn.yaml (gate 17 vs 21 mm); review_20261113_rev3_release.md m-18

## 5. PT-7: door zip proof

Why: CORRIE door

Specimen: sewn bag

Steps

1. Zip on a 68 kg fill

2. 1.6 g walking load 30 min

3. 50 gloved open/close cycles

Pass: no tooth separation

Write in the results sheet (results_template.csv, column measured):

- PT-7-01 (1): passes the register criterion: no tooth separation. Type Y or N.

Defined in: designs/pack_5200/analysis/corrie_body.md §6

## 6. WP-4: rain-fly fit

Why: G-2

Specimen: F-01 over the packed shell

Steps

1. Pack the bag, wings closed, pouch on

2. Fit the fly

Pass: covers to the B-05 seam, no gap at the hinge band

Write in the results sheet (results_template.csv, column measured):

- WP-4-01 (1): passes the register criterion: covers to the B-05 seam, no gap at the hinge band. Type Y or N.

Defined in: designs/pack_5200/analysis/corrie_body.md §12.4

## 7. WP-5: TPU trim tick

Why: B-10 trims Hypalon -> TPU 840D

Specimen: ports, hose frames, cam-cup base

Steps

1. Walk with the trims against webbing

2. Listener at 1 m

Pass: no audible tick at 1 m

Write in the results sheet (results_template.csv, column measured):

- WP-5-01 (1): passes the register criterion: no audible tick at 1 m. Type Y or N.

Note: B-10 is in B.2: a fail adds 12.8 g back

Defined in: designs/pack_5200/analysis/corrie_body.md §12.4

## 8. HS-1: shoulder pressure: XLPE L200 vs EVA 45C J-strap load foam

Why: BL-037 H-1 (-22.3 g)

Specimen: J-strap pair: left with XLPE L200 5 mm load foam, right with EVA 45C 5 mm (comfort layer, shell, mesh as drawn); 2 FSRs per strap

| load | N | lb | kg on the scale |
|---|---|---|---|
| pack, run 1 | 294 | 66 | 30.0 |
| pack, run 2 | 667 | 150 | 68.0 |

Steps

1. Tape one FSR under each strap at the top of the shoulder and one 60 mm down the front

2. Calibrate each FSR on the puck first

3. 30 kg in the pack, walk 20 min, read the peaks at 5, 10, 15 and 20 min

4. Repeat at 68 kg

5. 3 testers

6. Ask each tester if either strap felt a hard spot (bottoming)

7. Swap sides for the third tester

Pass: XLPE peak / EVA peak <= 1.10 at 30 kg and 68 kg on all 3 testers; no bottoming felt; CS-1 passed

Write in the results sheet (results_template.csv, column measured):

- HS-1-01 to HS-1-06 (T1-30kg to T3-68kg): XLPE peak / EVA peak. -, pass if at most 1.10.

- HS-1-07 to HS-1-09 (T1 to T3): no bottoming felt. Type Y or N.

Note: BL-038: added (BL-037). Walking with 68 kg waits for T1-T4 (frame released)

Defined in: designs/pack_5200/analysis/weight_push.md 2 (H-1), 4, 7 XLPE-SHOULDER

## 9. ST-1: gloved one-hand sternum clip-on, 25 mm side-release

Why: BL-037 H-2 (-6.0 g)

Specimen: sternum strap with the 25 mm side-release buckle sewn on the harness

| load | N | lb | kg on the scale |
|---|---|---|---|
| pack | 147 | 33 | 15.0 |

Steps

1. Wear the loaded pack (15 kg)

2. Winter gloves on

3. Clip the sternum buckle with one hand, 10 tries

4. Time each

Pass: 10 of 10 one-handed (record the times)

Write in the results sheet (results_template.csv, column measured):

- ST-1-01 (1): one-hand gloved clip-ons, out of 10. Count, pass if at least 10.

Note: BL-038: added (BL-037); R-06 fit trial session

Defined in: designs/pack_5200/analysis/weight_push.md 2 (H-2)

## 10. HB-1: hipbelt pressure map, 3 testers, + front-strap sensor

Why: belt B fit (OD-HB1); BL-006 c2 (with HB-D2); front-strap pad (+42.9 g if it fails); BL-037 B-1 / B-3 (XLPE L200 wing load and lumbar foams, 4 mm wing comfort)

Specimen: belt B on 3 testers, 68 kg

Steps

1. FSRs under the belt at the crest and the ASIS, plus one FSR on the front strap 10-20 mm in front of the wing tip under the two 25 mm legs

2. Belt tension T_req read on the scale

3. Record peaks walking and standing

4. If the front-strap sensor fails, repeat with the 110 x 120 mm front-strap pad

Pass: every sensor <= 48.6 kPa (absolute pass, orchestrator call 2026-11-10); bony landmarks watched at 30 kPa; front-strap sensor <= 48.6 kPa without the pad; BL-037 build: wing worst <= 56.5 kPa, lumbar <= 48.6 kPa, hold not worse than belt B on the same testers

Write in the results sheet (results_template.csv, column measured):

- HB-1-01 (T1): wing worst. KPa, pass if at most 56.5.

- HB-1-02 (T1): lumbar worst. KPa, pass if at most 48.6.

- HB-1-03 (T1): front strap. KPa, pass if at most 48.6.

- HB-1-04 (T2): wing worst. KPa, pass if at most 56.5.

- HB-1-05 (T2): lumbar worst. KPa, pass if at most 48.6.

- HB-1-06 (T2): front strap. KPa, pass if at most 48.6.

- HB-1-07 (T3): wing worst. KPa, pass if at most 56.5.

- HB-1-08 (T3): lumbar worst. KPa, pass if at most 48.6.

- HB-1-09 (T3): front strap. KPa, pass if at most 48.6.

Note: crude estimate under the front-strap legs is 79-103 kPa (hipbelt_diet.md), so a fail is likely: then the pad pair (+42.9 g, loadpath_verification §5) is booked and HB-1 re-run with it. Old relative criterion (B <= current - 15 %) dropped; old belt not needed

Defined in: designs/pack_5200/analysis/hipbelt_fit.md §10; designs/pack_5200/analysis/hipbelt_diet.md (front strap); designs/pack_5200/analysis/loadpath_verification.md §5 (front-strap pad), §8 (PAD); docs/DECISIONS.md 2026-11-10 BL-016 and BL-006 rows

## 11. HB-D2: wing stiffener 0.75 mm roll-over (one wing each)

Why: BL-006 c2 (A_c2); BL-037 B-2 (0.5 mm stiffener)

Specimen: belt B wings side by side: 0.5 mm and 0.75 mm HDPE stiffener (and the 1.0 mm reference)

Steps

1. 68 kg, 30 min hike

2. Mark the crest line on both wings

3. Photograph the top edge at the crest every 10 min

4. HB-1 sensors on both wings

Pass: the thinnest stiffener that passes the HB-D2 roll-over criterion is kept (0.5 mm keeps BL-037 B-2)

Write in the results sheet (results_template.csv, column measured):

- HB-D2-01 (0.5mm): 0.5 mm wing passes roll-over. Type Y or N.

- HB-D2-02 (0.75mm): 0.75 mm wing passes roll-over. Type Y or N.

Note: fail: back to 1.0 mm (+9.8 g)

Defined in: designs/pack_5200/analysis/hipbelt_diet.md (c2, HB-D2)

## 12. HB-3: hipbelt slip / settle hike

Why: belt B

Specimen: complete pack 68 kg

Steps

1. 30 min hike

2. Mark the crest on the belt

3. Measure creep

Pass: <= 15 mm (ASSUMPTION)

Write in the results sheet (results_template.csv, column measured):

- HB-3-01 (1): passes the register criterion: <= 15 mm (ASSUMPTION). Type Y or N.

Defined in: designs/pack_5200/analysis/hipbelt_fit.md §10

## 13. HB-D1: no stabilizer straps: side-hill stability

Why: BL-006 b (A_b)

Specimen: complete pack, 68 kg sandbag, stabilizers fitted but slack vs cinched

Steps

1. 20 min side-hill traverse (about 25 deg) and a 10 min step-down set, stabilizers cinched, then fully slack

2. Same tester

3. Belt tension at T_req (luggage scale)

Pass: belt creep within HB-3's 15 mm both ways; no tester-reported roll or sway difference

Write in the results sheet (results_template.csv, column measured):

- HB-D1-01 (1): passes the register criterion: belt creep within HB-3's 15 mm both ways; no tester-reported roll or sway difference. Type Y or N.

Note: fail: refit the straps (+22.4 g); the frame anchor pads stay

Defined in: designs/pack_5200/analysis/hipbelt_diet.md (b, HB-D1)

## 14. HB-D3: pad-seat cord tie and 1000D sleeve retention

Why: BL-006 f1 + f2 (A_f)

Specimen: belt on the frame, pad seat with cord tie + cord lock and 1000D sleeve

Steps

1. Swing the pack off and set it down 20 times

2. Then hang the pack by the belt from the lumbar pad for 1 min with 10 kg in the bag

Pass: pad stays seated, cord lock holds, sleeve does not fray at the flange edge

Write in the results sheet (results_template.csv, column measured):

- HB-D3-01 (1): passes the register criterion: pad stays seated, cord lock holds, sleeve does not fray at the flange edge. Type Y or N.

Note: fail: strap + ladder lock and Hypalon sleeve back (+12.6 g)

Defined in: designs/pack_5200/analysis/hipbelt_diet.md (f1, f2, HB-D3)

## 15. FT-S2: no wings: 5 km side-hill + door cycles with the tripod loaded

Why: side wings deleted (A_wings)

Specimen: complete pack, no wings, tripod + spotter pod under W1/W2

Steps

1. Paint-pen mark tripod and pod against the gusset

2. 5 km of side-hill, both slope directions, measure shift every 1 km

3. Then open and close the door 5 times with the tripod loaded (unhook W1/W2, open, close, re-hook) and measure

Pass: shift <= 20 mm throughout; the tripod is never freed by a door cycle

Write in the results sheet (results_template.csv, column measured):

- FT-S2-01 (1): passes the register criterion: shift <= 20 mm throughout; the tripod is never freed by a door cycle. Type Y or N.

Note: door cycles added from BL-014 S7. Fail: a gusset tripod keeper (team B A2 32.2 g; BL-022 proposes a shock-cord keeper) or the wings back (+101.6 g)

Defined in: designs/pack_5200/analysis/strap_architecture.md (ASSUMPTION W1/W2); designs/pack_5200/analysis/corrie_f_body.md; docs/DECISIONS.md 2026-11-11 owner row; designs/pack_5200/analysis/review_20261111.md S7

## 16. MH-1: bag-on meat mode walk

Why: R-09

Specimen: complete pack

Steps

1. 30 kg sandbags in game bags high on B-01, IM1-IM3 cinched

2. 30 min on a slope

3. Time it gloved

Pass: no load shift you can feel; <= 60 s

Write in the results sheet (results_template.csv, column measured):

- MH-1-01 (1): passes the register criterion: no load shift you can feel; <= 60 s. Type Y or N.

Defined in: designs/pack_5200/BUILD_PLAN.md §5a.2

## 17. MH-2: frame-only meat haul (float / bag-off)

Why: R-08, float meat limit 53 kg

Specimen: frame + cradle, 36 kg sandbag quarter

Steps

1. IM1-IM3 + wraps

2. Walk 30 min

3. Measure slip

Pass: < 25 mm slip after 30 min

Write in the results sheet (results_template.csv, column measured):

- MH-2-01 (1): passes the register criterion: < 25 mm slip after 30 min. Type Y or N.

Note: stubs in step 1 became fairlead eyes; use the eyes

Defined in: designs/pack_5200/BUILD_PLAN.md §5a.4

## 18. MH-3: meat strap wash

Why: R-12

Specimen: IM straps

Steps

1. Unhook, machine-wash cold, dry

Pass: hooks and cams still work

Write in the results sheet (results_template.csv, column measured):

- MH-3-01 (1): passes the register criterion: hooks and cams still work. Type Y or N.

Defined in: designs/pack_5200/BUILD_PLAN.md §5a.5; designs/pack_5200/DESIGN.md R-12

## 19. R28-1: head-clearance photo

Why: R-28

Specimen: worn pack, collar unrolled

Steps

1. Side photo looking up

Pass: ~30 deg tilt before contact

Write in the results sheet (results_template.csv, column measured):

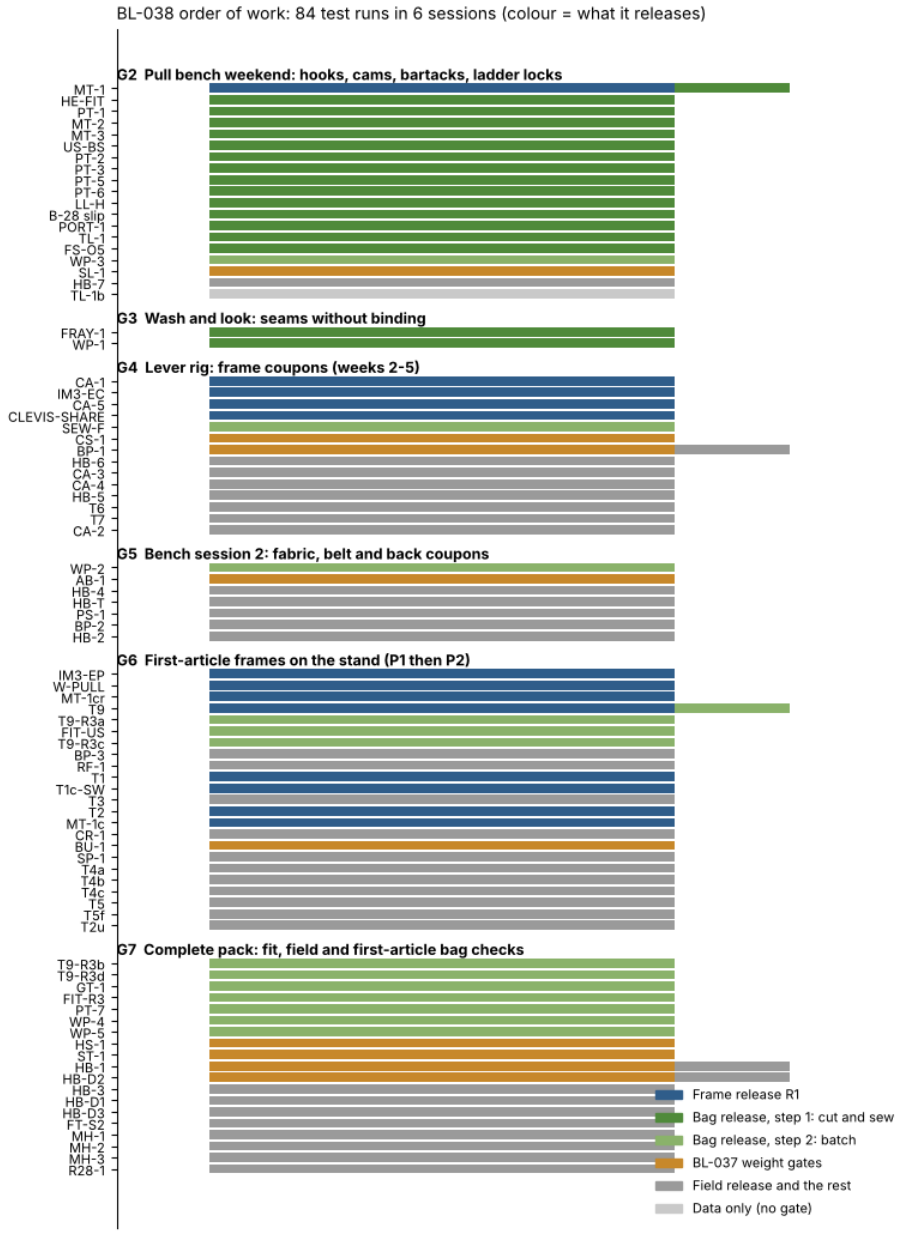
- R28-1-01 (1): passes the register criterion: ~30 deg tilt before contact. Type Y or N.

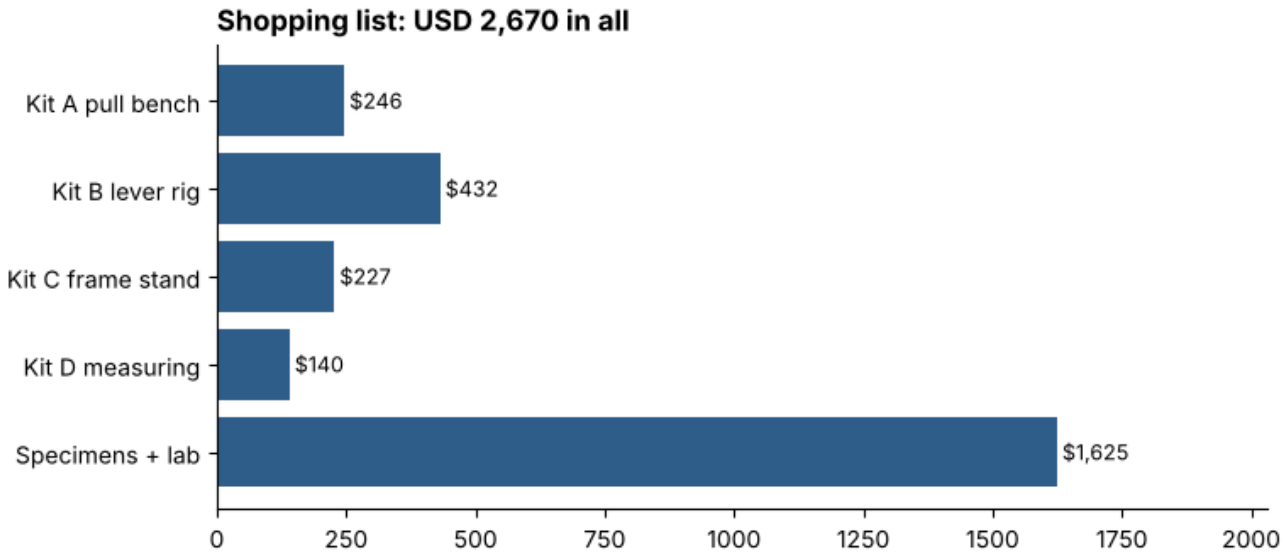
Defined in: designs/pack_5200/BUILD_PLAN.md §5a.6; designs/pack_5200/DESIGN.md R-28

## After the session

1. Fill measured and date for every row you ran; leave the rest empty.
2. Photograph every failed or doubtful sample; put the photo number in photo.
3. Send the sheet. The engineers read it back with huntkit.manufacturing.test_results.

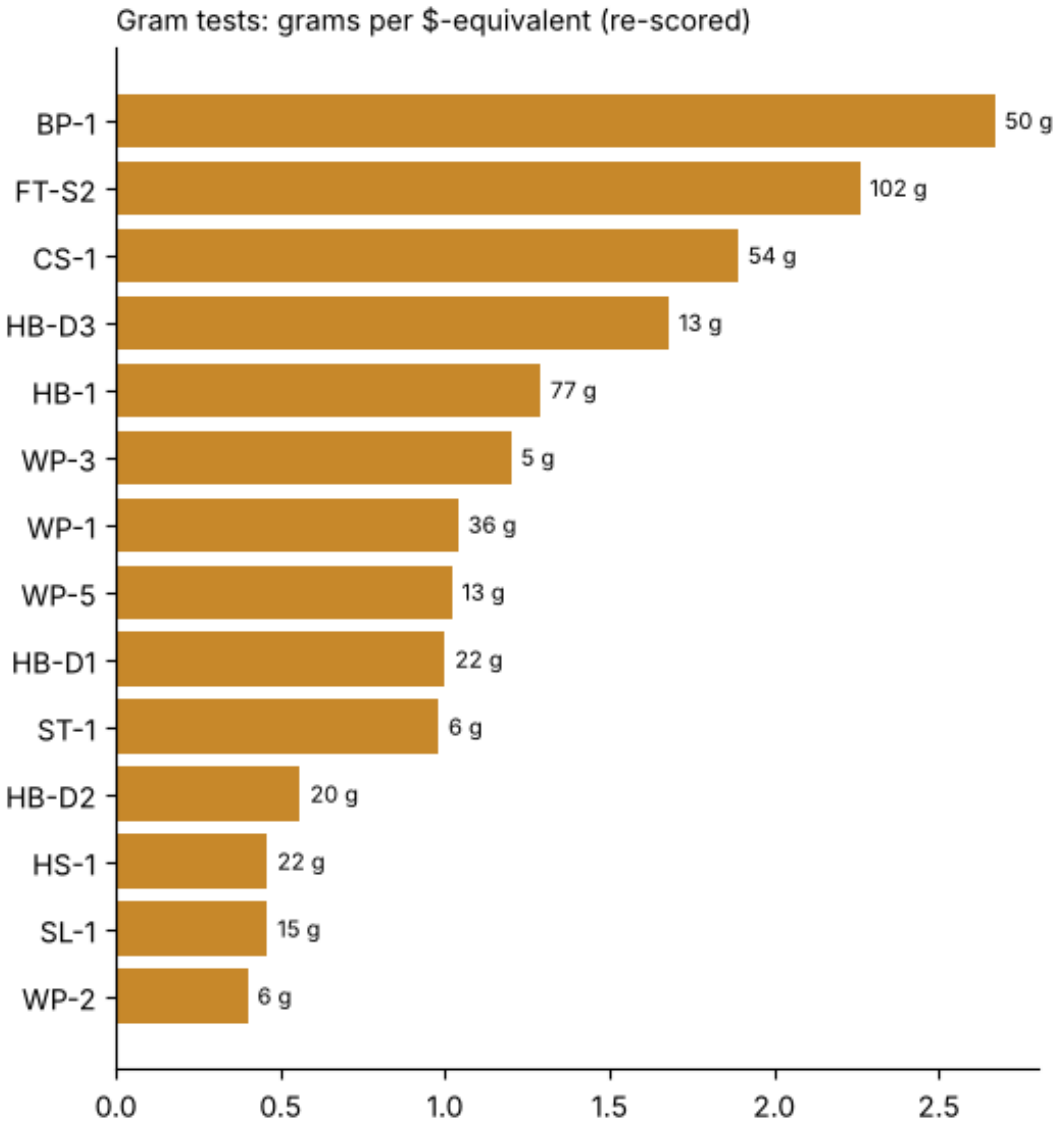
# Pictures


BL-038 order of work: 84 test runs in 6 sessions (colour = what it releases)
G2 Pull bench weekend: hooks, cams, bartacks, ladder locks
MT-1
HE-FIT
PT-1
MT-2
MT-3
US-BS
PT-2
PT-3
PT-5
PT-6
LL-H
B-28 slip
PORT-1
TL-1
FS-O5
WP-3
SL-1
HB-7
TL-1b
G3 Wash and look: seams without binding
FRAY-1
WP-1
G4 Lever rig: frame coupons (weeks 2-5)
CA-1
IM3-EC
CA-5
CLEVIS-SHARE
SEW-F
CS-1
BP-1
HB-6
CA-3
CA-4
HB-5
T6
T7
CA-2
G5 Bench session 2: fabric, belt and back coupons
WP-2
AB-1
HB-4
HB-T
PS-1
BP-2
HB-2
G6 First-article frames on the stand (P1 then P2)
IM3-EP
W-PULL
MT-1cr
T9
T9-R3a
FIT-US
T9-R3c
BP-3
RF-1
T1
T1c-SW
T3
T2
MT-1c
CR-1
BU-1
SP-1
T4a
T4b
T4c
T5
T5f
T2u
G7 Complete pack: fit, field and first-article bag checks
T9-R3b
T9-R3d
GT-1
FIT-R3
PT-7
WP-4
WP-5
HS-1
ST-1
HB-1
HB-D2
HB-3
HB-D1
HB-D3
FT-S2
MH-1
MH-2
MH-3
R28-1
Frame release R1
Bag release, step 1: cut and sew
Bag release, step 2: batch
BL-037 weight gates
Field release and the rest
Data only (no gate)




ORDER FIRST (long lead):

- cam buckle 25 mm aluminium (Austere type)  (MT-3, US-BS)
  LONG LEAD: RBTR all variants sold out, Extremtextil stock problems (2026-11-13). Order now; buy the pack's 5 cams in the same order
- YKK #10 coil slider, metal (cord pull fitted)  (SL-1)
  LONG LEAD: Rockywoods listed sold out (2026-11-13); Shelby Outdoor EU stocks the YKK 10C slider
- ROHACELL 71 IG-F core for the 2 mm strip  (BP-1)
  LONG LEAD: 2 mm is not a stock sheet (3 mm thinnest, back_system_integration.md); sand a 3 mm offcut or special order
- HyperD 300 printable, custom camo print  (AB-1, FRAY-1)
  LONG LEAD: custom print, lead time not published
- Martindale abrasion test, accredited lab (2 samples + order fee)  (AB-1)
  LONG LEAD: about 4 weeks after the samples arrive
- sacrificial P1 frame (IM3-EP, W-PULL, then T2u)  (IM3-EP, W-PULL, T2u)
  print time: allow 2 weeks with P2




Frame release R1

CA-1 at 0.40, 3/3 run-outs
(structural countersign,
adopted 2026-11-13)

AND MT-1c: swing <= 117 %
at mean <= 150 N per anchor

0.40 fails: table level for
the swing, >= 0.313

CA-1 at 0.40 is worth 9.9 g
(option F-1, not booked; was 60 g)
rank 18 of 20